

Prosociality and Layoffs*

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Abstract

Standard models treat layoffs as routine input adjustments; yet managers consistently describe them as the hardest and most painful decisions. Using novel U.S. data on layoffs, we document several patterns standard models struggle to explain, but that are consistent with empathy-driven prosocial motives. First, layoffs impose a personal health toll on managers, significantly reducing their lifespan. Second, CEOs' layoff aversion increases over the course of their tenure. Third, layoff aversion intensifies when the consequences for employees are more severe and when CEOs are socially or geographically closer to workers. Finally, layoff aversion is most pronounced among CEOs with empathy-related traits.

Keywords: Layoffs, CEOs, Managerial Prosocial Concerns, CEO Job Demands

JEL Codes: G30, M12, M50, D91

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Todos M

- remove same figure as right panel of figure 1 from appendix
- Make sure all appendix tables (with IV) have reasonable magnitudes
- Make sure MM is cited for LLM.

Todos Clint

- if we keep new ceo pay table, pls implement the IV version and add a sentence to the writeup

Todos Ulrike

- for acknowledgement, presented at more places right? e.g. bocconi?
- Review Matsa comments (referee?)? Review Ferreira comments (referee?)? How likely is either at AER?

Disclosure Statement
Prosociality and Layoffs

I have nothing to disclose.

Warm regards,
Marius Guenzel

Disclosure Statement
Prosociality and Layoffs

I have nothing to disclose.

Warm regards,
Clint Hamilton

Disclosure Statement
Prosociality and Layoffs

I have nothing to disclose.

Warm regards,
Ulrike Malmendier

In the standard firm investment model, the adjustment of human capital is treated like physical capital adjustment: each period, the firm optimizes over all factor inputs, subject to potential adjustment costs. In practice, however, CEOs describe their personal experiences with hiring and firing in starkly different terms. In a global survey of 2,000 executives, 60% report that hiring and firing was “the toughest decision” of their careers (Martin 2005). Similarly, senior managers of Fortune 500 firms interviewed in Hallock (2005) express that the “entire layoff process is gut wrenching” because it is “a human decision.” CEOs commonly describe the impact of layoffs on them personally as “something that took a heavy toll on me ... something I don’t ever want to do again.”¹ These first-hand accounts depict hiring and firing decisions as fundamentally different from the purchase, write-down, and divestment of machines.

Prior work on frictions in human capital choices emphasizes “classical” adjustment costs in hiring and firing, such as training new hires, job advertising and interviewing, and labor-market policies like severance pay and mandatory layoff notices (Hamermesh 1989).² In this paper, we propose a complementary, novel type of “behavioral adjustment costs” firms may face with respect to labor—but not capital—that captures layoffs being “a human decision.” *Managers feel empathy towards their employees, especially if they have known them for a while and are close to them. As a result, they dislike inflicting adversity on them and tend to shy away from layoffs.*

The proposed channel builds on prior evidence on managerial prosociality toward workers in both lab and field settings.³ While previous studies link CEOs’ prosocial attitudes to CEOs’ corporate social responsibility (Cheng, Hong and Shue 2023; Cronqvist and Yu 2017), we focus on the mass layoffs, where CEO decisions can negatively affect the lives of many employees. Such actions are widely viewed as among the most morally objectionable (Iliewa, Kempf and Spalt 2025).

We assemble a new, comprehensive dataset covering layoff announcements by S&P 1500 firms (and other firms in Execucomp) over the last two decades. Prior work typically proxies for layoffs by changes in net employment (e.g., Landier, Nair and Wulf 2009; Schmieder, von Wachter and Heining 2022). In contrast, we use a combination of natural language processing (NLP) and hand-coding to identify actual layoffs and their characteristics in news-like text data. We utilize the data on “Discontinued Operations and Downsizings” in the S&P Capital IQ Key Developments database. Unlike existing datasets, our data allows us to distinguish *involuntary* layoffs from employee reassignments or voluntary departures, such as early retirements. We also collect information on layoff severity and geography, including the number of employees laid off and layoff sites. Our final

¹ Quote from a former executive at Siemens AG in the 2016 documentary titled “Lonely at the Top: Top-Level Managers at Their Limit,” available (in German) at youtube.com/watch?v=FcRH3r0nEDE.

² Correspondingly, classical adjustment costs for physical capital include interruptions of the production process, installation costs, and site cleanup costs after disinvestment (Rothschild 1971, Doms and Dunne 1998, Teisberg 1993).

³ Chen and Li (2009) find that prosocial behavior is stronger toward in-group members. Fehr and List (2004) show that it is pronounced among experimental subjects who are real-life CEOs. Relatedly, Taylor (2010) estimates large personal disutility costs of board members from firing CEOs to explain low CEO turnover rates.

dataset covers more than 12,000 layoff announcements.

Our **empirical tests of the effects and determinants of layoff decisions** are guided by a simple conceptual framework that allows CEOs to experience a utility loss from the act of firing workers. The underpinning of this (potential) disutility cost comes from the psychology literature on *empathy*, i. e., the affective response to other people’s (actual or inferred) emotional state (Decety and Jackson 2004). Here, we hypothesize that CEO behavior can be interpreted as their affective response to workers’ distress from job loss, in the spirit of the introductory quotes. Empathy is known to deepen as attachment to others increases (“friends versus strangers”) and to depend on perceived closeness, similarity, or relatability (“self-other overlap”) (Meyer et al. 2013, Morelli, Lieberman and Zaki 2015, and Andries et al. 2025).⁴ Building on these insights, the theoretical framework predicts that CEOs’ layoff propensity decreases in (i) their tenure, as CEOs’ attachment to the workforce increases, (ii) in the disutility experienced by workers from a layoff, such as during recessions or Christmas, and (iii) in workers’ geographical and social proximity to the CEO.

In the first part of our empirical analysis, we estimate the personal health costs of layoffs for CEOs. We document a significant reduction in the longevity of CEOs who implement layoffs in response to industry distress shocks. The estimated effect size is large, roughly corresponding to a CEO becoming two years older, and is robust to controlling for the direct effect of industry distress (cf. Borgschulte et al. (2025)) as well as firm performance.

The estimated health costs are consistent with the proposed empathy-driven prosociality mechanism as well as other channels that make firing stressful. They are harder to reconcile with alternative explanations such as social-image concerns or a response to media- or stakeholder-induced social pressure, which reflect social considerations but no actual stress induced by the act of firing.

In the second part of the paper, we analyze how CEOs’ propensity to implement layoffs evolves with tenure as they form deeper social ties inside the firm. Guided by our theoretical framework, we proceed in three steps. In the first step, we contrast the firing decisions of newly appointed and incumbent CEOs in the same industry. We utilize plausibly exogenous CEO changes due to death or illness, which have been previously used as an instrument for CEO turnover (e.g., Fee, Hadlock and Pierce 2013). Comparing firms with and without exogenous turnover in the same industry over the five years before and the five years after treatment, we estimate a significant 8.7 percentage points (pp) increase in the propensity to lay off employees. These additional layoffs are accompanied by large, positive announcement returns, averaging more than 2%. These contrasting patterns suggest frictions in firing among incumbent CEOs, to the detriment of shareholders.

In the second step, we show that CEOs’ propensity to make layoffs declines sharply with tenure. An interquartile increase in CEO tenure (from 3 to 11 years) predicts an 11% reduction in

⁴ Older literature in psychology, cognitive science and sociology also includes Byrne (1961), McPherson, Smith-Lovin and Cook (2001), Echterhoff, Higgins and Levine (2009).

layoff propensity (1.5 pp) relative to the baseline annual layoff probability of 13%. The negative tenure-layoff relation is not explained by firm size, profitability, productivity, or fixed effects for year, industry, or firm, which rules out endogenous matching between long-tenured CEOs and firms that generally avoid layoffs. The finding is also robust to controlling for the *CFO*'s tenure.

Unlike the first analysis of newly appointed CEOs, the second analysis of layoffs over the course of a CEO's tenure does not exploit plausibly exogenous variation. To probe the causality of the estimated tenure-layoff link, we adopt the instrumental variables (IV) approach of [Altonji and Shakotko \(1987\)](#), also applied by [Graham, Kim and Leary \(2020\)](#). That is, we instrument tenure with the proportion of total tenure. As discussed in [Graham, Kim and Leary \(2020\)](#), this measure is correlated with tenure but orthogonal to fixed individual CEO characteristics and CEO-firm match quality, both of which can affect tenure length, since all CEOs begin at 0% and end at 100% of tenure completed. Thus, the instrument addresses the concern that the observed decline in layoff propensity might reflect inherently different layoff propensities of different CEO "types" (with long versus short tenure), or that long-tenured CEOs face different economic circumstances that also influence layoff propensity. All results replicate using instrumented tenure.

We further show that the tenure effect on employment changes operates asymmetrically: longer CEO tenure is strongly associated with reduced employment contractions (layoffs), but only weakly with employment expansions and hiring. This pattern is directly predicted by empathy-related prosociality but not by neoclassical or other prosocial mechanisms such as warm-glow or altruism.

In the third step, we show that internally hired CEOs, who have been at the firm for longer, are significantly less likely to lay off employees than externally hired CEOs. This finding is consistent with the tenure-layoff relation being driven by CEOs' attachment to employees.

We also show that the estimated decline in layoff propensity is distinct from prior evidence of "CEO investment cycles," which involve CEOs increasing investment and decreasing disinvestment over their tenure and which have been attributed to growing CEO power ([Pan, Wang and Weisbach 2016](#)). Even though the documented link to CEO power does not extend to employment growth, we test whether our estimates of declining layoff propensity are explained by existing measures of CEO power or entrenchment. Using a wide range of measures, including time-varying board co-option, CEO duality, and the [Bebchuk, Cohen and Ferrell \(2009\)](#) index, we reject such an explanation.

In summary, the second set of analyses shows that CEOs' layoff propensity declines over the course of their tenure, as they deepen connections inside the firm, consistent with increasing prosociality toward employees. While other stressors associated with firing, such as lawsuits by employees, could explain the first set of results on adverse CEO health effects, they cannot explain the second set of results, since such stressors do not tend to systematically increase with CEO tenure.

In a third part of the analysis, we implement a series of heterogeneity tests guided by the empirical evidence on the determinants of stronger or weaker empathy. One prediction is that layoff

decisions become more costly to the manager when their impact on employees is more severe—for example, during economic downturns, when adverse consequences for displaced workers are stronger (Jacobson, LaLonde and Sullivan 1993), or during the holiday season. We show that, over their tenure, CEOs become more averse to these “painful” layoffs and are significantly less likely to implement them, both unconditionally and conditional on making a layoff in a given year. The within-year layoff timing evidence is particularly difficult to reconcile with alternative explanations of the tenure-layoff link. The results are again robust to instrumenting tenure.

For another set of heterogeneity tests motivated by our framework, we identify layoffs that are likely more painful for the CEO to witness, holding workers’ disutility constant. One such type of variation is geographical closeness. We compare layoffs at or near the firm’s headquarters with more distant layoffs, where CEOs are less likely to witness the effects directly or to know the affected employees personally. Another type of variation is social closeness. Building on the above-cited evidence that similarity and relatability amplify empathetic activation and prosocial behavior, we compare CEOs’ propensity to lay off socially closer white-collar workers compared to blue-collar workers. Along both dimensions, we find that CEOs’ increasing reluctance to lay off workers over the course of their tenure is driven by employees who are geographically or socially close. Additionally, the social-closeness effect reverses for CEOs with a blue-collar formative background: they become less likely over time to lay off blue-collar workers, consistent with empathy-based models. As before, all results are robust to using instrumented tenure. These findings complement prior evidence showing that survey participants reduce support for layoffs when their geographical and social distance from affected workers is exogenously reduced (Iliewa, Kempf and Spalt 2025).

We also conduct an auxiliary analysis to examine how longer-tenured CEOs manage the financial costs of reduced layoffs, especially during times of economic distress. We find that the higher payroll burden appears to come at the expense of other firm activities, particularly non-personnel research and development (R&D). Longer-tenured CEOs reduce R&D investment during recessions when financing constraints are more likely to bind, with the effect fully driven by labor-nonintensive R&D industries, while simultaneously avoiding layoffs. We further document that longer CEO tenure is associated with lower compensation during periods of industry distress, consistent with longer-tenured CEOs sharing the financial burden of retaining employees during downturns.

As a final test of the proposed interpretation, we consider individual-level variation in CEOs’ dispositional tendency to develop empathy for others (cf. Davis 1983). Prior work on CEO personality assessments identifies empathy in contrast to overconfidence (resoluteness) as one of two key managerial personality dimensions and shows that the widely used *Longholder* overconfidence measure (Malmendier and Tate 2005) is negatively correlated with empathy scores (Kaplan, Sørensen and Zakolyukina 2022). Guided by this taxonomy, we show that the decline in layoff propensity over a CEO’s tenure is fully driven by the more-empathetic, non-overconfident CEOs.

Overall, all empirical results on frictions in layoff decisions are consistent with a “human” component of layoff avoidance in the spirit of the empathy literature: the link between layoffs and adverse CEO health outcomes, the link between CEO tenure (or attachment to their firm) and layoff avoidance, the heightened aversion to painful layoffs during economic distress, the holiday season, and among socially or geographically close employees, and finally the individual-level variation by empathy-proxy all point to an affective response to workers’ distress. Thus, our analysis suggests that managers’ empathetic concern for employees is likely to play a significant role in firm decision-making with implications for standard models of the firm’s optimization problem.

Literature. Our paper makes four main contributions to the literature. First, we connect the literatures on managerial preferences and prosocial behavior arising from empathy-related mechanisms (Allport 1954). We focus on empathy-related motives, which are related to but distinct from moral values and moral judgments emphasized in recent work (e.g., Haidt 2012, Enke 2024, Iliewa, Kempf and Spalt 2025): empathy is inherently relational and context-dependent, varying with proximity, relatability, and exposure. The economics and psychology literatures emphasize several key factors that affect the activation and strength of empathy, including individual variation and predisposition (Davis 1983, Eisenberg and Miller 1987), perceived similarity and relatability (Allport 1954, Cialdini et al. 1997, Andries et al. 2025), and the severity of others’ distress (Batson, Fultz and Schoenrade 1987, Decety and Jackson 2004). We bring new field evidence from managerial layoff decisions—an environment where empathy and empathetic capacity are difficult to observe directly, but where we propose new measures based on managerial behavior and formative background—that aligns with these predictions and is consistent with prosocial considerations within the firm. More broadly, our analysis builds on economic frameworks that allow for altruism and other-regarding preferences in decision-making (Becker 1974), and complements behavioral economics research that incorporates psychological mechanisms—such as empathy and responses to visceral states—into models of firm behavior and organizational choice (see Pope and Sydnor 2015). In turn, we contribute to research on nonstandard managerial decision-making rooted in nonstandard preferences (see Malmendier (2018) and Guenzel and Malmendier (2020) for recent literature surveys).

Second, by documenting CEO health effects from layoffs through the lens of prosocial disutility, we add to the literature on the effects of CEOs’ job duties and stressful work environments. Prior work documents the intense demands of the CEO role (Bandiera et al. 2018, 2020), the role of labor-friendly managerial preferences (Bertrand and Mullainathan 2003, Cronqvist et al. 2009), and the taxing nature of layoffs in particular (Hallock 2005). Most closely related to the result on layoffs and CEO mortality, Borgschulte et al. (2025) show that plausibly exogenous shocks to CEOs’ job demands affect aging and mortality. We extend this work by documenting that layoffs are a major

contributor to the stressors and resulting health costs of the CEO role which, in turn, shape CEOs' layoff decisions.

Third, we contribute to the literature on CEO tenure and its effects on firm policies. [Pan, Wang and Weisbach \(2016\)](#) document a link between tenure and investment and disinvestment driven by CEO power and entrenchment dynamics, while we focus on labor decisions and highlight managerial prosocial concerns as an underlying channel. Complementary evidence on labor decision-making comes from surveys ([Bertheau et al. 2022](#), [Hoffman and Tadelis 2021](#)) and from studies showing that CEOs' career background affects employment reductions, with external CEOs more likely to implement them ([Bai and Mkrtchyan 2023](#)). Unlike their focus on manufacturing, we study a broader set of firms and directly identify layoffs. Our findings extend this work by showing that internally promoted CEOs, who have stronger ties to their workforce, are systematically more reluctant to initiate layoffs.

Finally, our work contributes to the broader literature on social and institutional influences on corporate labor decisions. Prior research highlights how financial and institutional frictions shape layoffs—for example, financially constrained CEOs disproportionately dismiss recently hired workers with lower current but higher future productivity ([Caggese, Cuñat and Metzger 2019](#)), and the likelihood of binding constraints depends on CEO characteristics such as financial expertise ([Custódio and Metzger 2014](#); [Custodio, Mendes and Metzger 2025](#)). Other work emphasizes employee-related social considerations, including gender differences in managerial labor policies ([Matsa and Miller 2013](#); [Matsa and Miller 2014](#)), the role of geographic and social closeness ([Yonker 2017](#)),⁵ and the effects of unemployment insurance expansions, which by reducing the hardship of job loss are associated with larger employment reductions, particularly in firms led by internally promoted or Democratic CEOs ([Keum and Meier 2023](#)). Recent experimental evidence also shows that public **moral** concerns over layoffs are especially acute ([Iliewa, Kempf and Spalt 2025](#)). A related body of research documents how CEOs' education, personal traits, and social preferences shape broader corporate outcomes—for example, managers exposed to shareholder-value education reduce rent-sharing with workers ([Acemoglu, He and le Maire 2025](#)); boards and social norms influence managerial retirement ([Taylor 2010](#), [Jenter and Lewellen 2015](#)); and prosocial attitudes are linked to corporate social responsibility ([Cheng, Hong and Shue 2023](#), [Cronqvist and Yu 2017](#)).

Our findings also speak to a longer tradition that views managerial traits as potential commitment devices in the firm–worker relationship. Since [Azariadis \(1975\)](#), economists have emphasized the role of implicit contracts that shield workers from the “vicissitudes of the business cycle,” with later contributions exploring how firms sustain such commitments ([Harris and Holmstrom 1982](#), [Berk,](#)

⁵ **Related contemporaneous work links CEOs' geographic proximity to employees to employee-friendly workplace environments ([Amore, Bennedsen and Larsen 2025](#)); our focus instead is on layoffs and prosocial, empathy-based mechanisms.**

Stanton and Zechner 2010). More recent theoretical work (Rotemberg and Saloner 2000; Van den Steen 2005; Bolton, Brunnermeier and Veldkamp 2013) highlights how managers' intrinsic motives or vision can make their commitments to employees credible. While our empirical evidence points to clear ex post frictions from CEOs' reluctance to lay off workers, ex ante benefits—through sorting, specific human capital investment, or lower entry wages—may counteract these costs. Evaluating this ex ante perspective is beyond the scope of our paper, but our results suggest that managerial layoff aversion and preferences could extend beyond being a source of frictions to also serving as a possible commitment mechanism within firms.

The rest of the paper is structured as follows. Section 1 introduces a simple conceptual framework. Section 2 describes the data. Sections 3 to 6 present our results. Section 7 concludes.

1 Conceptual Framework

1.1 Framework and Testable Predictions

To fix ideas, we introduce a stylized conceptual framework that extends the classical model of firm production decisions to incorporate managerial prosociality. Details and proofs are provided in Appendix-Section A.

Setup. Consider a single firm in a simple three-period framework. In period 0, labor is exogenously determined, e. g., by a previous manager. In periods $t \in \{1, 2\}$, the firm's manager chooses capital k_t and labor n_t subject to expenditures (i. e., capital costs and wages) not exceeding available cash c_t . The firm follows a Cobb-Douglas production function with constant returns to scale.⁶ The firm faces i.i.d. stochastic shocks to cash on hand c_t at the beginning of each period.

Layoffs are reductions in employment ($n_t < n_{t-1}$). We denote worker i 's period- t utility as $u_{t,i}^s$, where s denotes the employment status, $s \in \{emp, unemp\}$, with utility higher when employed than unemployed ($u_{t,i}^{emp} > u_{t,i}^{unemp}$).

A standard manager's objective is to maximize total profits, by choosing $\max_{n_1, k_1} \pi(n_1, k_1) + \mathbb{E}_1[\max_{n_2, k_2} \pi(n_2, k_2)]$. By contrast, a prosocial manager j also places weight on the utility of the firm's employees,⁷ and maximizes

$$\max_{n_1, k_1} \pi(n_1, k_1) + \mathbb{E}_1 \left[\max_{n_2, k_2} \pi(n_2, k_2) + \int_0^{n_1} \phi(j, i; u_{i,2}) di \right], \quad (1)$$

where $\phi(\cdot)$ captures the degree of prosociality. Guided by the idea that layoff aversion arises from managers' vicarious experience of employees' distress, we follow the literature on empathy and its established core mechanisms (cf. the related discussions in the introduction and Section 1.2) to

⁶ All predictions hold for any standard production functions with decreasing marginal returns to capital and labor.

⁷ Broader formulations could include job candidates, likely future employees, or other stakeholder groups. The more restrictive formulation above best aligns with managers' survey responses and quotes.

specify the prosocial utility component as a function of (i) relatability and (ii) the presence and magnitude of employee distress:

$$\phi(j, i; u_{i,t}) = \underbrace{r_{ji,t}}_{\text{relatability}} \cdot \underbrace{\mathbf{1}\{s_{i,t} = \text{unemp}\}}_{\text{empathic activation}} \cdot \underbrace{(u_{i,t-1} - u_{i,t})}_{\text{utility loss}}, \quad (2)$$

where relatability $r_{ji,t}$ increases in two components: the manager’s empathetic disposition, α_j , and the relational proximity between manager j and employee i at time t , $\rho_{ji,t}$. We posit that the manager’s perceived relational proximity to employees increases with managerial tenure—reflecting the length of time the CEO has been responsible for the workforce—as well as bottom-up salience from socio-demographic similarity and geographic proximity. We parsimoniously capture the first effect in the framework by setting relational proximity to zero upon the CEO’s arrival (i.e., $r_{ji,1} = 0$) and positive thereafter (i.e., $r_{ji,2} > 0$), and the second effect by allowing for heterogeneous increases in the second period across employees. Accordingly, $r_{ji,2}$ increases more for employees that share more characteristics with the manager, and less for those that do not, because shared characteristics make employee outcomes easier for the manager to internalize.

This stylized framework allows us to illustrate the key predictions arising from managerial prosociality regarding labor decision-making.

Prediction 1 (CEO Layoff Disutility Cost). *A prosocial manager incurs a direct disutility cost from making layoffs.*

Prediction 1 follows directly from the model setup, but is worth highlighting. When managers consider layoffs, they face a trade-off between the benefits to firm value and their personal disutility. We test Prediction 1 empirically by examining the effect of undertaking layoffs on CEO mortality as a proxy for reduced utility.

Prediction 2 (CEO Turnover Effect). *For a given number of workers employed in the prior period, a new manager has a higher propensity to make layoffs than an incumbent prosocial manager.*

Note that, in Prediction 2 (as well as in the subsequent predictions), we compare identical firms, holding not only prior employment levels, but also production functions and budgets constant. In the empirical analysis, we explicitly control for the firm’s number of employees and incorporate various proxies for its financial condition, such as profitability and economic distress.

We test Prediction 2 by examining the layoff propensity of new CEOs (after an exogenously induced turnover) relative to incumbent CEOs within the same industry.

Prediction 3 (CEO Tenure Effect). *Compared to a shorter-tenured prosocial manager, a longer-tenured prosocial manager*

- (i) *has a lower propensity to make layoffs (extensive margin);*
 - (ii) *lays off fewer employees, conditional on a layoff (intensive margin),*
- holding constant the number of workers employed in the prior period.*

In the model, the contrast between shorter- and longer-tenured CEOs amounts to comparing their behavior at $t = 1$ versus $t = 2$. We test Prediction 3(i) by studying managers' layoff propensity over the course of their tenure as their longer tenure allows them to build stronger connections within the firm. In additional tests, we also inspect differences between internally and externally hired CEOs.⁸ Similarly, we test Prediction 3(ii) by examining the fraction of the workforce laid off each year as CEO tenure increases. Here, a prosocial manager opts for smaller layoffs (as their tenure increases) since the disutility cost of layoffs varies with the number of employees laid off.

Prediction 4 (Asymmetric Employment Effect). *Holding fixed the number of workers employed in the prior period, managerial tenure for a prosocial manager is more strongly associated with layoffs than with employment expansions (hiring).*

Prediction 4 highlights a key asymmetry in labor decisions implied by empathy-based underpinnings, in contrast to both certain forms of altruism-based mechanisms and standard frictions such as empire building. Equation (2) implies that prosocial behavior is primarily activated by negative consequences for individuals with whom the manager has existing relational ties. As a result, the model predicts a stronger role for managerial tenure in mitigating employment contractions, while any effects on employment expansions are comparatively muted and indirect. We test Prediction 4 empirically by examining how the association between CEO tenure and employment changes varies across the distribution of employment growth.

Prediction 5 (Employee Heterogeneity). *Compared to a shorter-tenured prosocial manager, a longer-tenured prosocial manager reduces their layoff propensity and layoff size more strongly*

- (i) *when layoffs affect employee utility more adversely and*
 - (ii) *for employees whose utility the manager weighs more heavily,*
- holding constant the number of workers employed in the prior period.*

Prediction 5 states that the more severe the adverse effects of layoffs on affected employees are or the more the manager cares about and can relate to affected employees, the more pronounced are the effects on the extensive and intensive margins. For our empirical tests of Prediction 5(i), we proxy for variation in the adverse effects on employees using the timing of layoffs relative to recessions—when displaced employees experience more severe long-term losses of earnings

⁸ @Ulrike just flagging whether this is the proper place for these cites (I think you had moved them here originally)? As Levitt and List (2007) discuss, social preferences are expected to be less pronounced among strangers, and more pronounced in the presence of social relations. Consistent with this, Bandiera, Barankay and Rasul (2005) find that social preferences among co-workers are more pronounced when friendship networks are stronger.

(Jacobson, LaLonde and Sullivan 1993)—and relative to the holiday season. For our empirical tests of Prediction 5(ii), we operationalize managerial relatability by allowing managers to place greater weight on employees who are “closer.” We examine geographical closeness (i.e., propensity to lay off employees near versus far from headquarters) as well as social closeness (i.e., the propensity to lay off white- versus blue-collar employees).

Prediction 6 (Labor-Capital Trade-Off). *Managers who avoid layoffs due to their prosocial motives will reduce their capital spending to offset the higher labor costs.*

Prediction 6 lays out the implication of managerial prosociality for capital spending. A prosocial manager’s reluctance to initiate layoffs leads to a higher payroll. To offset the higher labor cost, the manager must reduce capital spending. We test Prediction 6, in conjunction with 5(i), by examining the relationship between layoffs and R&D spending during periods of industry distress. In such periods, financing constraints are more likely to bind; however, as our empirical test of Prediction 5(i) shows, the difference in layoff propensity between short- and long-tenured prosocial managers widens during industry distress. We focus on R&D spending because it is a key component of investment and tends to be more responsive to financing constraints than long-term physical capital investments. We also examine managerial compensation as an additional adjustment margin through which prosocial managers may help absorb higher labor costs.

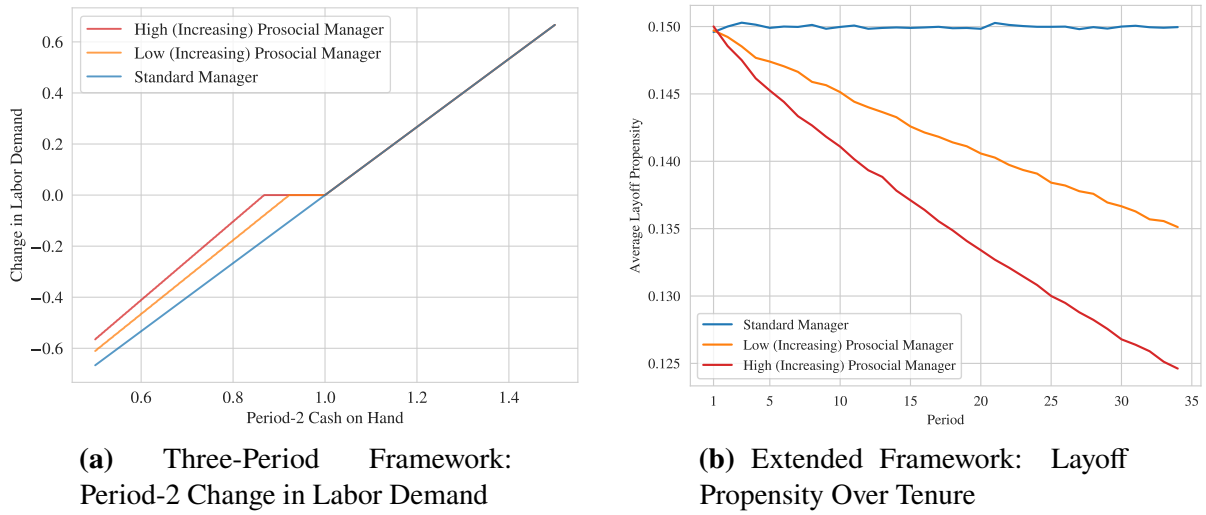
Prediction 7 (CEO Heterogeneity). *Higher prosociality over a CEO’s tenure predicts a steeper decline in layoff propensity and layoff size over time, holding constant the number of workers employed in the prior period.*

Intuitively, Prediction 7 states that the more a manager’s prosocial motive strengthens throughout their tenure, the steeper the corresponding decline both in layoff propensity and layoff size. Motivated by Kaplan, Klebanov and Sørensen (2012), who document that managerial overconfidence (resoluteness) is contrary to empathy—the unifying underpinning of prosociality in our setting—we test this prediction using empirical variation along this dimension.

Graphical Illustration. Figure 1a illustrates the relation between prosociality and labor choices graphically. It plots the change in labor demand (hiring and firing) between period 1 and period 2 as a function of the firm’s financial situation, i. e., cash available in period 2. The labor choices of standard managers are shown in blue, while those of managers with moderate and high prosociality (arising from variation in empathetic disposition and worker relatability) are shown in orange and red, respectively.

The figure illustrates three regimes. When cash on hand is high and the firm’s spending is less constrained, all manager types hire additional employees. At moderate levels of cash on hand, prosocial managers maintain their period-1 labor levels to avoid the disutility cost of layoffs, whereas

Figure 1. Key Model Insights



Notes: Panel (a) plots the change in labor demand from period 1 to period 2 for different realizations of period-2 cash on hand, c_2 , for the median realized value of period-1 cash on hand. Panel (b) plots the average simulated layoff propensity over tenure for managers in our extended framework. Additional details on the numerical solution are in Appendix-Section A.4.

standard managers continuously adjust labor downward. When cash on hand is even lower, all manager types engage in layoffs, but the magnitude of the workforce reduction is smaller for more prosocial managers. Thus, when a prosocial manager avoids or reduces layoffs, the manager has higher utility than under the standard labor choice, whereas firm profits are lower.⁹ The differences in prosocial weights in the figure map to our framework by representing newly appointed managers as not exhibiting prosociality, and incumbent managers as exhibiting positive (and potentially heterogeneous) levels of prosociality.

Extension to Multiple Periods. The simple three-period framework above serves as a proof of concept that isolates the mechanism of interest while holding other features of the firm fixed, and comparing otherwise identical firms as described. While this renders the model analytically tractable, it necessarily abstracts from richer dynamic considerations. To ensure our predictions extend to longer horizons, we therefore numerically solve and simulate a multi-period version of the model. We introduce an exogenous CEO exit process calibrated to match the 99th percentile of CEO tenure in our data (34 years) and calibrate the cash-on-hand process to generate an average layoff probability of approximately 15 percent, consistent with our empirical baseline. Figure 1b displays the resulting average layoff probability over time for the three manager types. While the

⁹ As alluded to, we abstract from potential ex ante benefits of managerial preferences, such as attracting talent, that could in turn affect profits.

standard manager exhibits a constant layoff probability, prosocial managers display a declining layoff propensity as tenure increases, with steeper declines when prosociality rises more sharply with tenure, matching the predictions of the three-period model. Additional details on the numerical calibration and simulation results are provided in Appendix-Section [A.4](#).

1.2 Underpinnings of Managerial Prosociality

Before turning to the empirical tests of our predictions, we discuss potential alternative explanations for the behavior attributed to managerial prosociality toward employees, as captured by the prosocial term $\phi(\cdot)$ in equation (1). While the framework is guided by empathy-based considerations, we consider a range of neoclassical interpretations as well as alternative mechanisms within the realm of prosocial preferences, before elaborating on empathy in greater detail.

Norms and Implicit Contracts. One possibility is that managerial social considerations arise from norms or implicit contracts between management and employees. [Gorton and Zentefis \(2020\)](#) argue that social norms form inside the firm and are directed by management to ensure a well-functioning organization. A mechanism based on implicit contracts and social norms is consistent with some, but not all of our evidence. For example, in line with our empirical results, establishing implicit contracts, e. g., by avoiding layoffs, might become more important later in a CEO’s tenure, and potentially in particular for white-collar employees. This mechanism does not predict, however, personal disutility cost from implementing layoffs. The social-contract mechanism predicts that the manager takes employee interests into account in the optimization but *if* layoffs are warranted in that process, the “act of firing” does not lead to disutility costs. By contrast, we will show in Section 3 that making layoffs predicts long-run CEO mortality, holding fixed the firm’s and industry’s overall condition.

Career Concerns. Another possibility is that CEOs avoid layoffs because of career concerns. Career-concern motives imply, however, that managers maximize their self-interested utility, also when making layoff decisions, which is again hard to reconcile with increased mortality. Further, one might argue that layoffs in good times should hurt reputation more than those in recessions, *since the latter may be viewed as unavoidable* (e.g., [Grenadier, Malenko and Strebulaev 2014](#)), whereas our heterogeneity tests find evidence consistent with stronger prosocial considerations during downturns. Additionally, career concerns predict that managers should be more likely to make layoffs closer to retirement when their career is near its end, which is the opposite of what we find. Relatedly, concerns about hiring in the future and risk aversion would point towards more layoffs closer to retirement.

Social Pressure. Alternatively, it is possible that CEOs display prosocial behavior as a result of social pressure from other stakeholders and agents, such as unions, (local) governments, or the media. Depending on the nature of such external pressures, this mechanism could be consistent with

some of our results. For example, it could explain geographic layoff heterogeneity, if pressures from local agents loom larger. However, it is again not easily reconcilable with our finding of personal health costs of firing for CEOs. Under a social-pressure mechanism, the CEO implements layoff decisions as “guided” or called for by others, effectively leaving little room for incurring personal disutility costs from such decisions.

Social Image. Further, managers’ prosociality could be driven by social-image concerns, which may increase with tenure, and which can generate time-varying and heterogeneous prosociality weights. To what extent a social-image mechanism is consistent with all our findings depends on the precise nature of social-image formation. On the one hand, it is plausible that social-image concerns might intensify during recessions, **for instance because employment decisions are more visible**. On the other hand, it is plausible that a manager’s social image might suffer more when laying off blue-collar workers as opposed to white-collar workers, which would predict amplified blue-collar layoff effects, whereas empirically we find stronger effects for socially-close white-collar employees.

Empathy. The mechanism most consistent with the full set of our results—**and which informs the formulation of the prosocial utility component in the previous section, generating the set of predictions that are all supported in the data**—is managerial empathy. Empathy has been described as “the affective experience of the other person’s actual or inferred emotional state ... feeling what another person is feeling, knowing what another person is feeling, and having the intention to respond compassionately to another person’s distress” (Decety and Jackson 2004). This characterization of empathy corresponds closely to the disutility **component** formulation in our conceptual framework (equation (1)).¹⁰

Various additional facets of empathy align with our empirical approach and findings, **and directly link to the conjectured components of the prosocial utility component in equation (2)** (see also the **introduction for additional references**). For one, psychologists emphasize a distinction between negative empathy (i.e., empathetic concern) and positive empathy (Morelli, Lieberman and Zaki 2015), implying in our setting that layoffs and new hires are unlikely to serve as “substitutes” for prosocial managers. Additionally, prior work in neuropsychology documents stronger empathetic responses when the degree of self-other overlap is larger, and that this effect grows over time as attachment to others increases (“friends-versus-strangers”). For example, Meyer et al. (2013) use neuroimaging-based techniques to show that when subjects observe friends suffering socially, pain-processing brain areas are activated that are associated with feeling pain and firsthand experiences of exclusion. When subjects observe strangers, activated brain areas instead are those associated

¹⁰ See also this Forbes article for a non-academic summary of the concept of empathy, emphasizing that empathy refers to “experiencing a fraction of [others’] emotions and feelings because you see things from their perspective” (forbes.com/sites/forbescoachescouncil/2018/08/13/the-critical-difference-between-sympathy-and-empathy/?sh=1262df5016a3).

with merely understanding others’ states and intentions.¹¹ Relatedly, Andries et al. (2025) provide supporting field evidence that increased perceived relatability magnifies empathetic responses and fosters prosocial behavior. The above findings are consistent with our evidence of amplified managerial concerns for socially-close white-collar employees on average, and the reversal of amplified concerns for blue-collar workers when managers have a blue-collar formative background. Our empirical distinction between overconfident and empathy-driven CEOs, along with evidence on resulting heterogeneity in layoff reluctance, also provides direct evidence supporting this microfoundation. Overall, a psychological underpinning of prosociality arising from managers’ empathetic concern for employees is consistent with all of our empirical findings.

2 Data

This section discusses data and descriptive statistics. Appendix-Section B contains further information on data sources and data processing, including more detail on the text-processing algorithms to identify layoffs, as well as variable definitions.

2.1 New Layoff Data

The starting point for the construction of our new dataset on layoff announcements is the Standard and Poor’s (S&P) Capital IQ Key Developments database. This database has reliable coverage starting in 2002 (Edmans et al. 2018; Cohn, Gurun and Moussawi 2020), and we restrict the sample period to 1/2002–2/2020, i.e., omit the COVID-19 period.

The database covers more than 250,000 companies worldwide. Since we rely on Execucomp for details about CEOs (see Section 2.2), we collect key developments for firms in Execucomp. We extract information for all observations where the type of announcement is categorized as “Discontinued Operations/Downsizings,” amounting to 18,019 key developments as of the download date (January 28, 2022).

Each key development entry in the database includes a date, headline, and a situation summary, which is a short news-like text description. The content is based on a variety of sources, including news aggregators, stock exchanges, regulatory websites, and company websites.

We leverage NLP techniques, augmented by hand-coding, to (1) classify each article as a layoff, closure (e.g., of a factory or branch), or neither; (2) extract the number or percentage of workers laid off; (3) identify the layoff location at the international, state, and city level; and (4) classify layoffs as blue-, pink-, or white-collar.

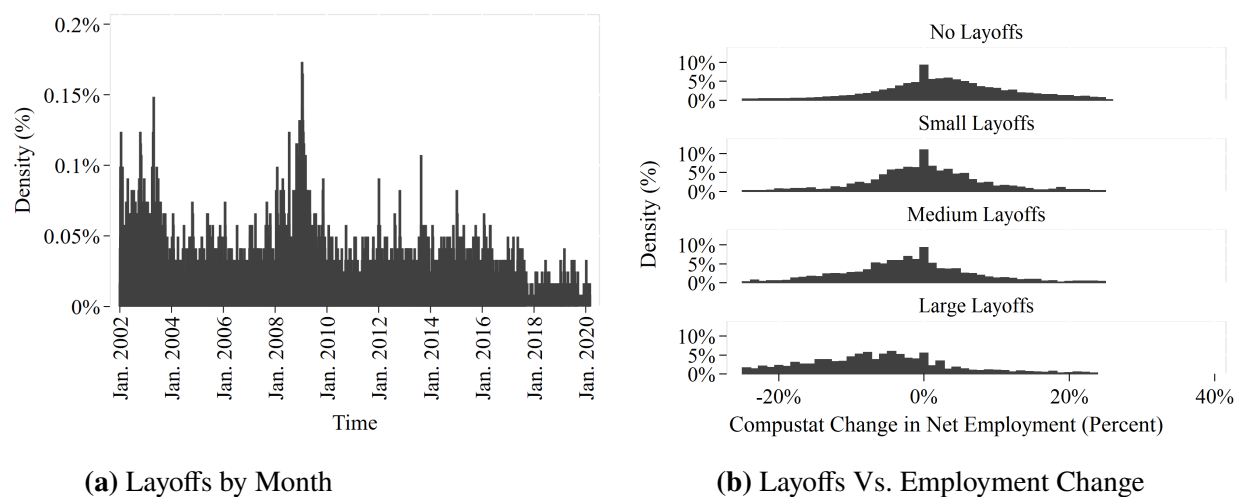
For (1), we first lemmatize the headline and situation summary, i. e., standardize verbs and nouns. For example, “cutting” becomes “cut” and “jobs” is reduced to “job.” We then create several word lists to aid in identifying layoffs and closures: (i) layoff verbs, such as “cut” and “slash,” (ii)

¹¹ The study concludes that their findings “support the idea that self-other overlap with a target may allow us to recruit brain mechanisms that indicate that we ourselves are experiencing the target’s situation.”

closure verbs such as “mothball” or “shut down,” and (iii) worker-related nouns, including words for synonyms of jobs like “position” or “role,” words for occupations such as “accountant” or “banker,” and collective job words, such as “headcount” or “workforce.” A key development is classified as a layoff if either the headline or a sentence in the summary contains both a layoff-related verb and a worker-related noun from the lists specified above. If it contains only one of these elements, it is tagged as a low-confidence layoff and flagged for manual review. A development is classified as a closure if it includes a closure-related verb and is not already classified as a layoff. If none of these criteria are met, the development is classified as neither.

All key developments are manually reviewed for accuracy, with special attention given to low-confidence observations. We consider the key developments for a given firm in the order they were announced, allowing us to identify and drop any duplicate layoff announcements. The manual review also allowed for using criteria that NLP could not easily capture. Since closures frequently result in layoffs, especially when entire plants or branches are closed, we treat them as layoffs *unless* the key development specifically mentions that a closure comes with no layoffs, e. g., due to employee reassignments. We also separate layoff announcements from business sales without effects on employees. Moreover, we distinguish involuntary layoffs from employee reassignments and voluntary departures such as early retirement.

Figure 2. Layoff Events



Notes: Panel (a) plots the number of all identified layoffs extracted from the S&P Capital IQ Key Developments database by month. Panel (b) plots the distribution of year-over-year net employment change from Compustat (in percent) for CEO-firm-years with no layoffs in the 2002-2020 S&P Capital IQ Key Developments data (top figure), as well as CEO-firm-years with layoff size data separated into small, medium, and large (bottom figure) layoff years, as defined by terciles of the proportion of the firm laid off.

Figure 2a plots the resulting distribution of all identified layoffs across time. The number of layoffs peaks during the Great Recession and is lower than average in the pre-COVID-19 years, when unemployment was very low (Petrosky-Nadeau, Valletta et al. 2019).

For (2), we extract the number and percentage of workers laid off. We use restricted text searches, commonly referred to as regular expressions, to capture all of the common ways to describe the count or proportion of jobs cut. For example, one pattern for percentages is *verb, other words, job words, other words, percent words*. The statement “reduce its workforce by 5%” would fall into this category. Corresponding patterns are used for “count” statements. These numbers are also manually reviewed.

The detailed count of layoffs represents a significant improvement over previous measures of layoffs, which have relied on proxies such as net employment changes. Figure 2b compares layoffs identified in our sample directly to firms’ net employment changes from Compustat. The figure plots the distribution of year-over-year net employment changes (in percent) for CEO-firm-years with no, small, medium, and large layoff activity, where the size of the layoff activity is determined by the proportion of the firm laid off.¹² With increasing layoff size, the distribution of net employment changes shifts to the left, implying that both variables are related. Yet, there remains a large dispersion in the distribution of employment changes within each layoff group, including a significant fraction of firm years with a shrinking workforce despite no layoffs, and another sizable portion growing despite high layoff activity. Employment changes that are not driven by involuntary layoffs can include, for example, voluntary departures as well as early retirement. Overall, changes in the net employment count appear to be a highly imprecise proxy of layoffs. Another source for layoffs is Worker Adjustment and Retraining Notification (WARN) Act filings. We examined the situation summaries for the key developments and found multiple direct references to WARN filings, which suggests that our dataset likely captures relevant WARN filings.

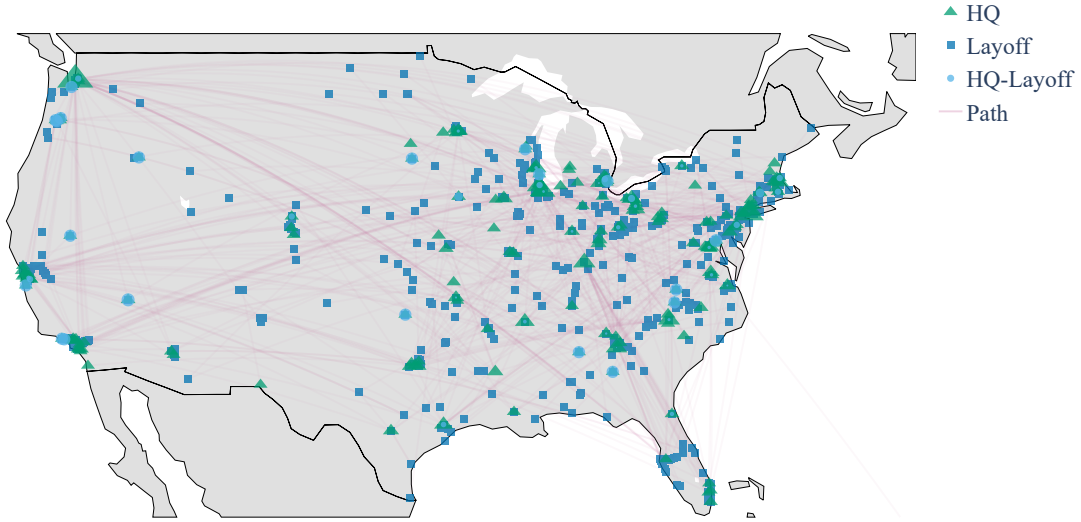
For (3), we collect the geographic location of the layoff up to the city level for U.S. layoffs and up to the country level for foreign layoffs. We also record layoffs that are specified as global, international, or similarly described, and review all geographic classifications manually.

Figure 3 shows the geography of layoffs for one year in our sample: 2009. The geographic separation between firms’ HQs and layoff sites is frequently substantial, indicated by the lines connecting HQs (green triangles) and layoff locations (dark blue squares). Light blue dots identify HQ-city layoffs, which are rare compared to more distant layoffs.

For step (4), we identify blue-, pink-, and white-collar layoffs based on a compilation of the

¹² Our sample contains 2,258 firm-years with more than one CEO. This can be due to co-CEOs or CEO turnover, for which we include observations for both the departing and the incoming CEO. Including both CEOs in the turnover year ensures that the layoff is correctly attributed to the CEO who made the decision. We calculate the proportion laid off as the sum of the proportions laid off in all layoffs in a given CEO-firm-year. Small, medium, and large activities are defined by terciles of the resulting distribution.

Figure 3. Layoffs Across Space



Notes: The figure plots layoffs in 2009 from the Firm–Layoffs dataset, whenever location data is available. Lines connect firms’ headquarters and layoff locations. Green triangles identify headquarters, dark blue squares layoff locations, and light blue dots headquarters-city layoffs.

500 most common words in the sample overall. Among these words, we classify keywords that indicate blue-collar (e.g., factory, plant), pink-collar (service-oriented jobs; e.g., retail, store), and white-collar (e.g., corporate, headquarters) layoffs and categorize our layoff events correspondingly. We verify that the resulting classifications are sensible using city-level data on occupation frequencies from the 2000 U.S. Census (see Appendix-Section B for additional details).

In our main empirical analyses, we focus on significant layoff events, namely, CEO-firm-years when a firm laid off at least 1% of the workforce or had at least three separate layoff events. For CEO-firm-years with fewer than three layoffs and that do not meet the 1% layoff size criterion but have missing information on layoff size for at least one of the layoff events, we manually search on the web and LexisNexis to obtain this information, to ensure precise measurement of variables. We are able to find the layoff size information for about 10% of these layoff events, suggesting that this information is usually included in the key development text when it is publicized. Our results are robust to using either the resulting indicator for significant layoff events or a layoff indicator based on *any* layoff activity in a given year. We use the indicator for *any* layoff activity when we study location-specific, month-specific, or employee-type-specific layoff activity in Tables 9 through 11.

2.2 Firm and CEO Data

We combine the new layoff dataset from Section 2.1 with standard firm and CEO data from CRSP, Compustat, and Execucomp.

Data on CEOs, CFOs, and their tenure comes from Execucomp. We identify exogenous CEO

changes by merging in the open-source database of CEO turnovers and dismissals by [Gentry et al. \(2021\)](#), which contains classifications for CEO turnovers. Data on firm characteristics, such as assets, employees, and profitability, comes from Compustat. We use log assets and log employees as proxies for firm size. Firm profitability is measured as operating income before depreciation divided by assets. We lag these variables in all analyses. Stock price data come from CRSP, and we calculate cumulative abnormal returns for our event study analysis based on the market-adjusted model.¹³ Similar to prior work ([Opler and Titman 1994](#); [Acharya, Bharath and Srinivasan 2007](#); [Babina 2020](#)), we define an industry-year as in distress if the two-year stock return of the median firm in a given SIC3 industry is -30% or less, where we consider both the prior two years and the prior and current year for the two-year horizon.¹⁴ We also use governance data from the Institutional Shareholder Services (ISS, formerly RiskMetrics), to calculate measures of co-opted directors, described further in Section 4.2. The resulting CEO-firm-year panel dataset, which we refer to as the *Firm-Layoffs Dataset* going forward, covers 2,779 unique firms.

2.3 Historical Layoff Data and CEO Mortality Data

To study long-term CEO health outcomes associated with implementing layoffs, we rely on an earlier layoff dataset from [Hallock \(1998\)](#), [Billger and Hallock \(2005\)](#), and [Farber and Hallock \(2009\)](#). This dataset covers all firms listed in the *Fortune 500* between 1970 and 1999 and provides a comprehensive list of layoff announcements collected from the *Wall Street Journal*.

We merge this layoff dataset with the mortality dataset from [Borgschulte et al. \(2025\)](#), which is an extension of the data in [Gibbons and Murphy \(1992\)](#) and covers all CEOs included in the *Forbes* Executive Compensation Surveys from 1975 to 1991. These surveys are based on corporate proxy statements and include the executives of the largest U.S. firms. The exact dates of CEOs' birth and death are sourced from ancestry.com, newspapers, and obituaries. CEOs who did not pass away by the cutoff date of October 1st, 2017 are censored.

After further merging with Compustat and CRSP data, the final *CEO Mortality Dataset* contains 31,917 CEO-year observations from 1,131 CEOs across 658 firms.

2.4 Descriptive Statistics

Table 1, Panel (a), presents summary statistics for the *Firm-Layoffs Dataset*. One observation corresponds to a CEO-firm-year. The average CEO tenure is 8 years, and the average CFO tenure is 5 years. 19% of observations are CEO-firm-years in industry distress, and 13% are classified as layoff-years.¹⁵

¹³ For the market return, we use the value-weighted return on the NYSE, AMEX, NASDAQ, and ARCA exchanges.

¹⁴ We find that this definition of distress is even more strongly associated with layoff activity in our data than an alternative definition based on two-year forward-looking stock price changes.

¹⁵ Appendix-Figure D.1 decomposes layoffs by industry. Layoffs are particularly likely in the retail trade industry, which features seasonal labor demand ([Anderson 1993](#)), and the manufacturing sector, which has been affected by

Table 1, Panel (b), presents summary statistics for the *CEO Mortality Dataset* at the CEO level. The median CEO was born in 1928 and, conditional on having passed, died at age 82. 11% of CEOs experienced layoffs during times of distress and 38% experienced layoffs during non-distressed times. 43% of CEOs experienced industry distress. The minimum stock performance, defined as the worst stock performance in any six-month period during a CEO’s tenure, is -30% on average in our sample.

3 Personal Cost of Firing for CEOs

We first examine personal long-term health consequences for CEOs associated with implementing layoffs as a direct way to detect firing-induced “disutility costs,” and specifically examine accelerated mortality. This analysis builds on [Borgschulte et al. \(2025\)](#), who document adverse aging and mortality outcomes for CEOs arising from more stressful work environments, including industry crises.

We estimate survival models to detect predictors of CEO mortality in the *CEO Mortality Dataset*. Our key variable of interest is whether the CEO implemented a layoff during their tenure. We allow for heterogeneous mortality effects depending on whether the CEO implements layoffs during industry distress or non-distressed times. Layoffs during economy-wide or industry downturns are particularly painful and damaging for affected workers ([Jacobson, LaLonde and Sullivan 1993](#), [Martin 2005](#), [Hallock 2005](#)). Downturns may also frequently force firms to lay off employees for reasons unrelated to their performance. Both aspects suggest particularly high and detectable CEO disutility costs associated with layoffs in times of crisis. Our main specification takes the form:

$$\begin{aligned} \text{Prob}(CEODeath_{i,t}) = & \alpha + \beta_1 \text{DistressLayoff}_{i,t} + \beta_2 \text{NonDistressLayoff}_{i,t} \\ & + \gamma_1 \text{IndustryDistress} + \mathbf{X}'_{i,t} \gamma_2, \end{aligned} \quad (3)$$

where $CEODeath_{i,t}$ is one if CEO i passes away in year t and zero otherwise. DistressLayoff and NonDistressLayoff are indicator variables for whether a CEO has experienced a distress or non-distress layoff, respectively, with the indicator remaining on once set to one. IndustryDistress controls for the direct mortality effect of experiencing industry distress, and $\mathbf{X}$ is a vector of control variables, including CEO age, proxies for firm size (assets and employees), firm profitability, and year (or, alternatively, year fixed effects). In addition, we include state and industry fixed effects, as discussed below.

We estimate equation (3) both as a logit model and as a hazard model. In the logit estimation,

offshoring and automation ([Slaper 2019](#)). We note that these statistics abstract from the distribution of firms within our sample. For example, 40% of firms in our sample are in the manufacturing sector which accounts for about 20% of years with a layoff. Meanwhile, only 8% of firms are in the retail trade sector which accounts for about 25% of years with a layoff.

shown in Table 2, we include an explicit event time control, i. e., control for the time that has passed since a CEO first assumed their position.¹⁶ As in [Borgschulte et al. \(2025\)](#), one observation corresponds to a CEO-year, and we continue to follow CEOs over time after they leave the CEO position until they pass away or the censoring date is reached.

In column (1), we include our main variables of interest, distress layoff and non-distress layoff experience, controlling for the direct effect of industry distress, CEO age, firm controls (assets, employees, and profitability), and time. We also include HQ state fixed effects and industry fixed effects as in [Borgschulte et al. \(2025\)](#). We cluster standard errors at the industry level, as this is the level at which distress experience is defined ([Abadie et al. 2017](#)).

We find that distress layoff experience significantly accelerates mortality, consistent with these layoff experiences leaving long-run marks on CEOs. The logit coefficient estimate of +0.294, significant at the 5% level, implies that the mortality effect of distress layoff experience corresponds to that of an increase in age by 1.85 years. This effect is large and of the same magnitude as the mortality effects associated with stressful job environments documented in [Borgschulte et al. \(2025\)](#). By contrast, we find no evidence for mortality effects associated with layoffs during non-distressed times. **We also note that once layoff experience is accounted for separately, the indicator for industry distress, accordingly capturing non-layoff distress, no longer enters significantly.** These results remain unchanged when replacing the linear time control with year fixed effects in column (2).

One potential concern regarding the estimates in columns (1) and (2) is heterogeneity in distress driving the results. That is, one might worry that an omitted variable reflecting the severity of distress experienced by a given firm drives both distress layoffs and mortality. To alleviate this concern, in column (3), we control for the worst stock performance in any six-month period during the CEO's tenure. This variable captures the severity of (possibly endogenous) firm distress over the course of a CEO's tenure and introduces variation in "CEO experience," holding fixed whether a CEO has experienced industry distress. The coefficient on Distress Layoff does not change substantially. This suggests that our results are driven by the experience of and responsibility for distress-induced layoffs, not simply by severe distress.

Robustness. We include several robustness tests in the Appendix. First, we re-estimate equation (3) using [Cox \(1972\)](#) hazard models instead of the logit estimations from Table 2. As expected, given the close link between logit models (with an explicit time control) and hazard models, we find very similar results (Appendix-Table D.1). Second, in Appendix-Tables D.2 and D.3, we repeat the logit and the hazard model estimations, adding a control for CEOs' cumulative tenure, in light of our results below that find longer-tenured CEOs are less likely to initiate layoffs. (We omit the CEO tenure control in our preferred specification, in line with [Borgschulte et al. \(2025\)](#), since distress

¹⁶ The time variable is created when the dataset is specified as a time-to-event dataset. See [Efron \(1988\)](#), [Jenter and Kanaan \(2015\)](#), and [Guenzel \(2025\)](#) for more details and finance applications of this approach.

and layoff experience may themselves affect tenure length.) In practice, adding CEO tenure as a control makes little difference. Longer tenure is associated with decreased mortality, suggesting that healthy, resilient CEOs select into serving as a CEO for longer. Regardless, we continue to estimate large adverse effects of distress layoff experience on CEO mortality.

4 Variation in Layoffs by CEO Time in Office (and Firm Attachment)

Having established a direct and personal cost of layoffs for CEOs, this section presents a series of empirical facts showing that CEOs with longer attachment to the firm are significantly less likely to lay off employees. We consider plausibly exogenous CEO turnover events, variation over the course of a CEO’s tenure, and comparisons between externally and internally hired CEOs.

4.1 Layoff Propensity Around Plausibly Exogenous CEO Turnovers

As a first step in identifying frictions in layoffs among CEOs who might be more attached to their employees, we examine how firms’ layoff propensity varies around plausibly exogenous CEO changes. Following prior work (e.g., [Fee, Hadlock and Pierce 2013](#)), we use in-office death or illness of the incumbent CEO to identify plausibly exogenous CEO changes. Note that the differential mortality patterns by CEO layoff experience documented in Section 3 are not inconsistent with this approach to identification as the former results only materialize in the long run, *after* CEOs have stepped down and retired. The median CEO in the *CEO Mortality Dataset* passes away at age 82, whereas the median CEO who is exogenously replaced due to a health shock is 60 years old at the time of replacement. We will also show that the identified exogenous CEO departures are not associated with any pre-trends in terms of key variables of interest (i.e., layoff propensity).

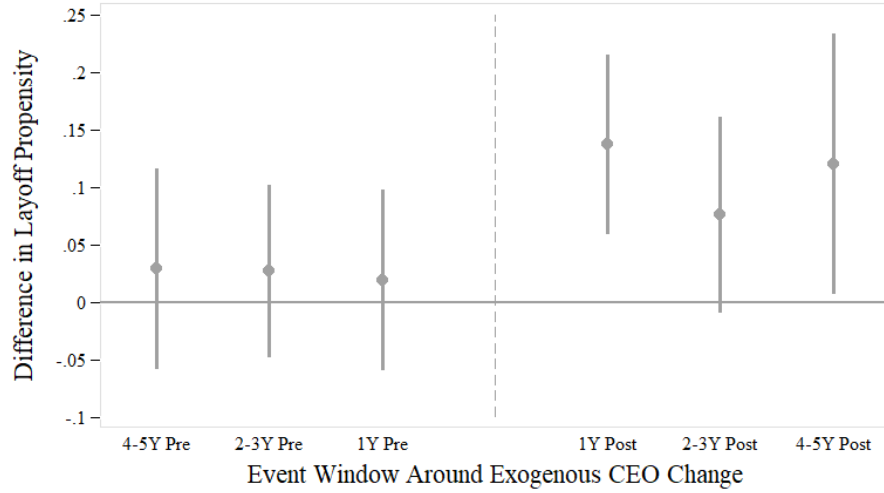
Stacked Difference-in-Differences Analysis. There are 115 CEOs in the *Firm–Layoffs Dataset* who depart due to death or illness based on the [Gentry et al. \(2021\)](#) data. Appendix-Table D.4 summarizes the reasons for exogenous CEO changes in our sample. The most commonly disclosed reason is cancer, followed by cardiovascular issues, predominantly heart attacks, and strokes. Since these events occur at different points in time, we implement a *stacked difference-in-differences* (DiD) analysis similar to [Gormley and Matsa \(2011\)](#) (cf. [Goodman-Bacon 2021](#)). For each year with an exogenous CEO change in firm j , we construct a cohort c of treated and untreated firms in the same three-digit SIC industry k . We follow these cohorts from five years before to five years after the year of the CEO’s exogenous departure, limiting the post-period to the tenure of the CEO who follows the turnover if it is shorter than five years. As in [Gormley and Matsa \(2011\)](#), we do not require firms to be in the data for all ten years around the event year. We then aggregate the cohort-event-time

data across cohorts into one panel and estimate:

$$\begin{aligned} Layoff_{j,k,c,t} = & \alpha + \alpha_{k,c} + \alpha_{c,t} + \beta_1 ExogCEOChange_{j,k,c} \\ & + \beta_2 ExogCEOChange_{j,k,c} \times Post_{c,t} + \varepsilon_{j,k,c,t} \end{aligned} \quad (4)$$

where $Layoff_{j,k,c,t}$ is an indicator for layoff-years, and $ExogCEOChange_{j,k,c}$ identifies treated firms with an exogenous death- or illness-driven CEO departure. $Post_{c,t}$ identifies post-treatment years. To reduce the residual variance and improve precision, we also include the firm control variables from Panel (a) of Table 1 and allow coefficients on the controls to vary between the pre- and post-period (Angrist and Pischke 2009). In addition, we include cohort-SIC3-industry fixed effects $\alpha_{k,c}$, which account for time-invariant differences between industries (within cohorts), and cohort-year fixed effects $\alpha_{c,t}$, which control for time trends (within cohorts) and also absorb the non-interacted $Post_{c,t}$ control. We now cluster at the firm level at which “treatment” is assigned (Abadie et al. 2017).

Figure 4. Layoff Propensity Around Exogenous CEO Changes



Notes: The figure plots the difference in layoff propensity around exogenous CEO departures relative to non-treated firms in the same industry cohort. Exogenous CEO changes are those occurring due to CEO death or illness of the incumbent CEO as identified in Gentry et al. (2021). The 95% confidence intervals are based on standard errors clustered at the firm level.

Figure 4 presents the stacked DiD results graphically, where equation (4) is slightly modified to allow for time-varying coefficients on the interaction term of interest (β_2) within the pre- and post-periods.

The figure shows that, consistent with the assumptions of no pre-trends and parallel trends in the counterfactual state, treated and control firms evolve similarly with respect to their layoff propensity in years prior to the CEO change. In the year after the CEO transition, the difference in layoff

propensity between firms with exogenously replaced CEOs and control firms jumps to +13.7 pp, becomes statistically significant at the 1% level, and remains elevated afterward. When we estimate equation (4), we find an average effect of 8.7 pp over the five-year period following the turnover, statistically significant at the 1% level (Appendix-Table D.5). Hence, newly appointed CEOs exhibit significantly higher layoff propensities than their longer-tenured predecessors.

These results are consistent with layoffs increasing when social connections between the CEO and employees are disrupted, and thus with incumbent CEOs' prosocial attitudes, combined with an average seven-year tenure before exogenous replacement, playing a role in shaping layoff decisions.

Checks for Potential Confounds. While health- and death-related turnover events have been established as the most stringent approximation of plausibly exogenous CEO changes, we consider a number of potential confounds. These concern the potential predictability of such departures, a lower match quality of the successor, and a potential shift in strategy by the successor, all of which might predict differences in layoffs. As for concerns about the possible predictability of CEO departures, we conduct several tests using the data from the pre-period. We find that none of the firm characteristics (assets, employees, or profitability) predict CEO death- and health-related departures.

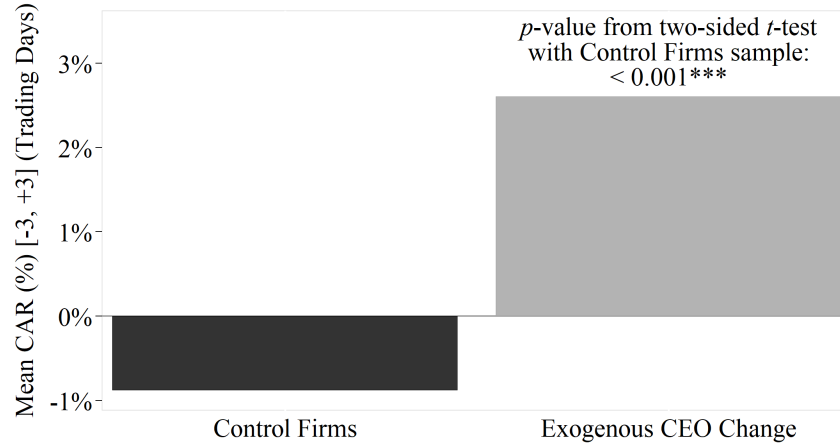
Turning to the concern that the exogenously replaced CEOs were superior matches for their firms, we estimate the market reaction to the announcement of the exogenous CEO departures. The average +/-3-day announcement return to the firm's stock is insignificantly positive in our sample (average CAR = 0.08%, $p = 0.87$), leaving little room for the proposed confound.

Regarding a shift in corporate strategy, we implement a [Romer and Romer \(2010\)](#)–style narrative approach of the layoff announcement descriptions after exogenous CEO turnovers. That is, we return to the key developments data and re-read the description of the key development for all layoffs implemented by the new CEO in our *Firm-Layoffs Dataset*. We find that not a single text mentions a shift in strategy due to the new CEO as a layoff reason. As a second test regarding a potential shift in corporate strategy, we analyze potential changes in other corporate outcome variables. Specifically, we implement a placebo test with firm leverage as the dependent variable and find neither pre- nor post-trends with respect to the financial side of the firm, as shown in Appendix-Figure D.2. This finding is consistent with [Fee, Hadlock and Pierce \(2013\)](#), who similarly do not find significant changes in leverage (nor in book asset growth or ROA) around exogenous CEO turnovers.

Market Reaction to Layoffs After Exogenous CEO Turnovers. We next study how investors react to layoffs made by newly appointed CEOs after exogenously determined turnovers. If the sharp increase in layoff propensity post-turnover reflects frictions prior to the CEO replacement, we would predict a significantly positive market reaction.

We calculate the +/-3-day announcement effects to layoff announcements by firms with exogenous CEO changes during the post-turnover period. We define the window in terms of trading days and require that return data is available for at least four of the seven days. The abnormal return is

Figure 5. Layoff Announcement Returns After Exogenous CEO Changes



Notes: The bar plots present the average CAR (in %) to layoff announcements at control firms (left bar) and firms that had an exogenous CEO change (right bar) during the five-year post period as specified in the DiD analysis. Exogenous CEO changes are due to the death or illness as reported in [Gentry et al. \(2021\)](#).

defined as the stock return minus the market return, the latter as defined in Section 2.2. As indicated by the right bar in Figure 5, these layoff announcements trigger a pronounced positive reaction (average CAR = +2.61%). For comparison, layoff announcements in the matched control firms from the stacked difference-in-differences analysis (i.e., same-industry firms with no exogenous CEO change) trigger a slight negative market reaction of -0.9% during the post-period on average, similar to the average announcement effect in the full sample (-0.3%). The difference in average layoff announcement returns between control and treated firms is highly statistically significant (two-sample *t*-test, *p*-value < 0.001). We can also reject that the announcement return samples have the same median (nonparametric equality-of-medians test, *p*-value < 0.001) or are from the same distribution (two-sample Wilcoxon rank-sum test, *p*-value < 0.001).

Overall, the evidence in Figures 4 and 5 points to sizable layoff frictions under incumbent CEOs that are detrimental to shareholder value. One possible interpretation is that incumbent CEOs have developed more pronounced social concerns towards workers. Upon replacement, these social connections are eliminated, and the new CEO's concern for employees, as captured by the $\phi(\cdot)$ term in the conceptual framework in Section 1 shrinks, leading to behavior that is closer to **shareholder** value-maximizing, including a higher propensity to make layoffs (cf. Prediction 2).

4.2 Layoff Propensity Over the Course of a CEO's Tenure

If CEOs' concerns for their employees increase over time, we would expect layoff activity to decrease over the course of CEOs' tenure (cf. Prediction 3 from Section 1).

We estimate the relationship between the layoff propensity of CEO i at the helm at firm j in year t and the CEO's tenure using the following specification:

$$Layoff_{i,j,t} = \alpha_j + \alpha_t + \beta CEOTenure_{i,j,t} + \mathbf{X}'_{i,j,t}\gamma + \varepsilon_{i,j,t}, \quad (5)$$

where $Layoff_{i,j,t}$ is an indicator equal to 1 in layoff-years, $CEOTenure_{i,j,t}$ is accumulated tenure of the CEO measured in years, and α_j and α_t are firm and year fixed effects. (Below we also estimate models with more no, fewer, and more granular fixed effects.) We continue to cluster standard errors at the firm level (Abadie et al. 2017).

Column (1) of Table 3 shows a strongly negative raw association between layoff propensity and CEO tenure. A one-year increase in tenure is associated with a reduction in the probability of a layoff in a given year by 0.427 pp. We expect this effect to be partially driven by “standard channels” such as a firm performance, i. e., better-performing firms seeing no need to either fire the CEO or lay off employees.

To account for this, we add a battery of firm- and industry-level controls to the empirical model, including lagged firm profitability and an indicator for industry-wide distress, as well as log assets, log employees, year fixed effects, location fixed effects, and Fama and French (1997) 49-industry fixed effects. (Using SIC-3 industry fixed effects produces very similar results.) As shown in column (2), the association between layoff propensity and CEO tenure remains strong, both economically and statistically, with these added controls and fixed effects. The size of the coefficient is reduced to 0.269 pp, though, while firm profitability emerges as another significantly negative predictor. Industry distress, instead, predicts a substantially higher layoff probability, as does firm size (number of employees or total assets).

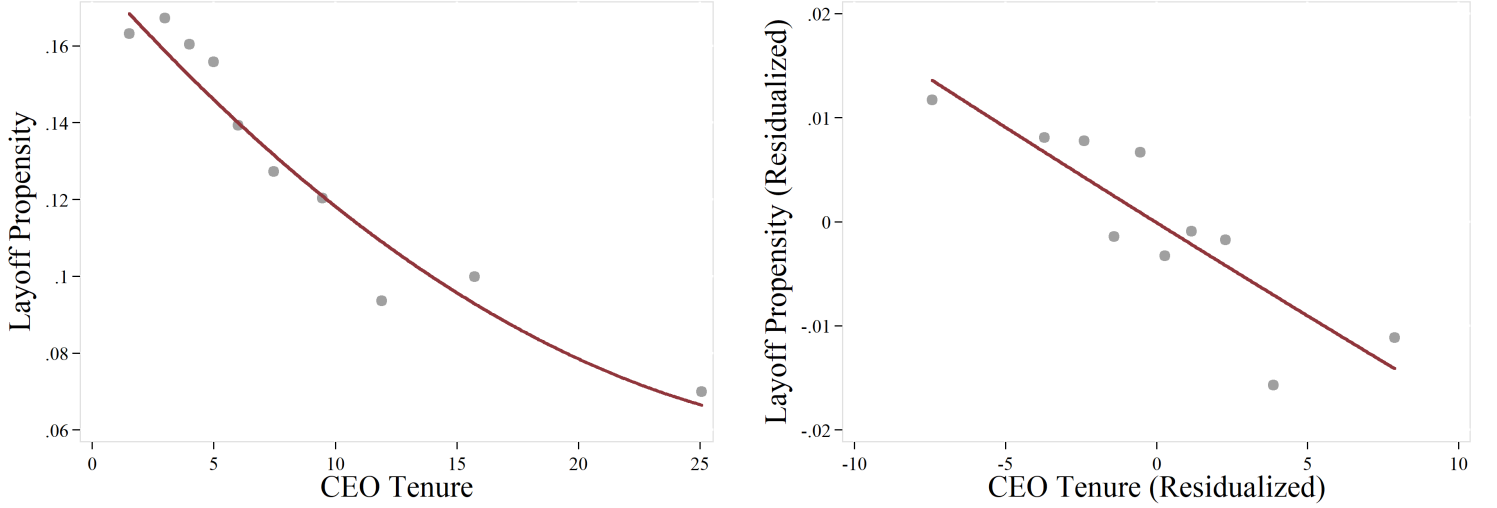
The layoff–CEO tenure relationship persists when we add firm fixed effects in column (3), albeit with a further reduction in magnitude to 0.187 pp. This robustness check accounts for any time-invariant firm characteristics and corresponding CEO matching that might affect the estimation results. For example, it rules out that we pick up an effect where firms with long-serving CEOs have a preference (firm culture) for long-term relationships with employees across hierarchy levels.

Our interpretation is further strengthened in column (4) when we additionally include the CFO's tenure at the firm. While CEO and CFO tenure are meaningfully correlated ($\rho = 0.20$, $p - \text{value} < 0.001$), the coefficient on CEO tenure remains almost unchanged after including CFO tenure as an additional explanatory variable, suggesting that there is a distinct CEO effect in this setting.

In terms of magnitudes, the estimates in column (4) imply that an interquartile increase in CEO tenure (8 years) predicts a reduction in layoff propensity by 1.5 pp, or by 11% relative to the baseline probability of a layoff of 13% (see Table 1). Figure 6 visualizes the relationship between layoff

propensity and CEO tenure graphically, plotting the raw relation on the left and the residualized relation (from column (4)) on the right.

Figure 6. Layoff Propensity and CEO Tenure



Notes: The figure shows binned scatterplots of the relation between firms' layoff propensity and CEO tenure. The left figure plots the raw data. The right figure plots the relationship after residualizing on the controls and fixed effects from column (4) of Table 3.

Robustness. The above results are robust to including various combinations of more granular fixed effects. In Appendix-Table D.6, we show the resulting estimates after including (1) state×year fixed effects to control for any state-specific economic fluctuations, (2) industry×year fixed effects to control for any industry-specific fluctuations, (3) both state×year and industry×year fixed effects, and (4) both state×year and industry×year fixed effects in addition to firm fixed effects. In all cases, the estimate on *CEO Tenure* remain very similar to the corresponding estimates from the baseline models in Table 3, both without and with firm fixed effects.

Alternative Explanations. Despite their robustness, the above results are correlational and not causal evidence of behavioral frictions. In the sections that follow, we aim to strengthen the plausibility of the proposed interpretation by conducting several heterogeneity tests that are implied by a model of prosocial, **empathy-driven** behavior, but **jointly** inconsistent with other explanations. Before turning to these analyses, we first directly assess several leading alternative explanations. We include a brief summary here, and provide a detailed treatment of alternative explanations in Appendix C. [finalize and move CEO power table to appendix](#)

First, one concern is that the estimated tenure effect reflects unobserved variation in firm performance or investment opportunities rather than managerial prosociality. To alleviate this, controlling for a rich set of measures of capital productivity, worker productivity, debt maturity, and

default risk leaves the estimated tenure coefficient virtually unchanged relative to the baseline in Table 3 (Appendix-Table D.7).

Second, the decline in layoffs with tenure may reflect increasing CEO power and empire-building motives (Pan, Wang and Weisbach 2016). However, measures of board co-option, CEO duality, and entrenchment do not explain layoff behavior once CEO tenure is included (Appendix-Tables D.8 and D.9).

Third, the results could be driven by special first-year effects, such as new CEOs undertaking “big bath” restructuring. Excluding the first year of each CEO’s tenure leaves the tenure effect intact and slightly stronger (Appendix-Table D.10).

Fourth, the tenure effect may reflect unobserved CEO characteristics correlated with long tenure. Year-by-year layoff propensities do not differ systematically across CEOs who ultimately achieve shorter versus longer tenure (Appendix-Figure D.3), and instrumenting tenure using the fraction of tenure completed following Altonji and Shakotko (1987) and Graham, Kim and Leary (2020) yields similar estimates (Appendix-Table D.11).

Layoff Size and CEO Tenure. Our detailed layoff data allow us to also take into account the number of employees laid off. Prediction 3(ii) states that prosocial considerations will not solely reduce layoff propensity but also reduce layoff size.¹⁷ Table 4 replaces the layoff indicator with the *fraction of employees laid off* in a given year, defined as zero for non-layoff years. CEO tenure also strongly predicts layoff size. As in Table 3, the CEO tenure coefficient is significant at the 1% level across specifications. Using the more stringent specifications with firm fixed effects, an increase in CEO tenure from the 10th to the 90th percentile is associated with a layoff size effect size about three-quarters as large as that corresponding to industry distress. These results also hold for the instrumented tenure variable, see Appendix-Table D.12, which replicates columns (2) and (3) of Table 4 using the IV approach.

Asymmetric CEO Tenure Effect on Employment Changes (Hiring vs. Firing). Prediction 4 states that managerial prosociality, at least when stemming from empathy-related motives, should affect labor demand asymmetrically: managerial tenure should matter most for preventing layoffs, not for stimulating hiring. Figure 1a illustrates this intuition: when cash on hand is low, the layoff response of a prosocial CEO diverges sharply from that of a standard CEO, whereas hiring responses, when firms are financially unconstrained, align.¹⁸

To test this prediction, we rely on recentered influence function (RIF) regressions, the state-

¹⁷ The results in Table 3 could, in principle, arise from new CEOs, who are presumably less familiar with many parts of the firm, making more frequent layoffs involving fewer employees per layoff, without the *layoff size* actually changing over the CEO’s tenure.

¹⁸ Appendix-Figure A.1 confirms that the prediction regarding asymmetric CEO tenure effects generalizes to the multi-period extension, where (unlike in the three-period version) prosocial considerations can also reduce hiring in order to lower the probability of future layoffs.

of-the-art approach for assessing how covariates affect different parts of an outcome distribution within a single framework (Firpo, Fortin and Lemieux 2009, Rios-Avila 2020). RIF regressions allow us to estimate how CEO tenure is associated with employment changes at various points of the employment-growth distribution, making it possible to distinguish whether tenure primarily mitigates sharp contractions, as predicted by our framework, or instead affects hiring more broadly. Intuitively, RIF regressions assign greater weight to observations that are most informative for each quantile of interest. While implementing this approach requires moving from our granular layoff measures to overall employment growth (cf. Section 2.1 for associated limitations), it enables a direct test of asymmetric effects across downturns and upturns in a unified specification.

Figure 7 shows that the effect of CEO tenure is concentrated in the lower tail of the employment-growth distribution, both economically and statistically, consistent with our framework. Longer-tenured CEOs significantly raise employment growth at low quantiles, corresponding to periods of firm-specific distress with reduced or avoided layoffs. The effect diminishes as employment growth approaches zero and remains flat across the upper tail. Overall, the distributional evidence supports the view that prosocial managerial concerns manifest asymmetrically, consistent with empathy-based microfoundations but not with neoclassical or other prosocial ones such as general altruism and warm glow, by predominantly protecting existing workers during downturns.¹⁹

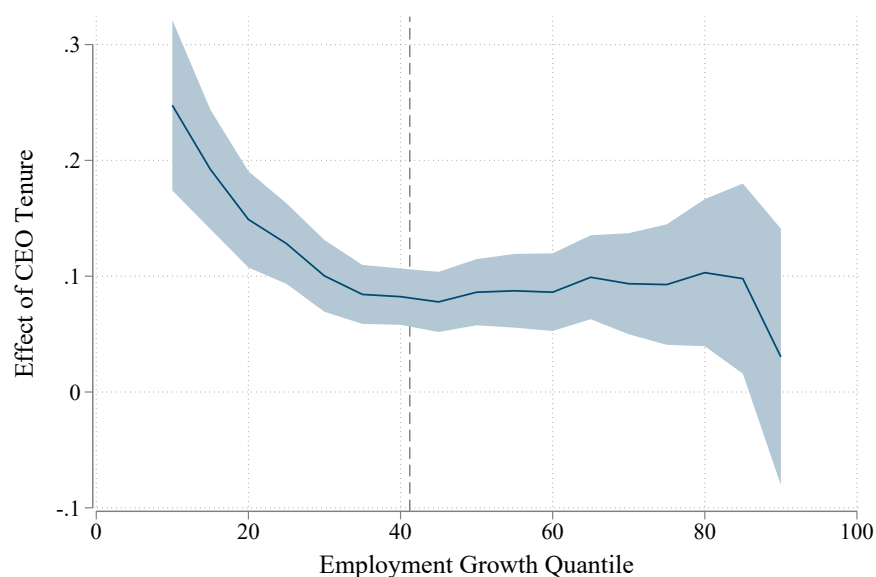
4.3 Layoff Propensity of Internally Versus Externally Hired CEOs

We conclude this section by comparing the layoff propensities of internally versus externally hired CEOs, capturing an additional source of variation in the attachment and relatability the CEO may have developed toward employees. Internally promoted CEOs were previously employed by the firm, have pre-existing ties within the firm, and may thus be more reluctant than externally hired CEOs to implement layoffs. In fact, Bai and Mkrtchyan (2023) show for manufacturing firms that, under external CEOs, more factories are closed or see employment reductions. We build on their evidence, analyzing a broader set of firms and layoff decisions. Of course, the selection of internal CEOs can reflect a wide range of determinants, and it is not possible to find sources of quasi-random variation, as we did for CEO turnover in Section 4.1. Nevertheless, this correlational analysis serves to complement the preceding and subsequent analyses.

One data limitation is that the Execucomp variable JOINED_CO, required to identify when a CEO joined the firm, is frequently missing (cf. Landier et al. 2013), reducing the usable sample

¹⁹ Two differences between Figure 7 and the model-implied patterns are worth noting (see Appendix A.4.3). First, the tenure effect flattens at a higher quantile in the data than in the model, reflecting that layoffs and net employment growth coincide one-to-one in the model but not in the data, where employment adjusts along additional margins (e.g., M&A, attrition). Second, tenure remains positively associated with hiring even at positive employment growth, whereas the model implies zero or negative effects away from the layoff margin. One plausible explanation are additional tenure-related empirical mechanisms that lie outside the empathy channel isolated in the model (e.g., CEO investment cycles; Pan, Wang and Weisbach 2016), designed to capture the incremental left-tail steepening of the tenure effect.

Figure 7. CEO Tenure Effect on Employment Growth



Notes: The figure plots the estimated effect of CEO tenure rescaled by dividing by 100 on employment-growth quantiles. Estimates are from RIF regressions run at 5-percentile increments from the 10th to the 90th percentile calculated following the procedure of [Rios-Avila \(2020\)](#). Shaded bands show 95% confidence intervals based on standard errors clustered by firm. The vertical line indicates the quantile at which employment growth equals zero in the data.

by more than two-thirds. Using this reduced sample, columns (1) and (2) of Table 5 reestimate columns (2) and (3) of Table 3 with an indicator for internally hired CEOs. Internal CEOs are 2.9 to 4.8 percentage points less likely to conduct a layoff (22–37% relative to baseline), depending on the fixed-effects specification. While these estimates remain correlational, the included fixed effects rule out that the type of firms selecting internal CEOs are simply firms with lower layoff propensities; moreover, the estimated effect of CEO tenure remains significant and becomes more negative, indicating that the tenure effect is not driven by the selection of internal CEOs. Columns (3) and (4) show that similar patterns hold when restricting the sample to non-manufacturing firms, extending the findings of [Bai and Mkrtchyan \(2023\)](#) beyond manufacturing.

5 Variation in Layoffs by Layoff Type: Employee Distress and Relatability

I have broken out the CEO heterogeneity as a separate section

We now turn to heterogeneity in the effect of CEOs on layoff decisions, guided by the predictions of the conceptual framework in Section 1 regarding the conditions under which managerial prosocial concerns for workers are likely to be particularly pronounced, as a function of the severity of being laid off and CEO–employee relatability.

5.1 Heterogeneity in the Consequences of Job Loss

We first focus on heterogeneity in how painful job losses are for workers and how this heterogeneity modulates the effect of CEO tenure on layoff propensity.

Layoffs During Recessions. Our mortality evidence in Section 3 shows that CEOs' disutility costs of implementing layoffs are particularly large for layoffs during crisis times. Consistent with prosocial concerns being more pronounced for "pain-inducing" layoffs during recessions, the long-term adverse effects of job loss are especially severe for workers displaced during downturns (Jacobson, LaLonde and Sullivan 1993). Considering this evidence, Table 6 tests whether CEOs' tenure-induced prosocial concerns for workers are particularly impactful in recession times (cf. Prediction 5(i)). We find that the effect of CEO tenure on layoff propensity is approximately twice as large in periods of recessions compared to tranquil times. This holds true across specifications, including when we interact all control variables with the distress indicator (in columns (3) and (4)). These results reinforce the prosocial-concerns interpretation of the CEO tenure effect from Table 3. Consistent with theory, it indicates heterogeneity in the degree of prosocial concerns based on how bleak the job loss implications for workers are. These findings are also generally robust when we use the instrumented tenure variable, particularly when firm fixed effects are included, as detailed in Appendix-Table D.13, which replicates Table 6 using the IV approach.

Layoffs During Recessions: R&D Expenditures. We next conduct an auxiliary analysis to examine how longer-tenured CEOs manage the financial costs of avoiding layoffs, particularly during periods of economic distress when financing constraints are more likely to bind. This analysis takes up Prediction 6 resulting from our framework, which states that prosocial managers who are reluctant to lay off workers must offset higher payroll costs by cutting other forms of investment.

We focus the empirical test on R&D investment because R&D expenditures are a major component of firm investment and are more sensitive to financing constraints than long-term physical capital (Brown and Petersen 2009, Brown, Fazzari and Petersen 2009). Table 7 shows that CEO tenure is not robustly associated with R&D spending on average, consistent with Pan, Wang and Weisbach 2016. At the same time, the interaction between CEO tenure and industry distress is significantly negative, indicating that longer-tenured CEOs reduce R&D expenditures during recessions. An interquartile increase in CEO tenure is associated with a 0.16 percentage point decline in the R&D rate during distress, corresponding to a 2% reduction relative to the mean. These patterns are again robust to instrumenting CEO tenure (Appendix-Table D.14) and support the interpretation that longer-tenured, prosocial managers offset forgone layoff-related cost savings by curtailing R&D expenditures during downturns.²⁰

²⁰ One concern is that observed reductions in R&D expenditures could reflect cuts to personnel-intensive R&D activities rather than substitution away from non-labor investment, which prosocial empathy-based mechanisms would not necessarily predict. To address this concern, we distinguish between industries with high and low labor intensity of

suggested new CEO pay writeup (for reference, old one is in Section 7.1

Layoffs During Recessions: CEO Pay. Beyond adjusting firm investment, another margin along which CEOs can absorb the costs of avoiding layoffs is their own compensation. From a standard compensating-differentials perspective, one might expect the opposite pattern: more demanding working conditions, such as the stress and adverse health effects associated with managing layoffs, should be accompanied by higher pay, possibly especially for longer-tenured CEOs with greater bargaining power. Instead, Table 8 shows that longer CEO tenure is associated with statistically and economically *lower* CEO compensation during periods of industry distress. These results are consistent with longer-tenured CEOs absorbing part of the cost of avoiding layoffs during downturns by curtailing their own compensation rather than shifting adjustment and “pain” onto employees, further corroborating the empathy-based prosocial interpretation.

if we keep this new ceo pay writeup, we should incorporate into prediction 6 and associated discussion in framework

Layoffs Around the Holiday Season. A second differentiator in how pain-inducing layoffs are is their timing within the year. The interviews with top managers in Hallock (2005) contain many responses that convey reluctance to implement layoffs around the holidays.²¹ Figure D.4 presents, for each calendar month, the percent of layoff-years in which a layoff occurs. Consistent with managers’ responses, we observe that December has the lowest percent of layoff-years with a layoff. January has the highest percent of layoff-years with a layoff, followed by October. This pattern could result from managers delaying layoffs to January or undertaking them in October before the holiday season begins. Table 9 tests whether CEOs’ tenure-driven prosocial considerations for workers are intensified around the holiday season (cf. again Prediction 5(i)). In columns (1) and (2), we examine how the unconditional probability of a December layoff depends on the CEO’s accumulated tenure. An interquartile shift in tenure is associated with a strong reduction in the probability of a December layoff by 0.5 percentage points, which is 25% of the baseline unconditional probability.

In columns (3) and (4), we examine how the occurrence of holiday layoffs varies with CEO tenure *conditional* on the firm announcing a layoff in a given year. Here, an interquartile tenure increase is estimated to lower the inclination of a holiday layoff by about 2.9 percentage points, or again 25% relative to the baseline conditional probability. These conditional results are particularly indicative of intensifying prosocial considerations over a CEO’s tenure, as they are derived from *when*—rather than *whether*—CEOs implement layoffs. That is, endogeneity concerns related to whether a CEO

R&D expenditures. Appendix-Table D.15 shows that the decline in R&D during recessions is concentrated in industries with below-median labor intensity, while no such effect is observed in labor-intensive R&D industries, consistent with CEOs curtailing non-personnel investment to avoid layoffs.

²¹ For example, one senior manager noted that they “didn’t want to have layoffs in December for emotional reasons...[not] at Christmas”, while another executive commented, “Is that appropriate to be laying people off around the holidays?”.

implements a layoff do not apply in this case.

The results are robust to extending the layoff window we condition on to include the first, or the first and second, quarters of the subsequent calendar year (cf. Appendix-Table D.16). The results also hold when using the instrumented tenure variable, as presented in Appendix-Table D.17, which replicates Table 9 using the IV approach.

Overall, these results imply that within-year layoff timing, i.e., the *choice* to avoid holiday layoffs, becomes more important to CEOs as they develop deeper connections within the firm over the course of their tenure.

Finally, we note that the decreasing December-layoff propensity with CEO tenure is *not* predicted by CEOs learning over time to “strategically” release bad news. On the contrary, investors appear *more* distracted during the December holiday season (Chava and Paraskar (2016)), and December layoff announcements do not come with significantly different announcement returns ($p = .21$).

5.2 Heterogeneity in Social or Geographic Proximity

We next focus on heterogeneity in “closeness,” as a proxy for the intensity of social interactions between CEOs and employees as well as the degree of CEO–employee relatability (cf. Prediction 5(ii) of the framework in Section 1). We consider both “social proximity,” as captured by the distinction of white-, blue-, and pink-collar employees, and geographic closeness, as captured by the geographic location of the layoff site relative to the headquarters.

Layoffs of Socially Close Employees. We first examine social proximity. The literature on other-regarding preferences shows that prosocial concerns tend to be more pronounced for in-group compared to out-group members (Chen and Li 2009). Further, as discussed in Section 1.2, neuroimaging-assessed empathetic responses are found to be stronger both toward individuals within one’s close social circle compared to strangers and when there is a greater degree of self-other overlap (Meyer et al. 2013).

We leverage our classifications of layoffs as blue-collar, pink-collar, and white-collar layoffs (see Section 2.1) to test whether, particularly as tenure increases, managers become more prosocial towards socially closer employees.²²

Table 10 presents the results. Columns (1) and (2) show that as CEOs’ tenure increases, the probability of laying off white-collar employees—who are socially closest for most CEOs in the sample—declines significantly. In column (2), an interquartile increase in tenure is associated with a 0.79 pp reduction in the probability of a white-collar layoff. This is a 25% drop relative to the baseline white-collar layoff probability of 3.2%, which is roughly double the estimated decrease from the baseline probability in Table 3 using all layoff types. Column (3) estimates the effect of CEO

²² The anecdotal evidence from the practitioners-oriented book by Martin (2005) is consistent with this hypothesis, with one respondent saying, “[a]lways, the most difficult decisions are those that affect families of people that have become friends.”

tenure on white-collar layoffs, conditional on the CEO-firm-year being one in which layoffs occurred. Conditionally, white-collar layoffs become substantially less likely as the CEO remains at the top for longer. For example, An interquartile increase in tenure is associated with a 2.2 pp decrease in the probability of a white-collar layoff in a year with layoffs. This constitutes an 11% reduction relative to the baseline probability of 19.4% that a CEO-firm-year with layoff activity involves white-collar layoffs. These results hold when using the instrumented tenure variable (Appendix-Table D.18), where the conditional estimate is directionally consistent but smaller.

CEO Blue-Collar Formative Background. One prediction of empathy-based models, including the framework in Section 1, is that prosocial effects are strongest for employee groups with whom managers can more readily relate. Unlike neoclassical models and other forms of prosocial behavior, empathy is inherently tied to the ability to place oneself in the position of others; empathetic concern is amplified when individuals experience greater self–other overlap and can more easily put themselves in others’ shoes. One plausible channel through which such relatability arises is similarity in lived experiences. If formative experiences shape whom managers can empathize with most strongly, then the social-proximity patterns documented above should reverse for CEOs whose backgrounds increase their attachment to blue-collar workers.

Our data allow us to test this prediction using white-collar layoffs as above and extending the analysis to blue-collar layoffs, while interacting CEO tenure with an indicator for blue-collar formative exposure, defined based on the occupational composition of the CEO’s birth county (see Appendix-Section B.2.2 for details).²³ Columns (4) and (5) of Table 10 report the results. We do not estimate a statistically significant interaction between CEO tenure and blue-collar formative exposure for white-collar layoffs (column (4)); although the point estimate is positive and opposite in sign to the main tenure effect, it is not statistically distinguishable from zero. By contrast, the interaction between CEO tenure and blue-collar exposure is negative and statistically significant for blue-collar layoffs (column (5)), indicating that CEOs with a blue-collar background become increasingly reluctant to lay off blue-collar workers as their tenure lengthens.²⁴

In other words, the pattern documented above for white-collar employees reverses for CEOs with greater blue-collar relatability, who instead become increasingly less likely to lay off blue-collar workers as their tenure increases. The heterogeneity in layoff behavior across employee groups based on managers’ formative experiences is consistent with differences in managerial relatability

²³ For the blue-collar analysis, we exclude the Trading and Real Estate industries, where the limited incidence of blue-collar layoffs leaves relatively little scope for variation in this outcome. The results are similar, though as expected somewhat weaker, without this restriction.

²⁴ The main effect of blue-collar formative exposure reflects differences in layoff propensities at the start of a CEO’s tenure. Our empathy-based framework does not make a prediction about this baseline difference; rather, it concerns how shared background interacts with tenure, as empathetic attachment requires both relatability and identification with the workforce over time. The opposing signs of the conditional white- and blue-collar estimates also persist when using the instrumented tenure variable.

shaping how empathy-based preferences translate into layoff decisions over time.

Layoffs Close to Firm’s Headquarters. As a second way to capture heterogeneity in social interactions and *relatability*, we examine the geographic, rather than social, proximity of laid-off employees, leveraging the geographic detail of our layoff data. Column (1) of Table 11 shows that layoff activity in the state of the firm’s HQ declines as CEO tenure increases. An interquartile increase in tenure reduces the propensity of HQ-state layoffs by 0.6 percentage points, or 15% relative to the baseline probability of HQ-state layoffs of 4%. Column (2) shows that longer CEO tenure is also associated with fewer HQ-city layoffs. Here, an interquartile tenure increase is associated with a 0.38 percentage point reduction in same-city HQ, or 21% relative to the baseline probability of just 1.8%.

Similar to above, columns (3) and (4) condition on years with layoff activity. Conditional on a year involving layoffs, an interquartile tenure increase is associated with a 1.4 percentage point reduction in the propensity of HQ-city layoffs, or 13% relative to the baseline conditional probability of 11%. Conditional on a year involving layoffs in the HQ-state, an interquartile tenure increase is associated with a 3.7 percentage point reduction in the layoff propensity of HQ-city layoffs, or 8% relative to the baseline conditional probability of 45%. However, standard errors increase substantially in this most narrowly defined subsample, and the point estimate is statistically insignificant ($p = 0.118$). Appendix-Table D.19 re-estimates the first three columns of Table 11 using the IV approach. The results are economically similar across columns, although in the conditional specifications the instrumented tenure coefficient is not statistically significant.

Overall, the results in Tables 10 and 11 are consistent with a prosociality-based explanation of layoff decision-making and increasing social considerations of longer-tenured CEOs, especially in relation to socially and geographically close employees.²⁵

6 Variation in Layoffs by CEO Type: Empathy Vs. Overconfidence

The heterogeneity documented above—where layoff aversion varies with CEOs’ formative blue-versus white-collar backgrounds—already points to systematic differences in managerial relatability and empathy. Building on this, this last section confronts the data with a final prediction of empathy-based models: prosocial disposition should vary across managers themselves, with more empathetic CEOs exhibiting a stronger increase in layoff aversion as their attachment to the firm deepens over time (Prediction 7).

Empirically testing for such heterogeneity is challenging, as it requires proxies that meaningfully capture variation in empathetic disposition among high-profile individuals. Many natural candidates

²⁵ Across Tables 10 and 11, replacing industry and location fixed effects with firm fixed effects yields similar economic magnitudes, though statistical power declines in most conditional specifications restricted to layoff-years or finer subsamples.

are ill-suited in this context, especially for wealthy CEOs.²⁶ For our tests, we draw on prior work that directly assesses CEO personality traits and links them to observable behavior. Kaplan, Klebanov and Sørensen (2012) show that empathy and overconfidence (resoluteness) constitute a primary dimension along which U.S. CEOs differ. Building on this evidence, Kaplan, Sørensen and Zakolyukina (2022) document that the widely used *Longholder* overconfidence measure (Malmendier and Tate 2005) is strongly negatively associated with core empathy-related attributes, such as listening skills and commitment to others. Motivated by these findings, we use the *Longholder* measure to distinguish between more empathetic CEOs (non-Longholders) and less empathetic CEOs (Longholders).

Table 12 shows that the decline in layoff propensity over a CEO's tenure is entirely driven by more empathetic CEOs. An interquartile increase in tenure reduces the layoff propensity of empathetic CEOs by 22% relative to a baseline of 12% ($p < 0.01$), whereas the corresponding effect for less empathetic CEOs is close to zero; the difference between the two effects is statistically significant. Similar heterogeneity arises when examining the timing of layoffs during the holiday season, where tenure-induced avoidance of particularly painful layoffs is again concentrated among empathetic CEOs (Appendix-Table D.20).

A natural concern is that the *Longholder*-based distinction may capture other CEO characteristics—such as general ability or overconfidence—rather than empathetic disposition per se. Two features of our design help mitigate this concern. First, our identifying variation comes from the interaction between CEO tenure and the empathy proxy, rather than from level differences across CEO types. Second, in the analysis of holiday layoffs we condition on firm-years with layoff activity, which limits alternative explanations based on baseline differences in the propensity to lay off workers.

Taken together, the evidence from CEO-type heterogeneity complements the results on employee proximity and formative background.²⁷ Across employees, experiences, and managers, variation in empathy systematically predicts how layoff decisions evolve as managerial attachment to the firm deepens over time, consistent with empathy-based prosocial motives rather than neoclassical mechanisms or generic forms of prosociality.

²⁶ For example, while donation behavior may partly reflect prosocial motives, a large literature documents that giving is also driven by social image and tax considerations (Kumru and Vesterlund 2010, Yermack 2009), making it an unreliable proxy for empathetic concern.

²⁷ This interpretation is related to Acemoglu, He and le Maire (2025), who document a decline in the labor share following the appointment of business-educated (shareholder-value oriented) CEOs. Consistent with their evidence, we replicate the rising prevalence of business-educated CEOs, particularly MBAs, over time (Appendix Figure D.5). In our data, however, we do not find evidence that business-educated CEOs *differentially* increase layoffs upon appointment compared to non-business CEOs (Appendix Figure D.6), nor that controlling for business education alters the estimated relationship between CEO tenure and layoff avoidance (Appendix Table D.21). This suggests that changes in aggregate labor shares associated with CEO type need not operate through involuntary dismissals, but may instead reflect adjustment along alternative margins, such as voluntary turnover, compensation policies, or organizational changes that encourage worker-initiated exit.

7 Conclusion

This paper offers a new perspective on firms' human capital choices and examines the role of CEO prosocial considerations in layoff decisions. We establish a number of results that point to the “human component” of layoffs being more important than captured in prior theoretical and empirical models. First, CEOs responsible for layoffs during recessions incur substantial personal health costs, detectable even in the long run, in the form of accelerated mortality. Second, consistent with incumbent CEOs being shaped by prosocial concerns, exogenous CEO turnovers trigger increased layoffs accompanied by large, positive stock market announcement returns. This effect appears to deepen over time with more social interactions, as layoff propensity declines with CEO tenure. Third, the tenure effect is amplified in contexts where prosocial motives are likely to play a greater role—specifically, when job loss consequences are more severe, when CEO-employee social interaction is more intense, or when CEOs exhibit more empathy-related traits.

Our findings open several avenues for further research. One question is how the effects of managerial prosociality interact with institutional features. Is the mechanism weaker when employees are unionized, as the CEO-employee relationship becomes more contentious? Is it stronger in family firms where managers may have deeper relational ties to employees, and where employees who identify strongly with the firm may experience greater distress from termination?

Another promising direction concerns the delegation of layoff execution to third-party consultants or restructuring firms, which is common in corporate practice. This form of organizational outsourcing may serve to buffer or dilute managers' psychological cost of layoffs. As such, it may function to align managerial decisions with shareholder value by shifting the perceived responsibility for layoffs to agents less affected by prosocial concerns.

Our findings also carry new corporate governance implications. In line with [Cheng, Hong and Shue \(2023\)](#), our results suggest that from the perspective of shareholder-value maximization a broad range of managerial activities needs to be monitored. Monitoring activities should, of course, be concerned with traditional CEO pet projects and perks (see, e.g., [Yermack 2006](#), [Décaire and Sosyura 2022](#)), but may also need to extend to decisions that are influenced by managers' prosocial considerations. Considering our evidence on increasing prosocial tendencies over the course of a CEO's tenure, one could imagine shareholder activists proposing the introduction of CEO term limits—a common practice in the political realm but rare in corporate governance. Vice versa, it is also conceivable that more talented workers might be more attracted to more prosocial firms, which could ultimately be firm value-increasing in the long run (cf. [Agrawal and Matsa 2013](#)). This contrast also highlights that distinguishing between the ex-ante and ex-post effects of managerial prosociality is a promising direction for future research.

Finally, the long-term personal costs associated with the CEO role and the empirical absence of compensating wage differentials associated with difficult layoff decisions suggest that certain “types”

of individuals may be more or less likely to select into the CEO position. Our findings highlight the importance of accounting for the interplay between CEO (prosocial) preferences, job demands, and both labor market and personal outcomes.

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Table 1. Summary Statistics**(a) Firm–Layoffs Sample (Sections 2.1 and 2.2)**

	N	Mean	SD	P25	Median	P75
Layoff	34,683	0.13	0.34	0.00	0.00	0.00
Layoff (Unrestricted)	34,683	0.17	0.37	0.00	0.00	0.00
December Layoff	34,683	0.02	0.14	0.00	0.00	0.00
White-Collar Layoff	34,683	0.03	0.18	0.00	0.00	0.00
HQ-State Layoff	34,683	0.04	0.20	0.00	0.00	0.00
HQ-City Layoff	34,683	0.02	0.13	0.00	0.00	0.00
Frac. Laid Off	33,251	0.01	0.03	0.00	0.00	0.00
CEO Tenure	34,683	8.12	6.94	3.00	6.00	11.00
CFO Tenure	32,433	4.64	3.55	2.00	4.00	6.00
Industry Distress	34,683	0.19	0.39	0.00	0.00	0.00
Ln Assets	34,683	7.68	1.85	6.39	7.57	8.86
Ln Employees	34,683	1.44	1.77	0.26	1.46	2.62
Firm Profitability	34,683	0.11	0.11	0.06	0.11	0.17
<i>Exogenous CEO Turnovers:</i>						
CEO Age at Turnover	115	60.75	9.13	54.00	61.00	65.00

(b) CEO Mortality Sample (Section 2.3)

	N	Mean	SD	P25	Median	P75
Age Appointed CEO	1,131	53.49	6.45	49.00	54.00	58.00
Year Born	1,131	1928.12	8.50	1922.00	1928.00	1934.00
Year Dead/Censored	1,131	2010.33	9.58	2004.83	2015.00	2017.75
CEO Age at Death	680	80.80	9.72	75.00	82.00	88.00
Ever Distressed Layoff	1,131	0.11	0.32	0.00	0.00	0.00
Ever Non-Distressed Layoff	1,131	0.38	0.49	0.00	0.00	1.00
Ever Industry Distress	1,131	0.43	0.50	0.00	0.00	1.00
Ln Assets	1,131	7.35	1.66	6.33	7.38	8.42
Ln Employees	1,131	9.61	1.40	8.78	9.67	10.56
Firm Profitability	1,131	0.14	0.07	0.09	0.13	0.18
Minimum Stock Performance	1,131	0.70	0.15	0.60	0.71	0.80

Notes: Panel (a) shows summary statistics for the *Firm–Layoffs Dataset*. One observation corresponds to a CEO-firm-year. Panel (b) shows summary statistics for the *CEO Mortality Dataset*. One observation corresponds to a CEO. More details about the data are in Section 2 and Appendix-Section B.

Table 2. Layoffs and CEO Mortality

	(1)	(2)	(3)
Distress Layoff	0.294** (0.137)	0.296** (0.138)	0.287** (0.139)
Non-Distress Layoff	-0.031 (0.098)	-0.038 (0.098)	-0.052 (0.099)
Industry Distress	0.149 (0.110)	0.156 (0.111)	0.132 (0.118)
CEO Age	0.159*** (0.010)	0.161*** (0.010)	0.163*** (0.010)
Ln Assets	-0.137* (0.078)	-0.134* (0.079)	-0.130* (0.078)
Ln Employees	0.064 (0.074)	0.061 (0.075)	0.067 (0.075)
Firm Profitability	0.669 (0.688)	0.628 (0.697)	0.808 (0.728)
Year	0.009 (0.009)		
Minimum Stock Performance			-0.326 (0.349)
Year FE	No	Yes	Yes
State FE	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes
Observations	31,917	31,917	31,917

Notes: This table reports estimated coefficients from logit regressions where the dependent variable is an indicator that equals one if the CEO dies in a given year. Distress layoff is an indicator equal to one in year t if, prior to or in year t , a CEO has undertaken a layoff during industry-wide distress. Non-distress layoff is an indicator equal to one in year t if, prior to or in year t , a CEO has undertaken a layoff not during industry-wide distress. Industry distress is an indicator equal to one if the median firm of the industry had a stock price decline of at least 30% in the two years prior or the prior and current year. CEO age is the CEO's age at time t . Minimum stock performance is defined as the worst stock performance in any six-month period during the CEO's tenure. Ln assets and Ln employees from the first year that the CEO assumed the top office are included in all specifications. Firm profitability is included for all years that the CEO is in office, and then profitability from the CEO's final year in office thereafter. More details about the data are in Section 2 and Appendix-Section B. Standard errors, clustered by industry, are in parentheses. Significance levels are indicated by $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

Table 3. Layoff Propensity and CEO Tenure

	(1)	(2)	(3)	(4)
CEO Tenure / 100	−0.427*** (0.034)	−0.269*** (0.029)	−0.187*** (0.039)	−0.181*** (0.041)
Industry Distress		0.057*** (0.007)	0.046*** (0.007)	0.040*** (0.007)
Ln Assets		0.029*** (0.003)	0.028*** (0.007)	0.029*** (0.007)
Ln Employees		0.026*** (0.003)	0.050*** (0.008)	0.055*** (0.008)
Firm Profitability		−0.280*** (0.023)	−0.251*** (0.030)	−0.255*** (0.032)
CFO Tenure / 100				−0.096 (0.071)
Year FE	No	Yes	Yes	Yes
State FE	No	Yes	No	No
Industry FE	No	Yes	No	No
Firm FE	No	No	Yes	Yes
Observations	34,683	34,683	34,565	32,301

Notes: This table reports coefficients from linear regressions where the dependent variable is equal to one for significant layoff events, defined as CEO-firm-years with at least three layoff events or where 1% of the firm is laid off and 0 otherwise. Industry distress is an indicator equal to one if the median firm of the industry had a stock price decline of at least 30% in the two years prior or the prior and current year. Ln assets is the natural logarithm of total assets. Ln employees is the natural logarithm of total employment. Firm profitability is operating income before depreciation divided by total assets. CEO and CFO tenure measure the number of years each executive has held their respective position. For ease of interpretation, tenure is rescaled by dividing by 100. Firm characteristics are from the beginning of the year. More details about the data are in Section 2 and Appendix-Section B. Standard errors, clustered by firm, are shown in parentheses. Significance levels are indicated by $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

Table 4. Layoff Propensity and CEO Tenure—Fraction of Employees Laid Off

	(1)	(2)	(3)	(4)
CEO Tenure / 100	−0.027*** (0.003)	−0.019*** (0.003)	−0.017*** (0.004)	−0.016*** (0.004)
Industry Distress		0.007*** (0.001)	0.006*** (0.001)	0.006*** (0.001)
Ln Assets		0.002*** (0.000)	0.002*** (0.001)	0.003*** (0.001)
Ln Employees		0.001** (0.000)	0.003*** (0.001)	0.003*** (0.001)
Firm Profitability		−0.021*** (0.002)	−0.019*** (0.004)	−0.021*** (0.004)
CFO Tenure / 100				−0.010* (0.006)
Year FE	No	Yes	Yes	Yes
State FE	No	Yes	No	No
Industry FE	No	Yes	No	No
Firm FE	No	No	Yes	Yes
Observations	33,251	33,251	33,130	30,941

Notes: This table reports coefficients from linear regressions where the dependent variable is the proportion of the firm's employees laid off in that CEO-firm-year. Industry distress is an indicator equal to one if the median firm of the industry had a stock price decline of at least 30% in the two years prior or the prior and current year. Ln assets is the natural logarithm of total assets. Ln employees is the natural logarithm of total employment. Firm profitability is operating income before depreciation divided by total assets. CEO and CFO tenure measure the number of years each executive has held their respective position. For ease of interpretation, tenure is rescaled by dividing by 100. Firm characteristics are from the beginning of the year. More details about the data are in Section 2 and Appendix-Section B. Standard errors, clustered by firm, are shown in parentheses. Significance levels are indicated by * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 5. Layoff Propensity and Internally Versus Externally Hired CEO

	Full Sample		Non-Manufacturing	
	(1)	(2)	(3)	(4)
Internal CEO	−0.029*** (0.010)	−0.048** (0.019)	−0.020 (0.012)	−0.053** (0.026)
CEO Tenure / 100	−0.369*** (0.056)	−0.247*** (0.095)	−0.368*** (0.073)	−0.228* (0.119)
Industry Distress	0.060*** (0.013)	0.046*** (0.013)	0.078*** (0.016)	0.064*** (0.016)
Ln Assets	0.027*** (0.005)	0.033** (0.014)	0.020*** (0.006)	0.010 (0.015)
Ln Employees	0.039*** (0.005)	0.060*** (0.017)	0.038*** (0.005)	0.052*** (0.017)
Firm Profitability	−0.327*** (0.044)	−0.270*** (0.063)	−0.251*** (0.063)	−0.247*** (0.085)
Year FE	Yes	Yes	Yes	Yes
State FE	Yes	No	Yes	No
Industry FE	Yes	No	Yes	No
Firm FE	No	Yes	No	Yes
Observations	10,952	10,872	6,280	6,224

Notes: This table reports coefficients from linear regressions where the dependent variable is equal to one for significant layoff events, defined as CEO-firm-years with at least three layoff events or where 1% of the firm is laid off and 0 otherwise. Internally hired CEOs are CEOs who have been with the firm for at least one year before assuming the CEO position. Industry distress is an indicator equal to one if the median firm of the industry had a stock price decline of at least 30% in the two years prior or the prior and current year. Ln assets is the natural logarithm of total assets. Ln employees is the natural logarithm of total employment. Firm profitability is operating income before depreciation divided by total assets. CEO tenure is the number of years the CEO has been in that position. We rescale tenure by dividing by 100. Firm characteristics are from the beginning of the year. More details about the data are in Section 2 and Appendix-Section B. Standard errors, clustered by firm, are shown in parentheses. Significance levels are indicated by * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 6. Layoff Propensity and CEO Tenure—Recession Periods

	(1)	(2)	(3)	(4)
CEO Tenure / 100	−0.229*** (0.029)	−0.141*** (0.039)	−0.239*** (0.028)	−0.150*** (0.039)
Industry Distress	0.074*** (0.010)	0.067*** (0.010)	0.018 (0.030)	0.005 (0.031)
CEO Tenure / 100 × Industry Distress	−0.221*** (0.078)	−0.261*** (0.075)	−0.170** (0.079)	−0.222*** (0.076)
Controls	Yes	Yes	Yes	Yes
Control Interactions	No	No	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
State FE	Yes	No	No	No
Industry FE	Yes	No	No	No
Firm FE	No	Yes	Yes	Yes
Observations	34,683	34,565	34,683	34,565

Notes: This table reports coefficients from linear regressions where the dependent variable is equal to one for significant layoff events, defined as CEO-firm-years with at least three layoff events or where 1% of the firm is laid off and 0 otherwise. Control variables are beginning-of-year firm profitability, Ln assets, and Ln employees. Control interactions are interactions between each of the control variables and industry distress. Industry distress is an indicator equal to one if the median firm of the industry had a stock price decline of at least 30% in the two years prior or the prior and current year. Ln assets is the natural logarithm of total assets. Ln employees is the natural logarithm of total employment. Firm profitability is operating income before depreciation divided by total assets. CEO tenure is the number of years the CEO has been in that position. We rescale tenure by dividing by 100. Firm characteristics are from the beginning of the year. More details about the data are in Section 2 and Appendix-Section B. Standard errors, clustered by firm, are shown in parentheses. Significance levels are indicated by * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 7. Research and Development Expenditures and CEO Tenure

	(1)	(2)	(3)	(4)
CEO Tenure / 100	0.066*** (0.026)	0.018 (0.012)	0.066*** (0.026)	0.018 (0.012)
Industry Distress	0.004 (0.003)	0.005*** (0.002)	0.013 (0.015)	0.005 (0.010)
CEO Tenure $\times$ Industry Distress	-0.065*** (0.024)	-0.035** (0.015)	-0.066*** (0.024)	-0.038** (0.015)
Controls	Yes	Yes	Yes	Yes
Control Interactions	No	No	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
State FE	Yes	No	Yes	No
Industry FE	Yes	No	Yes	No
Firm FE	No	Yes	No	Yes
Observations	15,052	14,994	15,052	14,994

Notes: This table reports coefficients from linear regressions where the dependent variable is research and development expenditures divided by assets at the beginning of the year. Control variables are beginning-of-year firm profitability, Ln assets, and Ln employees. Control interactions are interactions between each of the control variables and industry distress. Industry distress is an indicator equal to one if the median firm of the industry had a stock price decline of at least 30% in the two years prior or the prior and current year. Ln assets is the natural logarithm of total assets. Ln employees is the natural logarithm of total employment. Firm profitability is operating income before depreciation divided by total assets. CEO tenure is the number of years the CEO has been in that position. We rescale tenure by dividing by 100. Firm characteristics are from the beginning of the year. More details about the data are in Section 2 and Appendix-Section B. Standard errors, clustered by firm, are shown in parentheses. Significance levels are indicated by * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 8. CEO Pay and Tenure

	(1)	(2)	(3)	(4)
CEO Tenure / 100	−0.164 (0.156)	0.185 (0.149)	−0.160 (0.156)	0.191 (0.149)
Industry Distress	−0.008 (0.023)	−0.010 (0.019)	0.139* (0.080)	0.093 (0.070)
CEO Tenure / 100 × Industry Distress	−0.517*** (0.183)	−0.430*** (0.152)	−0.529*** (0.184)	−0.462*** (0.152)
Controls	Yes	Yes	Yes	Yes
Control Interactions	No	No	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
State FE	Yes	No	Yes	No
Industry FE	Yes	No	Yes	No
Firm FE	No	Yes	No	Yes
Observations	32,687	32,567	32,687	32,567

Notes: If we keep this table, we could / should(?) combine this table with the previous one and make them panel a and b, respectively, making sure everything fits on one page; would need a different/better title obviously This table reports coefficients from linear regressions where the dependent variable is the natural logarithm of the CEO's total compensation. Control variables are beginning-of-year firm profitability, Ln assets, and Ln employees. Control interactions are interactions between each of the control variables and industry distress. Industry distress is an indicator equal to one if the median firm of the industry had a stock price decline of at least 30% in the two years prior or the prior and current year. Ln assets is the natural logarithm of total assets. Ln employees is the natural logarithm of total employment. Firm profitability is operating income before depreciation divided by total assets. CEO tenure is the number of years the CEO has been in that position. We rescale tenure by dividing by 100. Firm characteristics are from the beginning of the year. More details about the data are in Section 2 and Appendix-Section B. Standard errors, clustered by firm, are shown in parentheses. Significance levels are indicated by * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 9. Layoff Propensity and CEO Tenure—Layoffs Around Holiday Season (December)

	Full Sample		Layoff-Years	
	(1)	(2)	(3)	(4)
CEO Tenure / 100	−0.066*** (0.009)	−0.064*** (0.015)	−0.319*** (0.072)	−0.407*** (0.129)
Industry Distress	0.009*** (0.003)	0.006* (0.003)	0.027** (0.013)	0.040** (0.017)
Ln Assets	0.007*** (0.001)	0.004* (0.002)	0.009 (0.006)	0.018 (0.021)
Ln Employees	0.006*** (0.001)	0.010*** (0.002)	0.008 (0.006)	0.010 (0.021)
Firm Profitability	−0.045*** (0.008)	−0.027*** (0.009)	−0.068 (0.050)	0.010 (0.085)
Year FE	Yes	Yes	Yes	Yes
State FE	Yes	No	Yes	No
Industry FE	Yes	No	Yes	No
Firm FE	No	Yes	No	Yes
Observations	34,683	34,565	5,722	5,174

Notes: This table reports coefficients from linear regressions where the dependent variable is equal to one for years with at least one layoff in December and zero otherwise. Layoff-years are CEO-firm-years with at least one layoff. Industry distress is an indicator equal to one if the median firm of the industry had a stock price decline of at least 30% in the two years prior or the prior and current year. Ln assets is the natural logarithm of total assets. Ln employees is the natural logarithm of total employment. Firm profitability is operating income before depreciation divided by total assets. CEO tenure is the number of years the CEO has been in that position. We rescale tenure by dividing by 100. Firm characteristics are from the beginning of the year. More details about the data are in Section 2 and Appendix-Section B. Standard errors, clustered by firm, are shown in parentheses. Significance levels are indicated by * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 10. Layoff Propensity and CEO Tenure—Social Proximity of Layoffs

	Baseline Analysis			CEO Blue-Collar Formative Background	
	Full Sample		Layoff-Years	Layoff-Years	
	(1) <i>WC</i>	(2) <i>WC</i>	(3) <i>WC</i>	(4) <i>WC</i>	(5) <i>BC</i>
CEO Tenure / 100	−0.091*** (0.014)	−0.099*** (0.020)	−0.272*** (0.092)	−0.153 (0.137)	0.088 (0.119)
Industry Distress	0.018*** (0.004)	0.014*** (0.004)	0.024 (0.015)	0.035 (0.022)	0.016 (0.017)
Ln Assets	0.013*** (0.001)	0.009*** (0.003)	0.021*** (0.008)	0.018 (0.012)	0.019** (0.010)
Ln Employees	0.010*** (0.001)	0.014*** (0.003)	0.015* (0.008)	0.026* (0.014)	0.014 (0.012)
Firm Profitability	−0.061*** (0.011)	−0.036*** (0.014)	−0.044 (0.069)	−0.119 (0.100)	−0.211** (0.100)
Blue Collar Formative Exposure				−0.024 (0.043)	0.083*** (0.031)
CEO Tenure × Blue-Collar Formative Exposure				0.163 (0.412)	−0.845** (0.410)
Year FE	Yes	Yes	Yes	Yes	Yes
State FE	Yes	No	Yes	Yes	Yes
Industry FE	Yes	No	Yes	Yes	Yes
Firm FE	No	Yes	No	No	No
Observations	34,683	34,565	5,722	2,755	2,679

Notes: This table reports coefficients from linear regressions where the dependent variable for columns (1) to (4) is equal to one for years with at least one white-collar layoff and zero otherwise and for column (5) is equal to one for years with at least one blue-collar layoff and zero otherwise. Column (5) excludes the Trading and Real Estate industries, where the limited incidence of blue-collar layoffs leaves relatively little scope for variation in the outcome. Layoff-years are CEO-firm-years with at least one layoff. Industry distress is an indicator equal to one if the median firm of the industry had a stock price decline of at least 30% in the two years prior or the prior and current year. Ln assets is the natural logarithm of total assets. Ln employees is the natural logarithm of total employment. Firm profitability is operating income before depreciation divided by total assets. CEO tenure is the number of years the CEO has been in that position. We rescale tenure by dividing by 100. Firm characteristics are from the beginning of the year. More details about the data are in Section 2 and Appendix-Section B. Blue-collar formative exposure is an indicator equal to one if the fraction of employed persons in blue-collar occupations in a CEO's birth county—measured in the 1970 Census, when CEOs in our sample were young—is above the 90th percentile of the cross-sectional distribution (approximately 50 percent). Standard errors, clustered by firm, are shown in parentheses. Significance levels are indicated by * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 11. Layoff Propensity and CEO Tenure—Geographic Proximity of Layoffs

	HQ-State Layoff	HQ-City Layoff		
	(1)	(2)	(3)	(4)
CEO Tenure / 100	−0.079*** (0.015)	−0.048*** (0.011)	−0.179** (0.089)	−0.458 (0.293)
Industry Distress	0.013*** (0.004)	0.008*** (0.003)	0.019 (0.012)	0.031 (0.038)
Ln Assets	0.014*** (0.002)	0.007*** (0.001)	0.018*** (0.007)	0.012 (0.021)
Ln Employees	0.010*** (0.001)	0.004*** (0.001)	−0.006 (0.007)	−0.008 (0.024)
Firm Profitability	−0.087*** (0.013)	−0.042*** (0.009)	−0.044 (0.058)	0.016 (0.209)
Sample	Full	Full	Layoff-Years	HQ-State Layoff-Years
Year FE	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes
Observations	34,683	34,683	5,722	1,406

Notes: This table reports coefficients from linear regressions where the dependent variables are equal to one if a firm makes at least one layoff at the respective geographic level in a given year and zero otherwise. Layoff-years are CEO-firm-years with at least one layoff. HQ-state layoff-years are CEO-firm-years with any layoff in the state of the firm's headquarters. Industry distress is an indicator equal to one if the median firm of the industry had a stock price decline of at least 30% in the two years prior or the prior and current year. Ln assets is the natural logarithm of total assets. Ln employees is the natural logarithm of total employment. Firm profitability is operating income before depreciation divided by total assets. CEO tenure is the number of years the CEO has been in that position. We rescale tenure by dividing by 100. Firm characteristics are from the beginning of the year. More details about the data are in Section 2 and Appendix-Section B. Standard errors, clustered by firm, are shown in parentheses. Significance levels are indicated by * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 12. Layoff Propensity and CEO Tenure—Heterogeneity in Prosociality

	Full Sample	CEO Empathy / OC Sample	
	(1)	(2)	(3)
CEO Tenure / 100	−0.187*** (0.039)	−0.155*** (0.057)	
CEO Tenure × Empathy			−0.330*** (0.111)
CEO Tenure × Overconfidence			−0.084 (0.065)
CEO Overconfidence		−0.006 (0.010)	−0.018 (0.012)
<i>p</i> -value for equality of coefficients on CEO Tenure × Empathy and CEO Tenure × Overconfidence			0.0549
Controls	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes
Observations	34,565	25,391	25,391

Notes: This table reports coefficients from linear regressions where the dependent variable is equal to one for significant layoff events, defined as CEO-firm-years with at least three layoff events or where 1% of the firm is laid off and 0 otherwise. Control variables are industry distress, beginning-of-year firm profitability, Ln assets, and Ln employees. CEO Overconfidence is an indicator variable defined based on the Longholder measure of managerial overconfidence (Malmendier and Tate 2005, Malmendier, Pezone and Zheng 2023). CEO Empathy is an indicator variable equal to one if the CEO is not classified as overconfident, motivated by Kaplan, Klebanov and Sørensen (2012) and Kaplan, Sørensen and Zakolyukina (2022) that identify overconfidence and empathy are primary distinguishing characteristics of managers. Industry distress is an indicator equal to one if the median firm of the industry had a stock price decline of at least 30% in the two years prior or the prior and current year. Ln assets is the natural logarithm of total assets. Ln employees is the natural logarithm of total employment. Firm profitability is operating income before depreciation divided by total assets. CEO and CFO tenure measure the number of years each executive has held their respective position. For ease of interpretation, tenure is rescaled by dividing by 100. Firm characteristics are from the beginning of the year. More details about the data are in Section 2 and Appendix-Section B. Standard errors, clustered by firm, are shown in parentheses. Significance levels are indicated by * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 13. Layoffs and CEO Pay

	(1) Log TC, t	(2) Log TC, t	(3) Log TC, $t + 1$	(4) Log TC, $t + 1$
Layoff	−0.003 (0.015)	−0.011 (0.012)	−0.044*** (0.016)	−0.036*** (0.012)
CEO Tenure / 100	−0.256* (0.152)	0.114 (0.143)	−0.324** (0.163)	0.060 (0.166)
Industry Distress	−0.054*** (0.018)	−0.047*** (0.014)	−0.068*** (0.019)	−0.048*** (0.015)
Ln Assets	0.356*** (0.012)	0.216*** (0.019)	0.350*** (0.013)	0.138*** (0.021)
Ln Employees	0.051*** (0.011)	0.062*** (0.020)	0.051*** (0.012)	0.084*** (0.021)
Firm Profitability	1.003*** (0.099)	0.751*** (0.086)	0.916*** (0.100)	0.485*** (0.087)
Year FE	Yes	Yes	Yes	Yes
State FE	Yes	No	Yes	No
Industry FE	Yes	No	Yes	No
Firm FE	No	Yes	No	Yes
Observations	32,686	32,568	27,639	27,490

Notes: This is the old CEO Pay table – I’m suggesting to get rid of it and replace with newly added Table 8 This table reports coefficients from linear regressions where the dependent variable is the natural logarithm of the CEO’s total compensation. Layoff is an indicator variable equal to one if there are at least three layoff events or 1% of the firm laid off in that CEO-firm-year and zero otherwise. Industry distress is an indicator equal to one if the median firm of the industry had a stock price decline of at least 30% in the two years prior or the prior and current year. Ln assets is the natural logarithm of total assets. Ln employees is the natural logarithm of total employment. Firm profitability is operating income before depreciation divided by total assets. CEO tenure is the number of years the CEO has been in that position. We rescale tenure by dividing by 100. Firm characteristics are from the beginning of the year. More details about the data are in Section 2 and Appendix-Section B. Standard errors, clustered by firm, are shown in parentheses. Significance levels are indicated by $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

Online Appendix

Prosociality and Layoffs

Not For Publication

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A Conceptual Framework: Proofs and Numerical Solution

A.1 Additional Framework Details

We assume that managers maximize the sum of current and future utility without discounting, for simplicity. Utility is the sum of firm profits and a prosociality term that depends on employees' utility:

$$u(n_t, k_t) = \pi(n_t, k_t) + \int_0^{n_{t-1}} \phi(j, i; u_{t,i}) \, di,$$

subject to the spending constraint, $wn_t + pk_t \leq c_t$, where c_t is the firm's cash on hand in period t , w is the wage, and p is the rental cost of capital. c_t is drawn from an independent and identically distributed (i.i.d.) atomless distribution with a strictly positive density over support $[c_l, c_h]$ with the probability density function (PDF) denoted as $q(\cdot)$. Firm production is Cobb-Douglas with constant returns to scale, so the firm's profit is given by

$$\pi(n_t, k_t) = n_t^\nu k_t^\gamma - wn_t - pk_t,$$

where ν and γ are both positive and sum to one. There is an exogenously assigned employment level $n_0 \in (\nu c_l, \nu c_h)$. Layoffs are reductions in employment ($n_t < n_{t-1}$). We denote worker i 's period- t utility as $u_{t,i}^s$, where $s \in \{\text{emp}, \text{unemp}\}$ denotes employment status, with utility higher when employed than unemployed ($u_{t,i}^{\text{emp}} > u_{t,i}^{\text{unemp}}$). As specified in Section 1, the prosociality term $\phi(j, i; u_{t,i})$ captures the manager's empathetic response to employee distress. Therefore, we can express ϕ as

$$\phi(j, i; u_{t,i}) = r_{t,i} \mathbb{1}\{s_{t,i} = \text{unemp}\} (u_{t,i}^{\text{unemp}} - u_{t,i}^{\text{emp}}),$$

where $r_{t,i} > 0$ specifies the manager's relatability towards employee i in period t . To ensure the manager chooses to operate the firm, we assume that $r_{t,i}$ is sufficiently small that the manager's utility is always positive (assuming an outside option of zero). We also assume that the marginal products of labor and capital are sufficiently high relative to costs that cash on hand is always fully utilized.

For all proofs for predictions, we use a three-period framework. Unless otherwise stated, we assume $r_{2,i}$, $u_{2,i}^{\text{emp}}$, $u_{2,i}^{\text{unemp}}$ are the same for all i and thus we can drop the subscript i . This allows us to simplify the notation, as in this case $\int_0^{n_1} \phi(j, i; u_{2,i}) \, di = r_2(u_2^{\text{emp}} - u_2^{\text{unemp}}) \min\{n_2 - n_1, 0\}$. Only in the proof of Prediction 5 will we consider heterogeneity across employees. We define a standard manager as a manager with $r_t = 0$ for all t . We define a prosocial manager as a manager with $r_1 = 0$ and $r_2 > 0$. Define $\tilde{r}_t \equiv r_t(u_t^{\text{emp}} - u_t^{\text{unemp}})$ and $\Psi_t = -\frac{\partial}{\partial n_t} \text{E}_t[u(n_{t+1}^*, k_{t+1}^*) | n_t]$ where $\text{E}_t[u(n_{t+1}^*, k_{t+1}^*) | n_t]$ is the expected utility in period 2 assuming the optimal employment and capital levels n_2^* and k_2^* are chosen for c_2 given the choice n_1 . Note that $\Psi_2 = 0$ by construction.

A.2 Lemmas

In this subsection, we provide the proofs for the lemmas that are used in the proofs of the predictions.

Lemma 1. *The general solution for the optimal choice of labor is given by*

$$n_t = \begin{cases} H(\Psi_t | c_t) & \text{if } n_{t-1} \leq H(\Psi_t | c_t) \\ n_{t-1} & \text{if } H(\Psi_t | c_t) < n_{t-1} \leq H(\Psi_t - \tilde{r}_t | c_t) \\ H(\Psi_t - \tilde{r}_t | c_t) & \text{if } n_{t-1} > H(\Psi_t - \tilde{r}_t | c_t) \end{cases}$$

where $H(\cdot | c_t) \equiv h^{-1}(\cdot | c_t)$ and

$$h(x | c_t) \equiv \frac{x^{\nu-1} \left(\frac{c_t - wx}{p} \right)^{\gamma-1} (\nu c_t - wx)}{p}.$$

Proof of Lemma 1. The Lagrangian for the prosocial manager in period 1 is

$$\begin{aligned} \mathcal{L} = & n_t^\nu k_t^\gamma - wn_t - pk_t + \lambda_t(wn_t + pk_t - c_t) \\ & + \tilde{r}_t \min\{n_t - n_{t-1}, 0\} + E_t[u(n_{t+1}^*, k_{t+1}^*) | n_t], \end{aligned}$$

where λ_t is the Lagrange multiplier for the spending constraint in period 1. The three first-order conditions can be combined, by substituting out λ_t and k_t , to give the following expression for the optimal employment level in the second period:

$$\frac{n_t^{\nu-1} \left(\frac{c_t - wn_t}{p} \right)^{\gamma-1} (\nu c_t - wn_t)}{p} = \tilde{r}_t \mathbb{1}_{n_t < n_{t-1}} + \Psi_t. \quad (\text{A.1})$$

Using the above definition for h , we can rewrite the above equation as $h(n_t | c_t) = \Psi_t + \tilde{r}_t$. Note that

$$\frac{\partial}{\partial x} h(x | c_t) = \frac{c_t^2 (\nu - 1) \nu x^{\nu-2} \left(\frac{c_t - wx}{p} \right)^\gamma}{(c_t - wx)^2} < 0$$

for $x \in (0, \frac{c_t}{w})$ as $(\nu - 1) < 0$. Monotonicity of h implies a single-valued inverse function $H \equiv h^{-1}$ exists. Further, by the inverse function theorem, $H(x | c_t)$ is strictly decreasing in x and thus $H(\Psi_t | c_t) < H(\Psi_t - \tilde{r}_t | c_t)$. Thus, there is no case where both interior solutions are valid. If $n_{2,\text{standard}}^* < n_1 < n_{2,\text{layoff}}^*$, then an interior solution does not exist, and the boundary solution n_1 is the optimal choice for n_2 . This follows from the concavity of the optimization problem, which implies that the marginal utility is increasing from both sides of the boundary solution towards it. Thus, $H(\Psi_t | c_t)$ or $H(\Psi_t - \tilde{r}_t | c_t)$ is the optimal solution if it is greater or less than n_{t-1} , respectively. If neither of these conditions is satisfied, then n_{t-1} is the optimal solution. $\square$

Lemma 2. Ψ_1 is weakly positive.

Proof of Lemma 2. Define $\hat{c}$ as the value of c that satisfies the equation $H(0 | \hat{c}) = n_1$. Define $\hat{c}$ as the value of c that satisfies the equation $H(-\tilde{r}_2 | \hat{c}) = n_1$. The expected value for period 2 in period

1, given n_1 , can thus be written as

$$\begin{aligned} E_1 [u(n_2^*, k_2^*) | n_1] &= \int_{c_l}^{\hat{c}} q(c) (\pi(H(-\tilde{r}_2 | c), (1/p)(c - wH(-\tilde{r}_2 | c))) + \tilde{r}_2(H(-\tilde{r}_2 | c) - n_1)) dc \\ &\quad + \int_{\hat{c}}^{\hat{c}} q(c) \left(\pi \left(n_1, \frac{c - n_1 w}{p} \right) \right) dc \\ &\quad + \int_{\hat{c}}^{c_h} q(c) (\pi(H(0 | c), (1/p)(c - wH(0 | c)))) dc. \end{aligned}$$

Taking the derivative with respect to n_1 , yields

$$\begin{aligned} \frac{\partial}{\partial n_1} E_1 [u(n_2^*, k_2^*) | n_1] &= - \int_{c_l}^{\hat{c}} q(c) \tilde{r}_2 dc \\ &\quad + \int_{\hat{c}}^{\hat{c}} q(c) \left(\frac{n_1^{\nu-1} \left(\frac{c - n_1 w}{p} \right)^\gamma (n_1 w - c\nu)}{n_1 w - c} \right) dc \leq 0. \end{aligned}$$

Note that while $\hat{c}$ and $\hat{c}$ both depend on n_1 , the terms that arise from applying the Leibniz rule cancel out exactly in this derivative. The first term is clearly negative as long as $\hat{c}$ is in the support of q and 0 otherwise. The second term is negative as long as $\hat{c}$ is in the support of q and 0 otherwise because the denominator is clearly negative and by the definition of $\hat{c}$, $\forall_{c < \hat{c}} n_1 w - c\nu > 0$. Thus, we have shown that

$$\frac{\partial}{\partial n_1} E_1 [u(n_2^*, k_2^*) | n_1] \leq 0.$$

This implies that $\Psi_1 = -\frac{\partial}{\partial n_1} E_1 [u(n_2^*, k_2^*) | n_1] \geq 0$. $\square$

A.3 Proofs

Proof of Prediction 1. For the prosocial manager, the utility in period 2 is given by $\pi(n_2, k_2) + \tilde{r}_2 \min\{n_2 - n_1, 0\}$. It is clear that $\tilde{r}_2 \min\{n_2 - n_1, 0\}$ is a manager-specific disutility cost of layoffs that does not affect firm value or managers who are not prosocial. If $n_2 < n_1$ a layoff occurs and the manager suffers a direct disutility cost $\tilde{r}_2(n_2 - n_1) < 0$. $\square$

Proof of Prediction 2. Suppose that both the new and incumbent managers have the initial level of labor from the previous period $\bar{n}$ and the same cash on hand $\bar{c}$. By Lemma 1, a new manager will choose $n_1 = H(\Psi_1 | \bar{c})$ by and not consider n_0 as $r_1 = \tilde{r}_1 = 0$. An incumbent manager will choose $n_2 \geq H(0 | \bar{c})$. By Lemma 2 and the monotonically decreasing property of H noted in the proof of Lemma 1, $H(0 | \bar{c}) \geq H(\Psi_1 | \bar{c})$. Thus for any values of $\bar{n}$, such that $H(0 | \bar{c}) \geq \bar{n} \geq H(\Psi_1 | \bar{c})$, the new manager will choose a layoff and the incumbent manager will not. $\square$

Proof of Prediction 3. Part (i) is equivalent to the statement in Prediction 2 that a new, alternatively shorter-tenured, prosocial manager has a higher propensity to make layoffs than an incumbent, alternatively longer-tenured, prosocial manager given the manager only makes decisions for two periods.

We thus only need to prove part (ii). Let the level of cash-on-hand be given by $\bar{c}$. By Lemma 1, conditional on a layoff by a shorter-tenured prosocial manager and longer-tenured prosocial manager, the shorter-tenured manager will choose $n_1 = H(\Psi_1 | \bar{c})$ and the longer-tenured manager will choose $n_2 = H(-\tilde{r}_2 | c_2)$. By the strictly decreasing property of h noted in the proof of Lemma 1,

$H(-\tilde{r}_2 | c_2) > H(\Psi_1 | c_1)$. Thus, the reduction in labor from the initial/previous period level will be smaller for the longer-tenured ($t = 2$) prosocial manager since $\bar{n} - n_2 \leq \bar{n}n_1$ which is part (ii) of Prediction 3. $\square$

Proof of Prediction 4. Recall that $\Psi_2 = 0$ by construction; therefore, Lemma 1 implies that in period 2, if $n_1 \leq H(0 | c_2)$, both the prosocial manager and the standard manager choose an identical level of employment, $n_2 = H(0 | c_2)$. As employment expansions only occur when $n_1 \leq H(0 | c_2)$, managerial tenure has no effect on employment expansions for a prosocial manager.

In contrast, if $n_1 > H(0 | c_2)$, the standard manager will undertake a layoff and choose $n_2 = H(0 | c_2)$. In comparison, the prosocial manager will choose either n_1 or $H(-\tilde{r}_2 | c_2)$. Since $n_1 > H(0 | c_2)$ by assumption and $H(-\tilde{r}_2 | c_2) > H(0 | c_2)$ by the strictly decreasing property of H noted in the proof of Lemma 1, it is clear that the prosocial manager will choose a higher level of employment than the standard manager.

Thus, the effect of managerial tenure operates exclusively through the layoff margin. $\square$

Proof of Prediction 5. We begin with the statement regarding the extent to which layoffs affect employee utility. Note that an increase in the adverse effect of being laid off on employee utility (i.e., an increase in $u_2^{\text{emp}} - u_2^{\text{unemp}}$) is an effective increase in $\tilde{r}_2$. Thus, we need to show that an increase in $\tilde{r}_2$ for a prosocial manager to $\tilde{r}_2'$ for a more prosocial manager leads to a larger reduction in layoff propensity and layoff size.

First, we consider layoff size. Conditional on a layoff for both managers, the prosocial manager will choose $n_2 = H(-\tilde{r}_2 | c_2)$ and the more prosocial manager will choose $n_2 = H(-\tilde{r}_2' | c_2)$. By the strictly decreasing property of H noted in the proof of Lemma 1, $H(-\tilde{r}_2' | c_2) > H(-\tilde{r}_2 | c_2)$. Thus, the reduction in labor from the initial/previous period level will be smaller for the more prosocial manager.

Next, we consider layoff propensity. The lowest realization of c_2 , denoted $\hat{c}$, before a prosocial manager will undertake layoffs is characterized by $H(-\tilde{r}_2 | \hat{c}) = n_1$. This is

$$n_1^{\nu-1} (c_2 - n_1 w)^{\gamma-1} (c_2 \nu - n_1 w) = -p^\gamma \tilde{r}_2.$$

We then apply the implicit differentiation technique to find the derivative of c_2 with respect to $\tilde{r}_2$ to get

$$\frac{dc_2}{d\tilde{r}_2} = \frac{-p^\gamma}{\frac{\partial}{\partial c_2} (n_1^{\nu-1} (c_2 - n_1 w)^{\gamma-1} (c_2 \nu - n_1 w))}.$$

Since $\frac{\partial}{\partial c_2} (n_1^{\nu-1} (c_2 - n_1 w)^{\gamma-1} (c_2 \nu - n_1 w)) > 0$, we can conclude that $\frac{dc_2}{d\tilde{r}_2} < 0$. This implies that the lowest value of c before layoffs are undertaken is decreasing in $\tilde{r}_2$, meaning that a manager with a greater $\tilde{r}_2$ parameter is weakly more likely to conduct layoffs. The difference is only weak as there may be no c_2 realizations low enough to necessitate a layoff for the considered $\tilde{r}_2$ values.

Now, we turn to the statement regarding heterogeneity across employees in the weights that the manager assigns to employee utility. We aim to show that, compared to a short-tenured prosocial manager, a long-tenured prosocial manager reduces layoff propensity and layoff size more strongly for employees whose utility the manager weighs more heavily. In order to do so, we allow $r_{2,i}$ to vary across i . For simplicity, suppose that there are two types of employees, l and h , for which

$r_{2,l} < r_{2,h}$. Among the mass n_1 time $t = 1$ employees, let mass $\frac{n_1}{2}$ be of either type. In this setting,

$$\begin{aligned} \int_0^{n_1} \phi(j, i; u_{2,i}) di &= \int_0^{n_1} \mathbb{1}_{\{s_{2,i}=\text{unemp}\}} r_{2,i} (u_2^{\text{unemp}} - u_2^{\text{emp}}) di \\ &= r_{2,l} \int_0^{n_1/2} \mathbb{1}_{\{s_{2,i}=\text{unemp}\}} (u_2^{\text{unemp}} - u_2^{\text{emp}}) di + r_{2,h} \int_{n_1/2}^{n_1} \mathbb{1}_{\{s_{2,i}=\text{unemp}\}} (u_2^{\text{unemp}} - u_2^{\text{emp}}) di \end{aligned}$$

At time $t = 2$, the long-tenured manager solves

$$\begin{aligned} \max_{n_2, k_2} \pi(n_2, k_2) + r_{2,l} \int_0^{n_1/2} \mathbb{1}_{\{s_{2,i}=\text{unemp}\}} (u_2^{\text{unemp}} - u_2^{\text{emp}}) di \\ + r_{2,h} \int_{n_1/2}^{n_1} \mathbb{1}_{\{s_{2,i}=\text{unemp}\}} (u_2^{\text{unemp}} - u_2^{\text{emp}}) di \end{aligned}$$

Consider a corresponding solution (n_2^*, k_2^*) . Fixing the solution, it is clear that, if $n_2^* < n_1$ (i.e., if a layoff occurs), the manager will optimally lay off mass $\min(\frac{n_1}{2}, n_1 - n_2^*)$ employees of type l and mass $\max(0, \frac{n_1}{2} - n_2^*)$ of type h . Intuitively, the manager will always prefer to lay off a type l employee over a type h employee. Hence, if c_l is the cash threshold below which a layoff involving type l employees occurs, the corresponding threshold for type h employees c_h must satisfy $c_h < c_l$. So, any reduction in layoff propensity for type l employees implies a further reduction in layoff propensity for type h employees. Similarly, for any layoff-inducing n_2^* , the mass of type l employees laid off is weakly greater than the mass of type h employees laid off: $\min(\frac{n_1}{2}, n_1 - n_2^*) \geq \max(0, \frac{n_1}{2} - n_2^*)$. Hence, any reduction in layoff size for type l employees implies at least as large of a reduction in layoff size for type h employees (with equality only if $n_2^* = 0$). $\square$

Proof of Prediction 6. Recall that the budget constraint holds with equality, and thus given n_t and c_t , $k_t = \frac{c_t - wn_t}{p}$. Thus, the proof for Prediction 3, which finds higher n_2 for a longer-tenured prosocial manager than a shorter-tenured prosocial manager, implies that k_2 is lower for the longer-tenured prosocial manager given the same value of cash-on-hand, thus proving the prediction. $\square$

Proof of Prediction 7. Note that a greater increase in relatability over a CEO's tenure (i.e., a greater $r_2 - r_1$) is equivalent to a larger $\tilde{r}_2$, given that we assume $r_1 = 0$. Hence, we can prove the prediction by showing that an increase from $\tilde{r}_2$ to $\tilde{r}_2'$ leads to a larger reduction in layoff propensity and layoff size. This was shown directly in the proof of Prediction 5. $\square$

A.4 Numerical Results

This subsection describes how the framework presented in Section 1 is solved numerically. We first describe how we solve the three-period model, which generates the results displayed in Figure 1a. We then conduct a calibrated analysis in a multi-period setting to verify that our predictions are robust to longer decision-making horizons, which generates the results displayed in Figure 1b.

A.4.1 Three-Period Framework

We solve the three-period model by iterating backwards on a discretized state and action space. Specifically, we discretize both the cash-on-hand distribution and the labor choice variable using 1,001 grid points each. For the three-period model, cash-on-hand is distributed independently and identically with a uniform distribution from .5 to 1.5. Once we discretize the key variables, we then solve for the optimal choice of n_2 for all possible combinations of period-1 labor, n_1 , and period-2 cash-on-hand, c_2 , realizations. Figure 1a displays the resulting optimal change in labor demand as a function of period-2 cash-on-hand for three manager types: a standard manager with $r_2 = 0$, a low prosocial manager with $r_2 = 0.2$, and a high prosocial manager with $r_2 = 0.4$. The difference in utility between states is $u_{2,i}^{emp} - u_{2,i}^{unemp} = 1$ for all employees. The figure illustrates that prosocial managers reduce labor demand less aggressively than standard managers when cash-on-hand declines, and the effect is more pronounced for managers with higher relatability.

We assume a Cobb-Douglas production function with labor share $\nu = 2/3$ and capital share $\gamma = 1/3$, consistent with standard macroeconomic calibrations. We set the rental rate of capital $p = 0.2$ and the wage $w = 0.5$.

A.4.2 Extension to Multiple Periods

The three-period framework is analytically tractable, but precludes analysis of managerial decision-making over longer horizons. To ensure our findings are robust to a extended time horizons, we simulate a version of the model with many periods and numerically solve it using backward induction. All parameters not specified below are the same as in the three-period model.

For the multi-period model, we introduce a per-period drop-out rate to allow for CEO turnovers as in the data. We model drop out through a factor $\rho < 1$ applied to future expected utility. We set $\rho = 0.01^{1/34} \approx 0.876$, implying that a manager's probability of remaining at the firm through period 34 is approximately 1 percent, matching the data. We solve for 50 periods by backward induction to ensure there is always a future time horizon, but only simulate 34 periods. We allow managers to account for this drop-out probability when making their employment decisions, and simulate outcomes for the same number of managers in all periods.

We model cash-on-hand using a discontinuous triangular Markov process that is then discretized. The transition probabilities are chosen such that, conditional on current cash-on-hand, there is a 15 percent probability of transitioning to a lower cash-on-hand and an 85 percent probability of transitioning to an equal or higher cash-on-hand. As in the three-period model, we consider cash-on-hand values between .5 and 1.5. The probabilities within each region follow a triangular distribution where the mode is the lower boundary for the upper part of the distribution, and the upper boundary for the lower part of the distribution. This calibration generates an average layoff probability of approximately 15 percent for the standard manager, consistent with the average layoff probability of 13% in our empirical data. For $t = -1$, all managers have a cash-on-hand value of

1.35, the steady-state mean of the cash-on-hand distribution, and labor equal to the mean labor for all simulated types of managers from period 34.

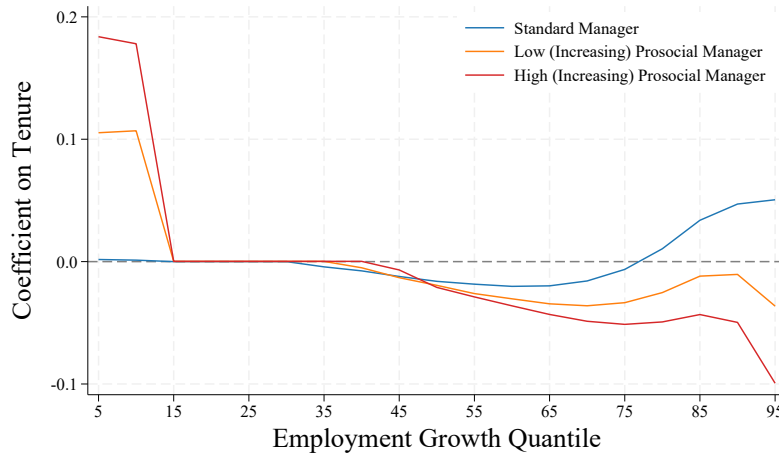
We consider three manager types that parallel those in the three-period analysis. The standard manager has $r_t = 0$ for all periods. The low (increasing) prosocial manager has r_t increasing linearly from 0 in period 1 to 0.2 in period 50. The high (increasing) prosocial manager has r_t increasing linearly from 0 in period 1 to 0.4 in period 50. The difference in utility between states is $u_{t,i}^{emp} - u_{t,i}^{unemp} = 1$ for all employees in all periods. We simulate for 10 million managers for each type.

Section 1.1 of the main text discusses the results of the extended framework, matching the predictions of the three-period model.

A.4.3 Recentered Influence Function Regression

To further facilitate a direct comparison between the predictions of the extended model and the data, we apply the same recentered influence function (RIF) methodology as in Section 4.2 to the simulated employment outcomes. To this end, we regress the RIF of employment growth at various quantiles on tenure and lagged log labor. Appendix-Figure A.1 displays the coefficient on tenure across the employment growth distribution for each manager type.^{1A.1} The simulated coefficients display a pronounced left-tail steepening, with tenure raising employment growth in contraction states and having negligible effects in the upper tail, closely matching the asymmetric pattern observed in Figure 7. See Section 4.2 for further discussion of the economic interpretation and the comparison to the empirical estimates.

Appendix-Figure A.1. RIF Regression Coefficients on Tenure Across Manager Types



Notes: The figure plots the estimated effect of tenure on employment-growth quantiles for three manager types from the multiple-period simulation. Estimates are from RIF regressions run at 5-percentile increments from the 5th to the 95th percentile calculated following the procedure of [Rios-Avila \(2020\)](#). The estimation controls for lagged log labor. Results are based on 10 million simulated observations for each manager type.

^{1A.1} Standard errors are extremely small due to the large sample size and, therefore, confidence intervals would not be visible if plotted.

B Data Appendix

B.1 Firm–Layoffs Dataset: Construction

Background. This section provides more details about the Standard and Poor’s (S&P) Capital IQ Key Developments database, our data processing using NLP, and manual review. The goal is to identify all significant layoff events in each of our sample firms, as well as detailed information about those events (size, location, type of employees laid off). The default definition of significant layoff events (and the main dependent variable in our empirical analysis) includes all cases where a firm has either a layoff affecting 1% or more of the firm’s employees or has at least three layoff events in a given year. (Our results are robust to removing these restrictions.)

Data Description. Each key development in the Capital IQ Key Developments data entry includes the announcement date, the date entered into the database, the last date the entry was modified, a short headline, a situation summary, a type, a source, a company role, and other identifiers. We use the announcement dates, headlines, and situation summaries. The headlines are typically a phrase, and the summaries are generally between one sentence and a small paragraph. According to S&P, the content is based on a variety of sources including “news aggregators, stock exchanges and regulatory websites as well as company websites.” For our analyses, we use and classify all key developments given the “Discontinued Operations/Downsizings” type, which includes layoff announcements and closure announcements.

Natural Language Processing. As discussed in the main text, we leverage NLP techniques to extract key data points from the text. Specifically, we automate four functions: (1) classifying each article as a layoff, closure, or neither, (2) extracting the number of workers laid off and/or the percentage of workers laid off, (3) identifying relevant geographies at the international, state, and city levels, and (4) classifying layoffs as blue-, pink-, or white-collar.

Classification. For (1), we first lemmatize the headline and situation summary and construct three lists of words to aid in identifying layoffs and closures: (i) layoff verbs, such as “cut” and “slashed,” (ii) closure verbs such as “mothball” or “shut down,” and (iii) worker-related nouns, including (iii.α) job words for synonyms of jobs like “position” or “role,” (iii.β) occupations words for industry-specific jobs such as “accountant” or “banker,” and (iii.γ) collective job words referring to jobs generally, such as “headcount” or “workforce.” A key development is classified as a layoff if the headline or sentence in the summary includes a layoff-related verb as well as a noun from one of the three worker-related lists specified above. Key developments are identified as low-confidence layoffs if either a layoff-related verb or worker-related noun is present, but not both. If the key development does not contain a layoff-related verb or worker-related noun, but contains a closure-related verb, then it is classified as a closure. Additionally, we eliminate any layoff or closure that is a speculation or consideration. We create a list of words that indicate uncertainty, such as “pondering” or “considering.” If the sentence contains the relevant layoff-related verb, worker-related noun, and/or closure-related verb and also includes a word indicating uncertainty, then the key development is not classified as a layoff or closure.

Layoff Size Extraction. Next, for (2), we extract the number and/or the percentage of workers laid off when available.^{1A.2} In addition to the restricted text searches (regular expressions) for percentages and numbers discussed in the main text, we also use these patterns to extract the

^{1A.2} If only the percent of a subset of workers being laid off is referenced, this number is not collected since we only have the number of total employees at the firm in Compustat.

percentages or numeric values that are not expressed as numbers. We use a list of common number and proportion words for fractions like “one half” or “two-thirds” and numbers like “thirty”, and convert them to their numerical integer values. For general quantities like “hundreds” or “thousands,” we encode them as the largest possible value, i.e., 999 or 9999, respectively. If none of the patterns were present, proportions or counts from sentences with a layoff-related verb or a worker-related noun were still extracted, but denoted as low confidence for the manual review.

Location Extraction. For (3), the geographic location of the layoff, we collect the location up to the city level for US layoffs and up to the country level for foreign layoffs. At the international level, we identify foreign locations in the text of each key development based on lists of country names, demonyms for people from the respective countries, and names of large global cities, such as Tokyo. If no specific country can be identified, we record words that would imply layoffs outside the US (e.g., foreign, international, etc.). We also note if the US or a US state is mentioned, since this might imply that at least some employees were laid off domestically. At the domestic level, we identify states by their names or variants (e.g., Alaska, Alas., AK). We identify cities from constructions of the form “city, state” (e.g., “Buffalo, NY”) and from a list of the 1,000 largest cities in the US based on 2013 Census data.^{1A.3} If only a city name was identified, then the NLP would include a list of all states with a city of the same name in parentheses ordered by population.

Manual Review. After NLP-based steps (1) to (3), we conduct a manual review of the prepopulated data for all key developments, regardless of how they are classified by NLP, with the goal of ensuring accuracy, checking low-confidence cases, adding missing information, applying more complex screening criteria and calculations, and eliminating duplicates. Reviewers were given comprehensive instructions about how to classify each case in both text and video form, consistent with the description here. The key development data was put into Microsoft Excel along with the classifications, layoff sizes, and locations from NLP. Low-confidence classifications were highlighted, and it was made clear to reviewers that they should pay close attention to these cases. Reviewers examined all of the output from NLP and updated the relevant data columns to make them accurate. Reviewers were instructed to classify a key development as a layoff if it described a layoff where “it is clear people lost their jobs.” Thus, business sales, including exits via sales where no jobs were lost, and temporary furloughs were not coded as layoffs. If workers were having their jobs terminated, but being reassigned to other positions within the company or taking early retirement, this was not considered a layoff. If some portion of the workers laid off were reassigned or retired early, they were subtracted from the total layoff size. A key development was classified as a closure if it described a closure (and not also a layoff) and classified as neither if it did not describe either a layoff or a closure. Additionally, precise instructions were given to not consider any layoffs and closures that were only speculative or under consideration, as neither a closure nor a layoff.

Manual Review of Location. In the case, that only a US city name was listed without a specific state, the reviewer would make their best guess for the state in which the layoff occurred. If the city name was identified in the top 1,000, the NLP algorithm would list the state where the city of that name had the highest population first, making this a straightforward best guess if no additional information was available. In some situations, it may be clear or likely that the layoff occurred in a specific state based on the context from other key developments.

Web Search for Location. Reviewers did not use outside information generally; however, reviewers were told to use additional research to determine the state if it was otherwise unclear

^{1A.3} We exclude York and Knoxville to avoid confusion with places/names that may contain these names.

and could quickly be determined (e.g., through a Google search with the city name and company name). We conducted additional searches (e.g., by using a search engine) for identifying locations if there was a place listed with a clear location (e.g., a very specific mall) where the city was not included in the text, but could be easily identified (e.g., via a Google search of a specific mall's name). Examples include searches for states when only the city was specified, and searches for cities when only non-city locations such as business complexes were specified.

Handling of Duplicates. As part of the manual review, duplicate layoffs/closures are noted when possible, e. g., when the summary notes that the announced layoff was part of an earlier announcement or the execution of an alternative announcement, or when it is obvious from context, e. g., if the text specifies a certain factory being shut down in two announcements. Duplicates were specified in the instructions for reviewers as “when an observation is believed to be a duplicate of a previous observation.” In some cases, the announcement of a closure may take place separately from, and potentially earlier than, the announcement of a related layoff that is part of the closure. All duplicates are removed from the dataset to prevent double-counting of layoff events and the number of people laid off. If two duplicates contain the same information, then the earlier announcement is chosen. If one of the duplicates contains more information (e.g., number of people laid off, location), then that one is chosen. In such cases, the closure announcement is followed by a separate announcement, which specifies an associated layoff. In this case, the closure announcement is classified as a duplicate. We classify the closure announcement as a duplicate since, by definition, it does not contain details about the workers being affected.

Additional Review of Potential Errors. Lastly, in case of an error during manual review, we also double-checked certain cases noted as atypical or with formatting errors. We checked all key developments where the percentage of the firm laid off was entered as less than 1%. Further, if we identified more than two-thirds of the firm was laid off across all layoffs in a given firm-year, we checked all layoff entries for that firm. We also checked cases where we identified the percent laid off column had an entry recorded as a date in Excel or there were parentheses in a city or state entry for one of the key developments.

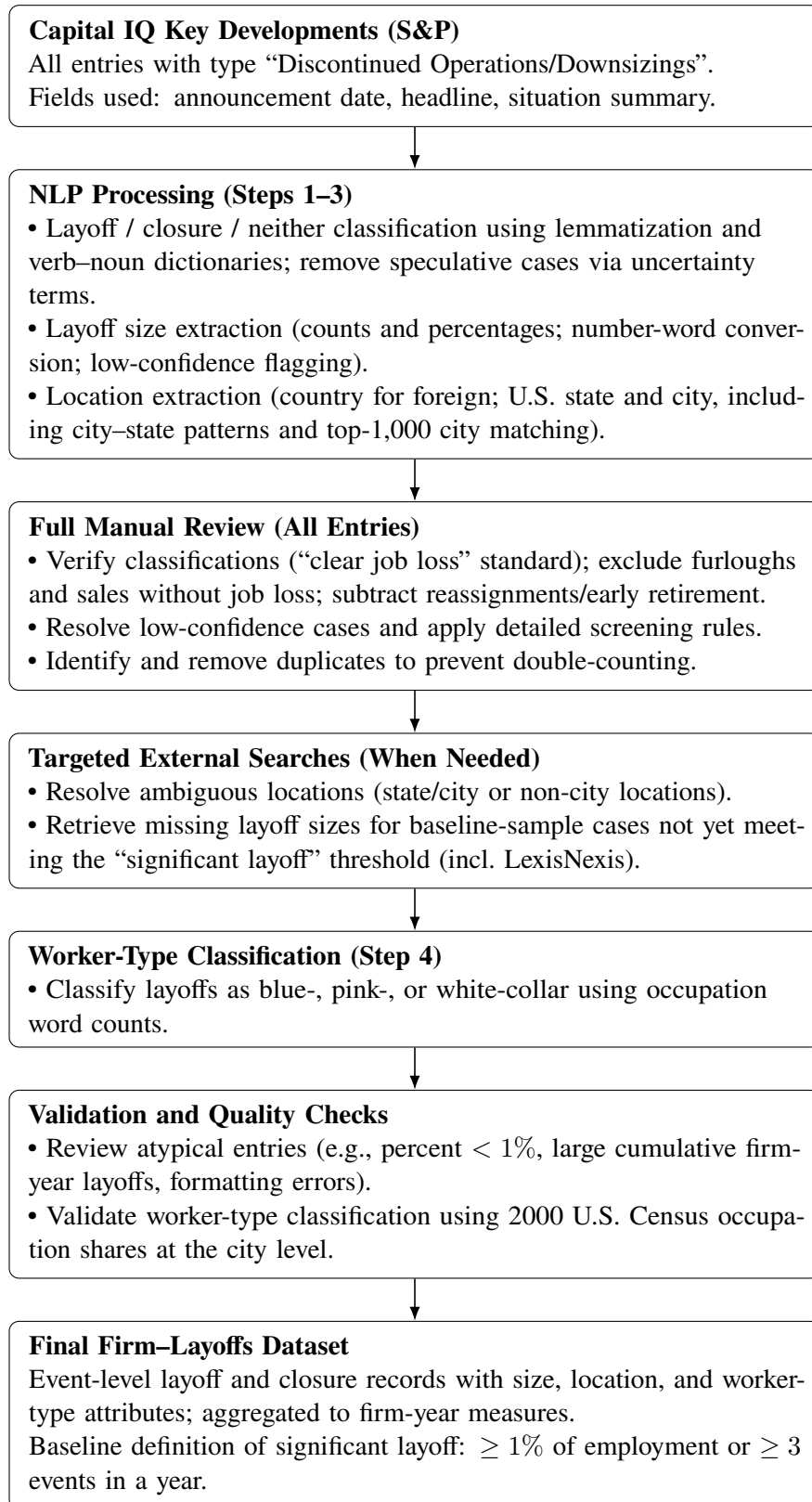
Web Search for Layoff Size. After constructing an initial data set with NLP and manual review, we conducted an additional search for data on all key developments in our baseline regression sample with missing information on layoff size in layoff-years that did not (yet) qualify as years with significant layoff events, i. e., that had fewer than three layoffs and less than 1% of firm employment laid off included so far. For this subset of layoffs, we conducted manual searches for layoff size using the web and LexisNexis. When the original news source is included in the data, that was the preferred source for the layoff size when found. If it could not be found or was missing, another source would be used if available. The additional searches generated the missing information on layoff size for about 10.67% of the key developments, suggesting that this information is usually included in the key development text when it is publicized.

White-Collar Layoff Classification. With these refinements and additions, we turn to NLP step (4) in order to classify layoff events into blue-, pink-, and white-collar layoffs. Here, we utilize the count of words from each category in the layoff description. If a description does not contain any blue-collar or pink-collar words and at least one white-collar word, we classify it as a white-collar layoff. We classify the remaining layoffs that contain at least one blue-collar word or pink-collar word as the type that it contains more words from, defaulting to blue-collar in the case of a tie. The reason for employing a higher hurdle towards white-collar classification is that, when a blue- or

pink-collar word appears in the description, we found it to be almost certainly not a white-collar layoff, as the latter would typically not discuss factories or retail stores.

Verification of White-Collar Layoff Classification. As mentioned in the main text, to verify our classification of layoff type leveraging the data on the locations of layoffs from the 2000 US Census downloaded from Social Explorer, a third-party data tool. Using the “Occupation For Employed Civilian Population 16 Years And Over” data at the geographic level of “Places,” which includes cities and townships, linked to our layoff location data, we construct a proxy for the percentage of predominantly blue-collar occupations, white-collar occupations, and pink-collar occupations in a given city (layoff location). There are 13 broad occupation categories, and we designate “Management, business, and financial operations occupations” and “Professional and related occupations” as white-collar, “Construction, extraction, and maintenance occupations” and “Production occupations” as blue-collar, and “Personal care and service occupations” and “Sales and related occupations” as pink-collar. We calculate the proportion of the employed population in the city in each of these categories. If the layoff or closure takes place in more than one city, we take the average of the proportions across the cities for all cities where data is available. We are able to construct this proxy for 80.4% of the layoff and closure events in our sample where we identify at least one city. We find that the proportion of the city, or average proportion of the cities, employed in white-collar jobs is 37.2% for the key developments text-based NLP classifies as white-collar compared to 33.5% for the other key developments. While this difference may not seem large initially, it is statistically significant with a p -value of less than 0.0001 for a two-sample t -test. We also conduct analogous two-sample t -tests for layoffs/closures classified as blue-collar and pink-collar and find p -values of less than 0.0001 in both cases as well.

Appendix-Figure A.2. Construction of the Firm-Layoffs Dataset



B.2 Firm–Layoffs Dataset: Variable Definitions

Layoff. We classify a CEO-firm year as a layoff if there are at least three separate layoff events or at least 1% of the workforce is laid off. For analyses focusing on location- or month-specific layoffs, we define a layoff as any layoff event occurring within the relevant geography or time period.

CEO Tenure. We identify CEOs using ExecuComp’s classification, where the CEOANN variable has the value “CEO”. CEO tenure is calculated using the “BECAMECEO” variable as the starting year when it is available. Otherwise, we calculate it using the Execucomp panel, which starts in 1992.

CFO Tenure. We identify CFOs using ExecuComp’s classification, where the CFOANN variable has the value “CFO” or a close variation such as “Chief Financial Officer.” Otherwise, we identify CFOs from the text of their title. Following [Jiang, Petroni and Wang \(2010\)](#), we employ the following order of titles: CFO, chief financial officer, treasurer, controller, finance, and vice president-finance. If more than one person in a year has the same title then the person with the higher age is chosen as the CFO.

CEO Power (% of New Directors). We construct the percentage of new directors appointed during a CEO’s tenure using data from the Institutional Shareholder Services (ISS). The ISS data covers S&P 1500 firms since 1996 and includes the directors in each year for each firm and the year they became a director. As with the Execucomp data, if no year is listed for the director’s first year, we use their first year in the data as the start year of their directorship. To match the ISS data between the current and legacy samples, we also rely on the director link between the current and legacy samples published as a part of, and described in, [Coles, Daniel and Naveen \(2014\)](#). The link is necessary as it fills in missing director IDs from surrounding years, corrects errors in the ISS data, and resolves inconsistencies between the current and legacy director identifiers.

Industry Distress. Similar to prior literature, we define an industry-year as in distress if the median firm’s stock price declined by at least 30% over a two-year horizon. For the context of layoffs, we consider both the prior two years and, alternatively, the prior and current year, and set the Industry Distress indicator equal to 1 if the firm is in distress over either horizon. Following [Babina \(2020\)](#), we (i) use SIC3 industry classes, (ii) restrict to single-segment CRSP/Compustat firms, i. e., drop firms with multiple segments in the Compustat Business Segment Database (CBSD), (iii) drop firms if the reported single-segment sales differ from those in Compustat by more than 5%, (iv) restrict to firms with sales of at least \$20m, and (v) exclude industry-years with fewer than four firms. We use firms’ modal SIC code across CRSP, Compustat, and CBSD, and the latter in case of a tie.

Ln Assets. Natural log of total assets (AT) from Compustat.

Ln Employees. Natural log of total employment (EMP) from Compustat.

Firm Profitability. Measured as $\frac{OIDBP}{AT}$, where $OIDBP$ is operating income before depreciation and AT is total assets, both from Compustat. This variable is winsorized at the 1st and 99th percentiles.

Blue-Collar Formative Exposure. Blue-collar formative exposure is an indicator variable equal to one if the fraction of employed persons aged 16 and over working in blue-collar occupations in the CEO’s birth county in the 1970 Census exceeded the 90th percentile of the cross-sectional distribution in our sample (approximately 50 percent).

Above Median Labor Intensity. Defined as an indicator variable equal to 1 if a firm’s labor intensity of R&D expenditures is above the median for our R&D sample calculated at the

CEO-firm-year level (or equivalently the firm level) and 0 otherwise. The data comes from the Business Enterprise Research and Development Survey produced by the National Center for Science and Engineering Statistics and Census Bureau. The data is published using NAICS codes. We merge this data with our panel using the NAICS codes provided by Compustat. We use the data for the most granular 6-digit NAICS codes when available, followed by 5-digit, 4-digit, 3-digit, and 2-digit codes to get the most granular estimate for the firm's industry available. We do not use the extremely broad estimates for all manufacturing firms and all non-manufacturing firms.

Market-to-Book Ratio. Defined as $\frac{PRCC_F \cdot CSHO}{AT}$, where $PRCC_F$ is the annual closing stock price from CRSP, $CSHO$ is the number of common shares outstanding from CRSP, and AT is the total assets from Compustat. This variable is winsorized at the 1st and 99th percentiles.

Tobin's Q. Defined as $\frac{DLC + DLTT + PRCC_F \cdot CSHO}{PPEGT}$, where DLC is debt in current liabilities from Compustat, $DLTT$ long-term debt from Compustat, and $PPEGT$ gross property, plant, and equipment from Compustat. We use the variable winsorized at the 5th and 95th percentiles and one-period lagged. We winsorize Q at the 95th percentile, as this measure is substantially more right-skewed than market-to-book, though our results do not depend on this winsorizing approach.

Employment Growth. Defined as $\frac{EMP_t - EMP_{t-1}}{EMP_{t-1}}$, where EMP is total employment from Compustat. This variable is winsorized at the 1st and 99th percentiles.

Debt Maturity. Defined as $\frac{DLTT - DD2}{DLC + DLTT}$, which is the fraction of total debt maturing in more than two years. $DD2$ is the fraction of debt maturing in two years. This variable is winsorized at the 1st and 99th percentiles.

P(Default). Defined as $\mathcal{N}(-\text{naïve } DD)$ from [Bharath and Shumway \(2008\)](#), where naïve DD is the naïve distance-to-default and $\mathcal{N}$ is the cumulative distribution function for the normal distribution. naïve DD is defined as $\frac{\ln[(E+F)/F] + (r_{it-1} - 0.5 \text{ naïve } \sigma_V^2)}{\text{naïve } \sigma_V}$, where E is the equity value of the firm, F is the face value of the debt, r_{it-1} is the stock return over the previous year, and σ_V is the volatility of firm value. E is defined as the fiscal year end share price times the number of shares outstanding from Compustat, $PRCC_F \cdot CSHO$. F is defined as the face value of short term debt plus one half the face value of long term debt from Compustat, $DLC + \frac{1}{2}DLTT$. r_{it-1} is defined as the proportion by which equity has increased (e.g., the ex-dividend return), approximated by the log difference in equity value from one year to the next, following the SAS code corresponding to the [Bharath and Shumway \(2004\)](#), a pre-publication version of [Bharath and Shumway \(2008\)](#). Intuitively, this is consistent with a simple model of firms that do not issue dividends. A firm that issues dividends suddenly reduces its value and thus moves closer to default. σ_V is defined as $\frac{E}{E+F}\sigma_E + \frac{F}{E+F}(0.05 + 0.25 * \sigma_E)$, where σ_E is the standard deviation of r_i which is the log difference in equity value from one year to the next.

B.2.1 CEO Overconfidence

We construct the Longholder proxy for overconfidence of [Malmendier, Tate and Yan \(2011\)](#), an option-based measure of overconfidence. See [Malmendier, Pezone and Zheng \(2023\)](#) for a detailed discussion of potential measures of overconfidence.

First, we link the “execid” identifiers from the ExecuComp data to the “personid” identifiers from Table 2 of the Thomson Insider Filing Data Files (Thomson data). While Wharton Research Data Services (WRDS) provides a linking table between the two identifiers, the linking table does not include many matches the authors deem unambiguous or highly probable. We therefore supplement

the linking table with our own matching procedure and the set of matches from [Malmendier, Pezone and Zheng \(2023\)](#).

Next, we construct our own linking table between the ExecuComp data and the Thomson data using token matching and manual review. Note that one execid can match multiple personid values as Thomson appears to make an effort have unique personid values for individuals, but they do not appear to always succeed at doing this nor do they claim to have a singular personid for one individual.^{1A.4}

For the next step, we start with the sample of CEOs from the ExecuComp data in our baseline regression for the 2002-2020 period, column (2) of Table 3, and the sample of CEOs from the Thomson data as defined by having a “rolecode1”, “rolecode2”, “rolecode3”, or “rolecode4” of “CEO”. We get the unique set of CEO identifier and 8-digit Committee on Uniform Securities Identification Procedures (CUSIP) number pairs for each data set. We then create a set of potential execid and personid matches by merging on the 8-digit CUSIP number. For CEO names in the ExecuComp data, we remove punctuation marks and apostrophes, strip any suffixes following a comma, and extract the first word of each last and first name. We keep all potential matches where the cleaned last name from the ExecuComp data matches any token, defined here as a string of letters separated from others by a space on one or both sides, in the cleaned name from the Thomson data. If there are multiple matches for a potential CEO from the ExecuComp data, but some also have a token match in the Thomson data name with the cleaned first name from the ExecuComp data, we keep only those matches and drop the matches with only a last name match for that CEO. This procedure yields 4,396 potential execid-personid matches. Then using the CEO name and company name from the ExecuComp data along with the cleaned name from the Thomson data, we manually review, leveraging google searches when necessary, the potential matches and keep only those that are unambiguous or highly probable to generate a set of personid and execid matches. This procedure yields 4,343 execid-personid matches, covering 4,262 unique CEOs.

The linking table from WRDS presents a set of matches between the ExecuComp data and the Thomson data. For each potential match, WRDS assigns a “match quality” score between 1 and 13 where 1 is deemed to be the highest quality match, a full name and CUSIP number match, and 13 is the lowest quality match, only a partial name match.^{1A.5} For each execid, we find the set of potential matches with the highest match quality score. If there is more than one match, we drop this execid from the linking table to avoid bad matches when it is unclear. Note that our previous merge should have already considered almost all relevant potential matches, and we evaluated which names seemed like plausible or likely matches in the case that there were multiple matches for one execid. Next, if one personid matched to multiple execids, we drop all potential matches for that personid. We then append this data to our previous linking table. After this step, we have 5,622 execid-personid matches, covering 5,500 unique CEOs.

To ensure our dataset is as large as possible and is consistent with [Malmendier, Pezone and Zheng \(2023\)](#), we then examine all of the potential execid-personid matches used by [Malmendier, Pezone and Zheng \(2023\)](#) for all CEOs, based on execid, in both samples that are not already in our combined linking table. We manually review these potential matches and add them to our linking table if they are unambiguous or highly probable. This procedure yields 5,676 execid-personid

^{1A.4} See documentation by WRDS for the Thomson data here: wrds-www.wharton.upenn.edu/pages/support/support-articles/lseg/insiders/unique-person-id-across-firms/.

^{1A.5} A partial name match can still be of very good quality because it can be the result of only one dataset using a nickname for the first name and/or including a middle initial.

matches, covering 5,547 unique CEOs.

Our calculation of the Longholder variable follows the exact same procedure as in [Malmendier, Pezone and Zheng \(2023\)](#) with one exception. The only difference is that [Malmendier, Pezone and Zheng \(2023\)](#) use only transactions where the variable “rolecode1” is “CEO” when calculating CEO overconfidence; however, we use all transactions where any of the four role code variables is “CEO”. This change only affects the number of transaction observations for 22 CEOs in our merged data. CEOs with no relevant transactions are dropped. This leaves us with coverage for 3,517 CEOs.

B.2.2 Blue-Collar Formative Exposure

This section describes how we generate a measure of blue-collar formative exposure based on the occupational demographics of the CEO’s birth county in 1970 (i.e., when CEOs in our sample were young). The measure is an indicator equal to one if the fraction of employed persons aged 16 and over working in blue-collar occupations in the CEO’s birth county exceeded the 90th percentile of the cross-sectional distribution within our sample (corresponding to a percentile cutoff of 50.2 percent).

We define blue-collar occupations using the following categories from the 1970 Census occupation classification: Craftsmen, Foremen, and Kindred Workers; Operatives, Except Transport; Transport Equipment Operatives; Laborers, Except Farm; Farmers and Farm Managers; and Farm Laborers and Farm Foremen. The county-level occupation data come from the 1970 Census on 2010 Geographies, obtained via Social Explorer.

To identify CEO birth counties, we use automated web searches via OpenAI’s GPT-5-nano model with high reasoning effort and web search enabled. For each CEO in our sample, we prompt the model with the CEO’s name and company affiliation using the following prompt:

I’d like to find out where [CEO Name], who is or was the CEO of [Company Name], was born. Please use clear, reputable sources (e.g., official bios, major publications, authoritative business profiles, Wikipedia). If the birthplace is found, then record your answer as:

City: (City Name)
County: (County Name)
State: (State Name)
Country: (Country Name)

Please include as many place identifiers as possible.

If the birthplace cannot be found after extensive research, please respond with “NA”.

Do not include any additional commentary or information beyond the birthplace details or NA if not found.

For example, if you learn that a CEO was born in Oakland, Alameda County, California, USA, your response should be:

City: Oakland
County: Alameda
State: California
Country: United States

The model was also provided with the following system message which the model treats as a guide for its behavior in responding to the prompt:

You are an AI assistant designed to conduct thorough and comprehensive internet research to accurately determine the birthplace of a CEO. You will explore a wide range of reputable sources—including Wikipedia, NNDB, trusted biographical databases, news archives, and additional reliable resources. Sources do not need to be included in the output. Please conduct thorough research across many reputable and diverse sources before concluding the information cannot be found.

We then match the returned county and state information to occupation data from the 1970 Census using the State, County, and City FIPS Reference Table from the U.S. Department of Transportation.^{1A.6} Prior to matching, we standardize county names returned by the model to align with the Census Bureau naming conventions. This included adjusting suffixes such as “County”, “Parish”, and “Municipio”, manually aligning names for independent cities, and other special cases.

Verifying Reliability. To verify the reliability of the large language model-based approach, we run the prompt three times for each CEO (Bastianello, Decaire and Guenzel 2025) and examine agreement across runs. Pairwise agreement rates are high—93.5% (Run 1 vs. Run 2), 93.2% (Run 1 vs. Run 3), and 93.9% (Run 2 vs. Run 3)—where agreement includes both cases in which the same county is returned and cases in which both runs return no county.

Across the three runs, 1,920 CEOs have a birthplace identified at least once. When all non-missing runs agree on the FIPS code, we use that result. Only 68 CEOs exhibit conflicting non-missing FIPS codes across runs. We manually review these cases and resolve each discrepancy by hand.

We also compare our data to the CEO birthplace data used in Bernile, Bhagwat and Rau (2017). Among the 549 CEOs with birth location information available in both samples, we find agreement on birth county in 88% of cases. The remaining discrepancies may reflect differences in coding rules—for example, one dataset may use the location where a CEO grew up, while the other records the location of birth.

B.3 CEO Mortality Data: Variable Definitions

Death. Based on hand-collected mortality data from Borgschulte et al. (2025). Censored at a cutoff date of October 1st, 2017.

Layoff. Layoffs are based on data from Hallock (1998), Billger and Hallock (2005), and Farber and Hallock (2009). This dataset provides a comprehensive list of layoff announcements collected from the *Wall Street Journal*.

Age. Based on hand-collected birth data from Borgschulte et al. (2025).

Industry Distress. The variable used in our analysis is equal to one if the CEO has experienced industry distress, as defined in Appendix-Section B.2.

Ln Assets. We define Ln assets as the natural logarithm of total assets from Compustat, variable code *AT*. We use assets as of the first year that the CEO has assumed the top position. When values are missing, we interpolate the missing values or, at the beginning and end of the sample, use the closest available year.

Ln Employees. We define Ln employees as the natural logarithm of total employment from Compustat, variable code *EMP*. We use employees as of the first year that the CEO has assumed

^{1A.6} Available at data.transportation.gov/Railroads/State-County-and-City-FIPS-Reference-Table/eeek5-pv8d/about_data.

the top position. When values are missing, we interpolate the missing values or, at the beginning and end of the sample, use the closest available year.

Firm Profitability. Firm profitability is measured as $\frac{OIDBP}{AT}$. We use profitability as of the first year that the CEO has assumed the top position. This variable is winsorized at the 1st and 99th percentiles. When values are missing, we interpolate the missing values or, at the beginning and end of the sample, use the closest available year.

Minimum Stock Performance. Minimum stock performance is defined as the worst stock performance in any six-month period during a CEO's tenure.

C Results Appendix

This section extends the discussion in Section 4.2 on alternative explanations for the relationship between CEO tenure and layoff propensity.

Firm (Under-)Performance. One alternative explanation for our results might be unobserved determinants of firm performance. That is, CEOs' layoff choices might not be explained by their concern about workers, but instead reflect fluctuations in firms' growth potential and investment opportunities. While such determinants are not necessarily increasing in CEO tenure, one might posit that they play an increasing role over the course of a CEO's tenure or that they are characteristic of CEOs who ultimately achieve long tenure.

To address such concerns, we extend our empirical model to include several additional measures of physical- and human-capital investment opportunities. Our baseline analysis in Table 3 already includes several measures related to firm productivity or performance, including firm profitability. In Appendix-Table D.7, we add several additional controls for employment growth, investment opportunities, debt maturity, and default risk. (See Appendix-Section B.2 for details on the construction of variables.) We consider two measures of capital productivity: lagged market-to-book ratio (Jovanovic and Rousseau 2002) and a more complex definition of Q from Livdan and Nezlobin (2021). As a measure of worker productivity, we use lagged net employment growth, based on Bilal et al. (2022). As a measure of debt maturity, we use long-term debt (debt with maturity greater than two years) as a percentage of total debt, following Chen, Xu and Yang (2021). As a measure of default risk, we use the naïve probability of default from Bharath and Shumway (2008), which they show outperforms the more computationally intensive application of Merton (1974) by Moody's KMV both in- and out-of-sample.^{IA.7}

As shown in Appendix-Table D.7, the tenure coefficient remains highly significant in all specifications and moves very little relative to the corresponding baseline specifications (without or with firm fixed effects) from Table 3. As expected, firms with higher labor and capital productivity, higher debt maturity, and a lower probability of default are less likely to implement layoffs. At the same time, the size and statistical significance of the coefficient on CEO tenure remains unaffected with the additional controls. Thus, measures of the marginal products of labor and capital, which are closely related to investment opportunities or debt maturity, do not appear to affect or drive our results.

CEO Power and Co-Opted Directors. Another alternative explanation relates to prior evidence on CEOs accumulating power over the course of their tenure. Pan, Wang and Weisbach (2016) document a positive association between CEO tenure and firm investment, and a negative association with disinvestment. They attribute the investment pattern primarily to increasing CEO power, measured by the fraction of the board appointed by the incumbent CEO.^{IA.8} While they do not link the decline in disinvestment to CEO power, we test for its role, in combination with empire building, as an underlying mechanism in our setting. To this end, we construct the percentage of new directors appointed during a CEO's tenure using data from the Institutional Shareholder Services (ISS), as well as additional indicator variables capturing a CEO's concentration of titles, specifically

^{IA.7} KMV, acquired by Moody's in 2002, developed a model based on Merton (1974) to calculate the probability that a firm will default on its debt within the next year. For more details, see Bharath and Shumway (2008).

^{IA.8} The idea is that director appointments during a CEO's tenure are, at least to some extent, under the CEO's influence, which empire-desiring CEOs may be able to use to their advantage, and increasingly so over their tenure. This idea was first formalized in Hermalin and Weisbach (1998); see also Coles, Daniel and Naveen (2014).

CEO–president and CEO–chairperson duality (Adams, Almeida and Ferreira 2005).

Table D.8 performs “horse race” regressions as in Pan, Wang and Weisbach (2016), with both CEO tenure and the CEO power measures as independent variables. In columns (1) and (3), we include only the new CEO power variables (the percent of co-opted directors as well as the duality measures) without including CEO tenure. In this case, we observe a negative link between new directors appointed and layoff propensity. However, when we add CEO tenure in columns (2) and (4), the CEO tenure effect dominates, whereas the CEO power effect becomes insignificant or changes sign. We conclude that increasing the power of empire-desiring CEOs is unlikely to be a primary mechanism of our findings.

Going beyond the measures employed in Pan, Wang and Weisbach (2016), we also consider the CEO entrenchment index of Bebchuk, Cohen and Ferrell (2009), which aggregates six governance provisions: staggered boards, limits to shareholder by-law amendments, poison pills, golden parachutes, and supermajority requirements for mergers and charter amendments. Since 2007, the first four measures above have been collected by ISS.^{1A.9} Using the ISS data, we construct a partial E-index, which ranges from 0 to 4, increasing by +1 for each provision. The resulting index is available for approximately two-thirds of our observations from our *Firm–Layoffs Data*. Columns (1) and (3) of Appendix-Table D.9 regress layoffs on an indicator for each level of the E-index from one to four, as well as the two duality controls. None of the E-index indicator variables have statistically significant coefficients. When we add back the CEO Tenure variable in columns (2) and (4), the estimated coefficients on *CEO Tenure* are similar to our baseline estimates, both in terms of magnitude and statistical significance.

Taken together, our results suggest that CEO entrenchment neither significantly drives layoff propensity, nor serves as a primary mechanism for our findings.

First-Year Effects. We also consider the possibility of endogenous “first-year effects,” such as new CEOs being hired with the explicit purpose of “taking a big bath” and implementing layoffs. While such effects cannot explain the post-turnover jump in layoff propensity from Figure 4, given the quasi-exogenous nature of the turnovers considered there, they might explain the behavior of other CEOs in their first year. More generally, CEOs may face less judgment for layoffs in the initial window of their tenure.

To address such concerns about special first-year effects, we re-estimate our empirical model excluding the first year of each CEO’s tenure. As shown in Appendix-Table D.10, the tenure coefficient estimates remain highly statistically significant in spite of the reduced sample size, and their magnitudes are, if anything, slightly larger and more stable than in our baseline regressions. In particular, they are even less affected by the introduction of firm fixed effects than the estimates on the full sample.

Other CEO Differences. A more general concern is that the estimated CEO tenure–layoff link might reflect other CEO traits that are correlated with tenure. To check for systematic differences in the propensity to lay off workers across CEOs who attain shorter versus longer tenure, we take a two-pronged approach.

First, we group our sample of CEOs into bins by their total tenure: short (up to five years), medium (above five and up to 10 years), and long (above 10 years). This split results in comparable subsample sizes in terms of the number of CEOs in each group ($N = 1,578$, $N = 1,426$, and

^{1A.9} The primary source of data in Bebchuk, Cohen and Ferrell (2009) is the Investor Responsibility Research Center (IRRC). However, the IRRC stopped producing this data after 2006.

$N = 1,674$, respectively). We then calculate the average layoff propensity tenure-year-by-tenure-year across CEOs within each subsample. As shown in Appendix-Figure D.3, the year-by-year layoff probabilities closely match each other across groups (i. e., up to 5 years for all groups, and up to 10 years for the groups with medium and long total tenure). That is, the graph suggests that the year-by-year propensities to lay off workers do *not* systematically differ across groups of CEOs who ultimately achieve shorter or longer tenure. Moreover, the graph confirms the strong association between cumulative tenure up to the layoff year and layoff probability, regardless of whether the CEO ultimately serves a shorter or longer overall tenure.

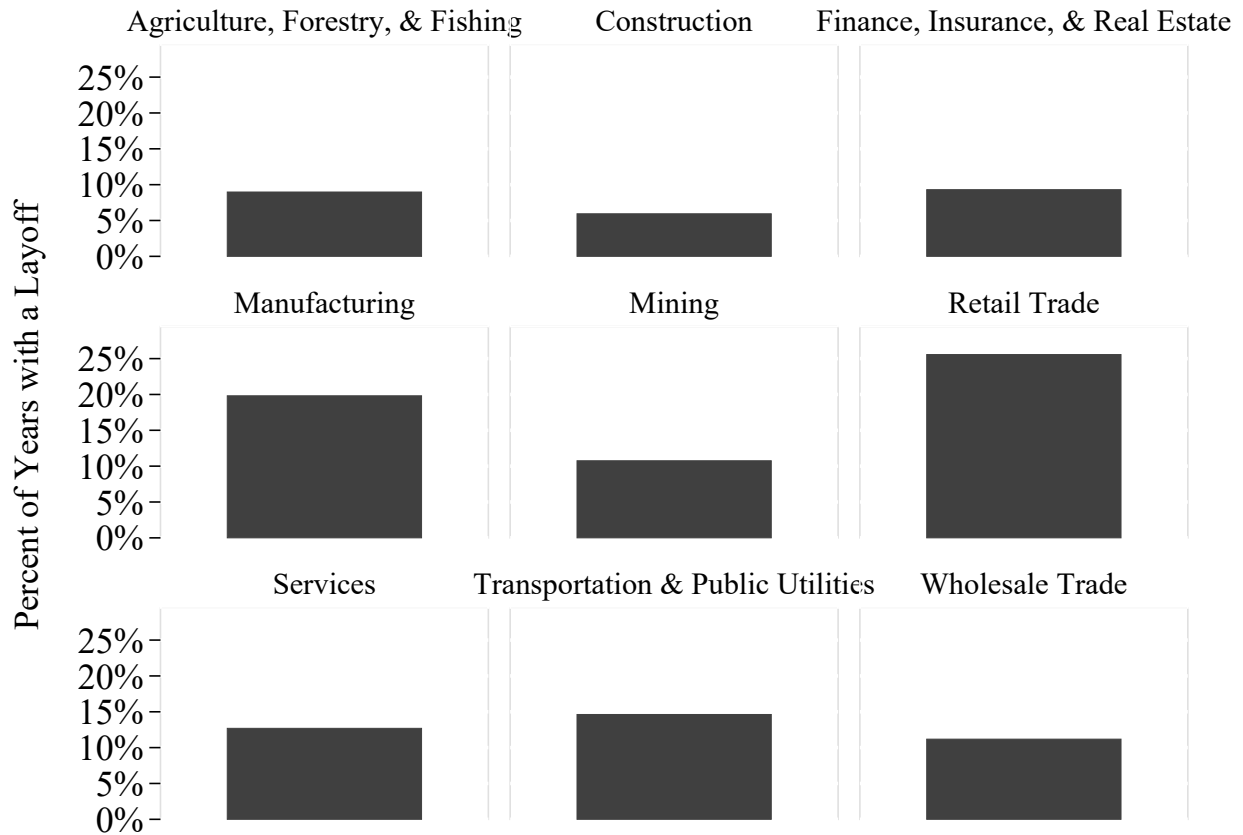
Going one step further in addressing concerns about CEO differences that might be correlated with tenure, we implement the approach of [Graham, Kim and Leary \(2020\)](#) and [Altonji and Shakotko \(1987\)](#), using the fraction of a CEO's total tenure completed so far as an instrument for tenure (measured in years).^{IA.10} The authors argue that the fraction measure serves as a plausible instrument, since the fraction of tenure completed is correlated with actual tenure (in years) but orthogonal to the CEO-firm match quality and other CEO characteristics that might affect tenure length. As they point out, all CEOs start at zero percent and complete one hundred percent of their tenure, which explains why the variation in the proportion of tenure completed cannot be correlated with CEO-firm pair-specific traits or other characteristics, such as match quality. Intuitively, this instrument purges the estimation of differences in average tenure across CEO-firm pairs.

Implementing this approach in Appendix-Table D.11, we estimate a strong first stage, both in the specification without firm fixed effects (column (1)) and in the specification with firm fixed effects (column (3)), with the F -statistic well above 10. In addition, the estimated effect of CEO tenure on the probability of a layoff remains significantly negative after instrumenting. The coefficients remain very similar in magnitude to the coefficients in Table 3 and are, in fact, larger after including firm fixed effects. These results lend additional support to the idea that the documented layoff-CEO tenure relation is not simply driven by cross-CEO differences.

^{IA.10} Specifically, the instrument is *CEO Tenure Prop* $\equiv (Tenure_{i,j,t} - \overline{Tenure_{i,j}}) / \max_t Tenure_{i,j,t}$.

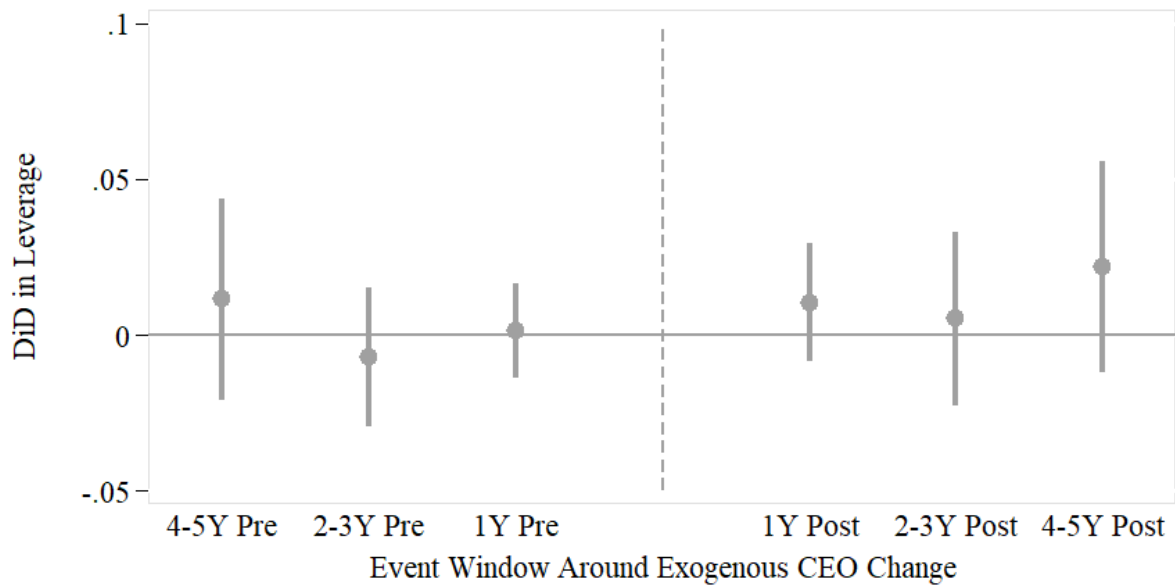
D Additional Figures and Tables

Appendix-Figure D.1. Layoffs by Industry



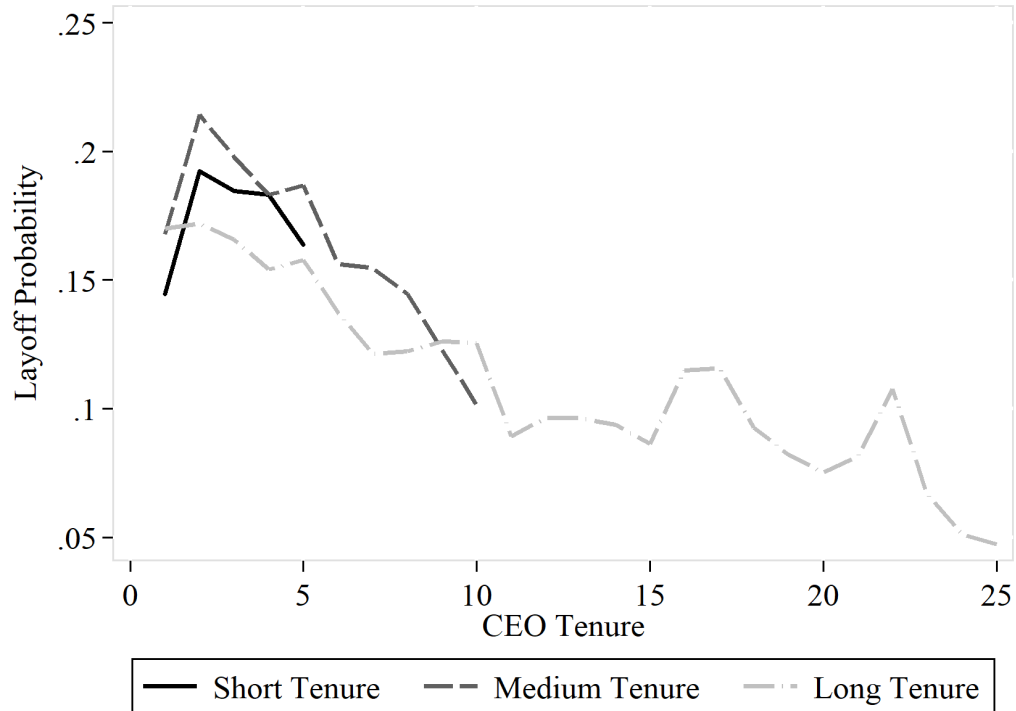
Notes: The figure plots the distribution of layoffs from our sample by industry classification based on SIC code. Data on layoffs assembled using NLP text processing and classification, as well as hand-coding of S&P Key Developments data as detailed in Appendix-Section B. SIC codes are provided by Compustat.

Appendix-Figure D.2. Placebo Test



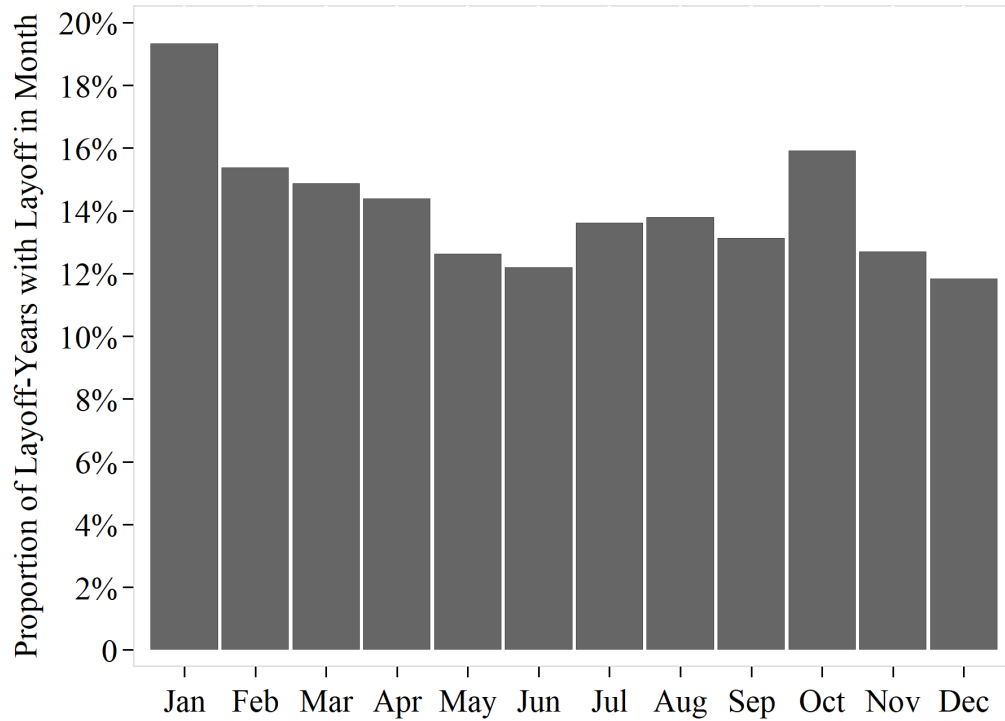
Notes: The figure plots the change in firm leverage (long-term debt over assets) around exogenous CEO departures relative to non-treated firms (stacked difference-in-differences analysis; see equation (4)). Exogenous CEO changes are defined as those occurring due to CEO death or illness of the incumbent CEO, and come from [Gentry et al. \(2021\)](#). See Section 4.1 for additional details. The figure also plots 95% confidence intervals based on standard errors clustered at the firm level.

Appendix-Figure D.3. Layoff Propensity and Tenure—Fixed CEO Differences?



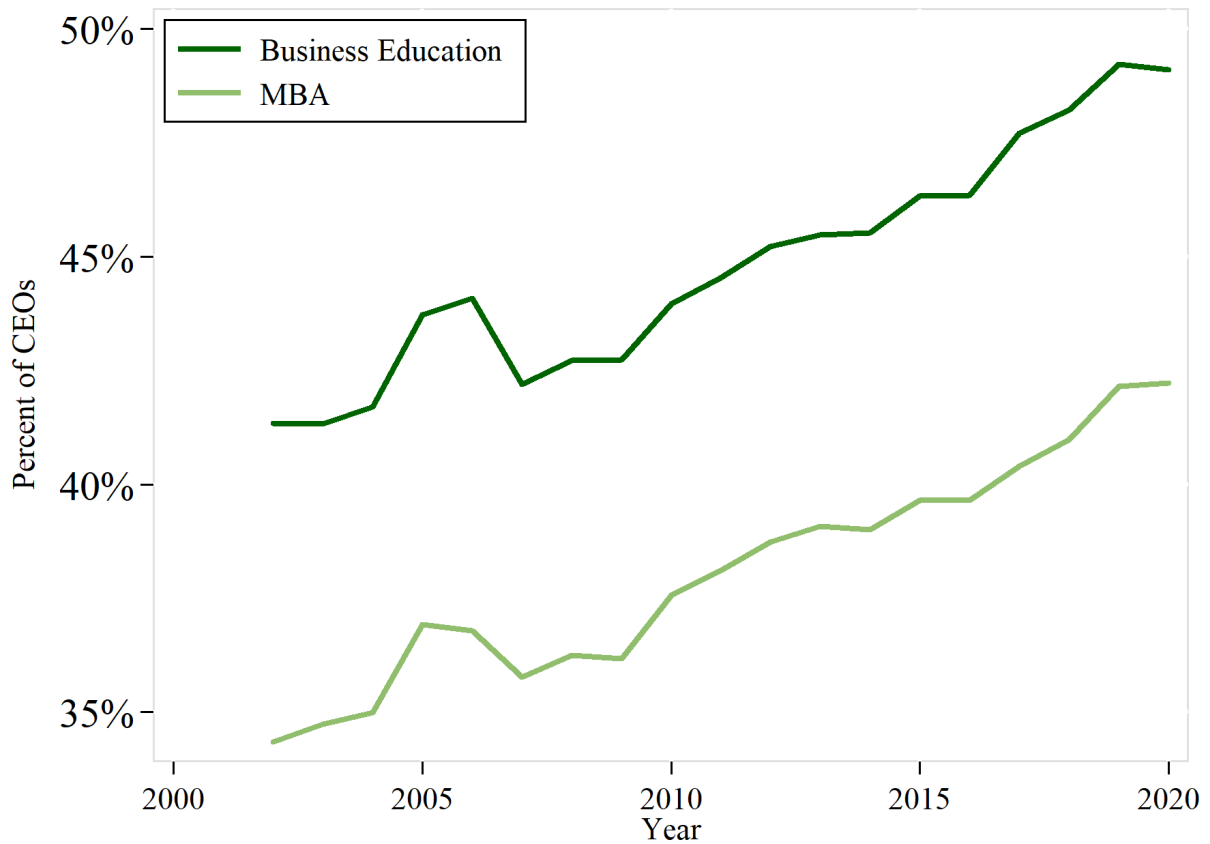
Notes: Layoff propensity by year of tenure plotted for three different CEO types based on overall/total tenure. Short tenure is up to five years. Medium tenure is more than five and up to ten years. Long tenure is more than ten years. Data on layoffs assembled using NLP text processing and classification, as well as hand-coding of S&P Key Developments data as detailed in Appendix-Section B. CEO tenure based on Execucomp data.

Appendix-Figure D.4. Layoffs By Month of the Year



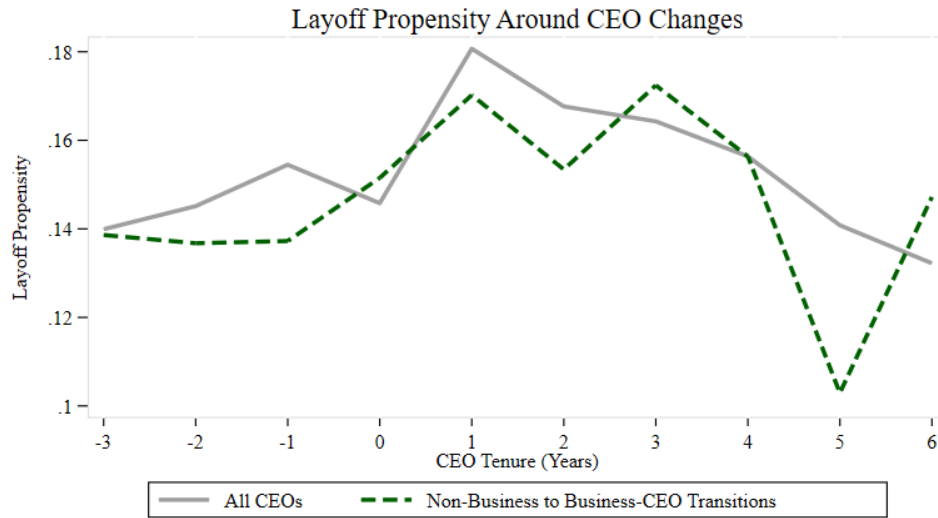
Notes: This figure plots the percent of layoff-years with a layoff in each calendar month. See Appendix-Section [B](#) for details on data construction.

Appendix-Figure D.5. CEOs with MBA and Business Education



Notes: The figure plots the proportion of CEOs in our sample with an MBA degree and the proportion with business education, meaning an MBA or any degree from a business school. Degree data is from BoardEx.

Appendix-Figure D.6. Layoff Propensity Around CEO Changes: All CEOs Vs. Business Education CEOs



Notes: The figure plots average layoff propensity around CEO changes for the full sample (gray) and for transitions from CEOs without business education to CEOs with business education. Year 0 denotes the first year of the new CEO's tenure. We construct event time relative to CEO turnover following [Pan, Wang and Weisbach \(2016\)](#), assigning each firm-year to a single event year. When consecutive CEOs' event windows overlap, a firm-year could otherwise be counted both as a post-turnover year and as a pre-turnover year. To avoid double counting, we classify such overlapping years as post-turnover years only, so that each firm-year enters the event study once.

Appendix-Table D.1. Layoffs and CEO Mortality—Stratified Cox Hazard Model

	(1)	(2)	(3)
Distress Layoff	0.292** (0.138)	0.287** (0.135)	0.283** (0.135)
Non-Distress Layoff	−0.103 (0.094)	−0.106 (0.092)	−0.112 (0.094)
Industry Distress	0.088 (0.115)	0.097 (0.114)	0.087 (0.122)
CEO Age	0.136*** (0.009)	0.138*** (0.009)	0.139*** (0.009)
Ln Assets	0.004 (0.073)	0.005 (0.071)	0.006 (0.071)
Ln Employees	−0.035 (0.070)	−0.038 (0.070)	−0.035 (0.072)
Firm Profitability	0.651 (0.697)	0.591 (0.700)	0.667 (0.725)
Year	−0.007 (0.009)		
Minimum Stock Performance			−0.145 (0.332)
Year FE	No	Yes	Yes
State FE	Yes	Yes	Yes
Observations	31,917	31,917	31,917

Notes: This table reports hazard ratios estimated from a [Cox \(1972\)](#) proportional hazards model. The dependent variable is an indicator that equals one if the CEO dies in a given year. Distress layoff is an indicator equal to one in year t if, prior to or in year t , a CEO has undertaken a layoff during industry-wide distress. Non-distress layoff is an indicator equal to one in year t if, prior to or in year t , a CEO has undertaken a layoff not during industry-wide distress. Industry distress is an indicator equal to one if the median firm of the industry had a stock price decline of at least 30% in the two years prior or the prior and current year. CEO age is the CEO's age at time t . Minimum stock performance is defined as the worst stock performance in any six-month period during the CEO's tenure. Ln assets and Ln employees from the first year that the CEO assumed the top office are included in all specifications. Firm profitability is included for all years that the CEO is in office, and then profitability from the CEO's final year in office is included thereafter. More details about the data are in Section 2 and Appendix-Section B. Survival models are stratified by firms' industry affiliation. Standard errors, clustered by industry, are in parentheses. Significance levels are indicated by $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

Appendix-Table D.2. Layoffs and CEO Mortality—CEO Tenure Control

	(1)	(2)	(3)
Distress Layoff	0.305** (0.139)	0.308** (0.140)	0.295** (0.142)
Non-Distress Layoff	0.034 (0.105)	0.023 (0.105)	0.007 (0.106)
Industry Distress	0.196* (0.119)	0.200* (0.120)	0.166 (0.126)
Cumulative CEO Tenure	−0.027* (0.014)	−0.026* (0.014)	−0.030** (0.014)
CEO Age	0.148*** (0.010)	0.151*** (0.011)	0.152*** (0.011)
Ln Assets	−0.158** (0.076)	−0.154** (0.077)	−0.152** (0.076)
Ln Employees	0.058 (0.075)	0.057 (0.076)	0.067 (0.075)
Firm Profitability	0.487 (0.704)	0.454 (0.715)	0.726 (0.735)
Year	0.007 (0.010)		
Minimum Stock Performance			−0.536* (0.325)
Year FE	No	Yes	Yes
State FE	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes
Observations	31,917	31,917	31,917

Notes: This table reports estimated coefficients from logit regression, where the dependent variable is an indicator that equals one if the CEO dies in a given year. The dependent variable is an indicator that equals one if the CEO dies in a given year. Distress layoff is an indicator equal to one in year t if, prior to or in year t , a CEO has undertaken a layoff during industry-wide distress. Non-distress layoff is an indicator equal to one in year t if, prior to or in year t , a CEO has undertaken a layoff not during industry-wide distress. Industry distress is an indicator equal to one if the median firm of the industry had a stock price decline of at least 30% in the two years prior or the prior and current year. Cumulative CEO tenure is CEO tenure while CEO and total tenure thereafter. CEO age is the CEO's age at time t . Minimum stock performance is defined as the worst stock performance in any six-month period during the CEO's tenure. Ln assets and Ln employees from the first year that the CEO assumed the top office are included in all specifications. Firm profitability is included for all years that the CEO is in office, and then profitability from the CEO's final year in office thereafter. More details about the data are in Section 2 and Appendix-Section B. Standard errors, clustered by industry, are shown in parentheses. Standard errors, clustered by industry, are in parentheses. Significance levels are indicated by $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

Appendix-Table D.3. Layoffs and CEO Mortality—Stratified Cox Hazard Model With Tenure Control

	(1)	(2)	(3)
Distress Layoff	0.313** (0.139)	0.306** (0.136)	0.296** (0.137)
Non-Distress Layoff	0.006 (0.104)	0.001 (0.103)	−0.014 (0.104)
Industry Distress	0.161 (0.127)	0.171 (0.126)	0.142 (0.133)
Cumulative CEO Tenure	−0.044*** (0.015)	−0.043*** (0.015)	−0.047*** (0.015)
CEO Age	0.118*** (0.010)	0.121*** (0.010)	0.122*** (0.010)
Ln Assets	−0.023 (0.071)	−0.023 (0.069)	−0.021 (0.069)
Ln Employees	−0.046 (0.072)	−0.047 (0.072)	−0.039 (0.072)
Firm Profitability	0.430 (0.738)	0.370 (0.742)	0.604 (0.751)
Year	−0.010 (0.009)		
Minimum Stock Performance			−0.484 (0.309)
Year FE	No	Yes	Yes
State FE	Yes	Yes	Yes
Observations	31,917	31,917	31,917

Notes: This table reports hazard ratios estimated from a [Cox \(1972\)](#) proportional hazards model. The dependent variable is an indicator that equals one if the CEO dies in a given year. Distress layoff is an indicator equal to one in year t if, prior to or in year t , a CEO has undertaken a layoff during industry-wide distress. Non-distress layoff is an indicator equal to one in year t if, prior to or in year t , a CEO has undertaken a layoff not during industry-wide distress. Industry distress is an indicator equal to one if the median firm of the industry had a stock price decline of at least 30% in the two years prior or the prior and current year. Cumulative CEO tenure is CEO tenure while CEO and total tenure thereafter. CEO age is the CEO's age at time t . Minimum stock performance is defined as the worst stock performance in any six-month period during the CEO's tenure. Ln assets and Ln employees from the first year that the CEO assumed the top office are included in all specifications. Firm profitability is included for all years that the CEO is in office, and then profitability from the CEO's final year in office thereafter. More details about the data are in Section 2 and Appendix-Section B. Survival models are stratified by firms' industry affiliation. Standard errors, clustered by industry, are in parentheses. Significance levels are indicated by $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

Appendix-Table D.4. Reasons Listed for Exogenous CEO Changes

Reason	Count
Accident	3
Cancer	21
Cardiovascular	8
Other specified illness	10
Stroke	6
Suicide	1
Unspecified illness/health	66

Notes: This table reports the reasons listed for exogenous CEO changes as identified in the notes provided in the CEO turnover data from [Gentry et al. \(2021\)](#). The categories were determined and classified by the authors.

Appendix-Table D.5. Layoff Propensity Around Exogenous CEO Changes

	(1)	(2)
Treat	−0.022 (0.029)	−0.002 (0.019)
T minus 5	−0.012 (0.049)	
T minus 4	0.062 (0.055)	
T minus 3	0.012 (0.047)	
T minus 2	0.040 (0.044)	
T minus 1	0.019 (0.040)	
T plus 1	0.137*** (0.040)	
T plus 2	0.103* (0.058)	
T plus 3	0.047 (0.050)	
T plus 4	0.120* (0.067)	
T plus 5	0.120 (0.073)	
Treat × Post		0.087*** (0.031)
Controls	Yes	Yes
Cohort-Year FE	Yes	Yes
Cohort-Industry FE	Yes	Yes
Observations	39,864	39,864

Notes: Stacked difference-in-differences analysis estimating the change in layoff propensity around exogenous CEO changes relative to non-treated firms. Controls are industry distress, Ln assets, Ln employees, and profitability. Exogenous CEO changes are those due to the death or illness of the incumbent CEO as identified in [Gentry et al. \(2021\)](#). The pre-period is the five years before the exogenous CEO change, and the post-period is the five years after the exogenous CEO change or the tenure of the CEO immediately following the exogenous turnover, whichever is shorter. Column (1) estimates equation (4) modified by using indicators for each pre- and post-year (instead of one indicator for Post) interacted with the treatment indicator. Column (2) estimates equation (4). Control variable coefficients are estimated separately for pre- and post-periods. Standard errors, clustered by firm, are shown in parentheses. Significance levels are indicated by * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Appendix-Table D.6. Layoff Propensity and CEO Tenure—Time-Varying Fixed Effects

	(1)	(2)	(3)	(4)
CEO Tenure / 100	−0.267*** (0.030)	−0.266*** (0.029)	−0.265*** (0.030)	−0.173*** (0.041)
Ln Assets	0.030*** (0.003)	0.029*** (0.003)	0.029*** (0.003)	0.024*** (0.007)
Ln Employees	0.026*** (0.003)	0.027*** (0.003)	0.027*** (0.003)	0.053*** (0.008)
Firm Profitability	−0.279*** (0.024)	−0.283*** (0.024)	−0.285*** (0.024)	−0.239*** (0.031)
Industry Distress	0.054*** (0.008)			
State FE	No	Yes	No	No
State-Year FE	Yes	No	Yes	Yes
Industry FE	Yes	No	No	No
Industry-Year FE	No	Yes	Yes	Yes
Firm FE	No	No	No	Yes
Observations	34,610	34,671	34,597	34,477

Notes: This table reports coefficients from linear regressions where the dependent variable is equal to one for significant layoff events, defined as CEO-firm-years with at least three layoff events or where 1% of the firm is laid off and 0 otherwise. Industry distress is an indicator equal to one if the median firm of the industry had a stock price decline of at least 30% in the two years prior or the prior and current year. Ln assets is the natural logarithm of total assets. Ln employees is the natural logarithm of total employment. Firm profitability is operating income before depreciation divided by total assets. CEO tenure is the number of years the CEO has been in that position. We rescale tenure by dividing by 100. Firm characteristics are from the beginning of the year. More details about the data are in Section 2 and Appendix-Section B. Standard errors, clustered by firm, are shown in parentheses. Significance levels are indicated by $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

Appendix-Table D.7. Layoff Propensity and CEO Tenure—Additional Controls

	(1)	(2)	(3)	(4)
CEO Tenure / 100	−0.265*** (0.036)	−0.252*** (0.036)	−0.172*** (0.049)	−0.163*** (0.050)
Industry Distress	0.057*** (0.009)	0.058*** (0.009)	0.050*** (0.009)	0.050*** (0.009)
Ln Assets	0.033*** (0.003)	0.038*** (0.004)	0.026*** (0.009)	0.035*** (0.009)
Ln Employees	0.022*** (0.003)	0.017*** (0.003)	0.053*** (0.009)	0.048*** (0.010)
Firm Profitability	−0.244*** (0.031)	−0.266*** (0.030)	−0.183*** (0.045)	−0.225*** (0.044)
Debt Maturity	−0.028*** (0.009)	−0.028*** (0.009)	−0.011 (0.009)	−0.010 (0.009)
$P(\text{Default})$	0.077*** (0.019)	0.081*** (0.020)	0.065*** (0.020)	0.070*** (0.020)
Employment Growth	−0.071*** (0.012)	−0.063*** (0.012)	−0.041*** (0.012)	−0.040*** (0.012)
Market-to-Book	−0.007*** (0.002)		−0.013*** (0.004)	
Q		−0.002*** (0.000)		−0.002*** (0.001)
Year FE	Yes	Yes	Yes	Yes
State FE	Yes	Yes	No	No
Industry FE	Yes	Yes	No	No
Firm FE	No	No	Yes	Yes
Observations	24,242	23,767	24,045	23,573

Notes: This table reports coefficients from linear regressions where the dependent variable is equal to one for significant layoff events, defined as CEO-firm-years with at least three layoff events or where 1% of the firm is laid off and 0 otherwise. Independent variables are defined in Appendix-Section B.2. More details about the data are in Section 2 and Appendix-Section B. Standard errors, clustered by firm, are shown in parentheses. Significance levels are indicated by $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

Appendix-Table D.8. Layoff Propensity and CEO Tenure—CEO Power

	(1)	(2)	(3)	(4)
CEO Tenure / 100		−0.407*** (0.053)		−0.213*** (0.081)
% of New Directors	−0.039*** (0.010)	0.030** (0.013)	−0.015 (0.012)	0.017 (0.018)
CEO and Chair	0.001 (0.007)	0.008 (0.007)	−0.001 (0.008)	0.002 (0.008)
CEO and President	−0.005 (0.006)	−0.007 (0.006)	0.006 (0.007)	0.004 (0.007)
Controls	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
State FE	Yes	Yes	No	No
Industry FE	Yes	Yes	No	No
Firm FE	No	No	Yes	Yes
Observations	23,621	23,621	23,425	23,425

Notes: This table reports coefficients from linear regressions where the dependent variable is equal to one for significant layoff events, defined as CEO-firm-years with at least three layoff events or where 1% of the firm is laid off and 0 otherwise. Controls are industry distress, Ln assets, Ln employees and firm profitability. % of new directors is between 0 and 1 and is calculated as the percent of directors appointed in the years following the CEO's appointment. CEO and Chair is an indicator equal to one if the CEO is also the chair of the board. CEO and President is an indicator equal to one if the CEO is also the president of the firm. Industry distress is an indicator equal to one if the median firm of the industry had a stock price decline of at least 30% in the two years prior or the prior and current year. Ln assets is the natural logarithm of total assets. Ln employees is the natural logarithm of total employment. Firm profitability is operating income before depreciation divided by total assets. CEO tenure is the number of years the CEO has been in that position. We rescale tenure by dividing by 100. Firm characteristics are from the beginning of the year. More details about the data are in Section 2 and Appendix-Section B. Standard errors, clustered by firm, are shown in parentheses. Significance levels are indicated by $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

Appendix-Table D.9. Layoff Propensity and CEO Tenure—CEO Entrenchment

	(1)	(2)	(3)	(4)
CEO Tenure / 100		−0.260*** (0.036)		−0.166*** (0.051)
E-Index = 1	0.018 (0.040)	0.021 (0.039)	0.038 (0.040)	0.036 (0.041)
E-Index = 2	0.020 (0.039)	0.021 (0.037)	0.024 (0.038)	0.021 (0.038)
E-Index = 3	0.019 (0.038)	0.019 (0.037)	0.035 (0.038)	0.033 (0.038)
E-Index = 4	0.012 (0.039)	0.012 (0.038)	0.027 (0.039)	0.025 (0.039)
CEO and Chair	−0.005 (0.006)	0.009 (0.006)	0.001 (0.007)	0.009 (0.007)
CEO and President	0.005 (0.005)	0.002 (0.005)	0.022*** (0.006)	0.018*** (0.006)
Controls	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
State FE	Yes	Yes	No	No
Industry FE	Yes	Yes	No	No
Firm FE	No	No	Yes	Yes
Observations	23,323	23,323	23,241	23,241

Notes: This table reports coefficients from linear regressions where the dependent variable is equal to one for significant layoff events, defined as CEO-firm-years with at least three layoff events or where 1% of the firm is laid off and 0 otherwise. Controls are industry distress, Ln assets, Ln employees and firm profitability. The E-index is defined as the count of four key governance provisions (staggered boards, limits to shareholder bylaw amendments, poison pills, and golden parachutes) that the firm has implemented. CEO and Chair is an indicator equal to one if the CEO is also the chair of the board. CEO and President is an indicator equal to one if the CEO is also the president of the firm. Industry distress is an indicator equal to one if the median firm of the industry had a stock price decline of at least 30% in the two years prior or the prior and current year. Ln assets is the natural logarithm of total assets. Ln employees is the natural logarithm of total employment. Firm profitability is operating income before depreciation divided by total assets. CEO tenure is the number of years the CEO has been in that position. We rescale tenure by dividing by 100. Firm characteristics are from the beginning of the year. More details about the data are in Section 2 and Appendix-Section B. Standard errors, clustered by firm, are shown in parentheses. Significance levels are indicated by * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Appendix-Table D.10. Layoffs and CEO Tenure—Drop First CEO-Year

	(1)	(2)	(3)	(4)
CEO Tenure / 100	−0.454*** (0.035)	−0.295*** (0.030)	−0.229*** (0.041)	−0.220*** (0.044)
Industry Distress		0.059*** (0.007)	0.046*** (0.007)	0.040*** (0.008)
Ln Assets		0.030*** (0.003)	0.029*** (0.007)	0.029*** (0.007)
Ln Employees		0.026*** (0.003)	0.049*** (0.008)	0.054*** (0.008)
Firm Profitability		−0.267*** (0.024)	−0.241*** (0.030)	−0.244*** (0.033)
CFO Tenure / 100				−0.054 (0.074)
Year FE	No	Yes	Yes	Yes
State FE	No	Yes	No	No
Industry FE	No	Yes	No	No
Firm FE	No	No	Yes	Yes
Observations	31,595	31,595	31,469	29,465

Notes: This table reports coefficients from linear regressions where the dependent variable is equal to one if there are at least three layoff events or 1% of the firm's employees are laid off in that CEO-firm-year and 0 otherwise. The sample is restricted to CEO-firm-years where the CEO has been in office for at least one year. Industry distress is an indicator equal to one if the median firm of the industry had a stock price decline of at least 30% in the two years prior or the prior and current year. Ln assets is the natural logarithm of total assets. Ln employees is the natural logarithm of total employment. Firm profitability is operating income before depreciation divided by total assets. CEO and CFO tenure measure the number of years each executive has held their respective position. For ease of interpretation, tenure is rescaled by dividing by 100. Firm characteristics are from the beginning of the year. More details about the data are in Section 2 and Appendix-Section B. Standard errors, clustered by firm, are shown in parentheses. Significance levels are indicated by * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Appendix-Table D.11. Layoff Propensity and CEO Tenure, IV Approach

	1st Stage	2nd Stage	1st Stage	2nd Stage
	(1)	(2)	(3)	(4)
CEO Tenure Proportion	0.128*** (0.002)		0.138*** (0.002)	
CEO $\widehat{\text{Tenure}} / 100$		-0.182*** (0.069)		-0.264*** (0.066)
Controls	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
State FE	Yes	Yes	No	No
Industry FE	Yes	Yes	No	No
Firm FE	No	No	Yes	Yes
Observations	34,683	34,683	34,565	34,565

Notes: This table reports coefficients from two-stage linear regression. The dependent variable in the first stage is CEO tenure. The dependent variable in the second stage is equal to one if there are at least three layoff events or 1% of the firm laid off in that CEO-firm-year and 0 otherwise. Controls are industry distress, Ln assets, Ln employees and firm profitability. CEO tenure proportion is defined as $(\text{Tenure}_{i,j,t} - \overline{\text{Tenure}_{i,j}}) / \max_t \text{Tenure}_{i,j,t}$ for CEO i at firm j in year t . The instrumented values of CEO tenure are used in the second stage. CEO tenure is the number of years the CEO has been in that position. We rescale tenure by dividing by 100. Control variables are from the beginning of the year and defined in Appendix-Section B.2. More details about the data are in Section 2 and Appendix-Section B. Standard errors, clustered by firm, are shown in parentheses. Significance levels are indicated by $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

Appendix-Table D.12. Fraction Laid Off and CEO Tenure, IV Approach

	1st Stage	2nd Stage	1st Stage	2nd Stage
	(1)	(2)	(3)	(4)
CEO Tenure Proportion	0.128*** (0.002)		0.139*** (0.002)	
CEO $\widehat{\text{Tenure}}$ / 100		-0.014** (0.006)		-0.027*** (0.006)
Controls	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
State FE	Yes	Yes	No	No
Industry FE	Yes	Yes	No	No
Firm FE	No	No	Yes	Yes
Observations	33,251	33,251	33,130	33,130

Notes: This table reports coefficients from two-stage linear regression. The dependent variable in the first stage is CEO tenure. The dependent variable in the second stage is the proportion of the firm laid off in that CEO-firm-year. Controls are industry distress, Ln assets, Ln employees and firm profitability. CEO tenure proportion is defined as $(\text{Tenure}_{i,j,t} - \overline{\text{Tenure}_{i,j}}) / \max_t \text{Tenure}_{i,j,t}$ for CEO i at firm j in year t . The instrumented values of CEO tenure are used in the second stage. CEO tenure is the number of years the CEO has been in that position. We rescale tenure by dividing by 100. Control variables are from the beginning of the year and defined in Appendix-Section B.2. More details about the data are in Section 2 and Appendix-Section B. Standard errors, clustered by firm, are shown in parentheses. Significance levels are indicated by $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

Appendix-Table D.13. Layoffs and CEO Tenure—Recession Periods, IV Approach

	(1)	(2)	(3)	(4)
CEO Tenure / 100	−0.137* (0.074)	−0.196*** (0.072)	−0.116 (0.074)	−0.177** (0.072)
CEO Tenure / 100 × Industry Distress	−0.260 (0.231)	−0.383* (0.222)	−0.406* (0.226)	−0.516** (0.217)
Controls	Yes	Yes	Yes	Yes
Control Interactions	No	No	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
State FE	Yes	No	Yes	No
Industry FE	Yes	No	Yes	No
Firm FE	No	Yes	No	Yes
Observations	34,683	34,565	34,683	34,565

Notes: This table reports coefficients from the second stage of two-stage linear regression. The dependent variables in the first stage regressions are CEO tenure and CEO Tenure interacted with industry distress. The dependent variable in the second stage is the proportion of the firm laid off in that CEO-firm-year. Controls are industry distress, Ln assets, Ln employees and firm profitability. CEO tenure proportion is defined as $(Tenure_{i,j,t} - \overline{Tenure_{i,j}}) / \max_t Tenure_{i,j,t}$ for CEO i at firm j in year t . The instrumented values of CEO tenure are used in the second stage. CEO tenure is the number of years the CEO has been in that position. We rescale tenure by dividing by 100. Control variables are from the beginning of the year and defined in Appendix-Section B.2. More details about the data are in Section 2 and Appendix-Section B. Standard errors, clustered by firm, are shown in parentheses. Significance levels are indicated by * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Appendix-Table D.14. R&D and CEO Tenure, IV Approach

	(1)	(2)	(3)	(4)
CEO $\widehat{\text{Tenure}} / 100$	0.008 (0.019)	0.037*** (0.013)	0.006 (0.019)	0.036*** (0.013)
CEO Tenure / $100 \times \widehat{\text{Industry Distress}}$	-0.097* (0.052)	-0.098*** (0.030)	-0.089* (0.051)	-0.089*** (0.029)
Controls	Yes	Yes	Yes	Yes
Control Interactions	No	No	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
State FE	Yes	No	Yes	No
Industry FE	Yes	No	Yes	No
Firm FE	No	Yes	No	Yes
Observations	15,052	14,994	15,052	14,994

Notes: This table reports coefficients from the second stage of two-stage linear regression. The dependent variables in the first stage regressions are CEO tenure and CEO Tenure interacted with industry distress. The dependent variable in the second stage is research and development expenditures divided by assets at the beginning of the year. Controls are industry distress, Ln assets, Ln employees and firm profitability. CEO tenure proportion is defined as $(\text{Tenure}_{i,j,t} - \overline{\text{Tenure}_{i,j}}) / \max_t \text{Tenure}_{i,j,t}$ for CEO i at firm j in year t . The instrumented values of CEO tenure are used in the second stage. CEO tenure is the number of years the CEO has been in that position. We rescale tenure by dividing by 100. Control variables are from the beginning of the year and defined in Appendix-Section B.2. More details about the data are in Section 2 and Appendix-Section B. Standard errors, clustered by firm, are shown in parentheses. Significance levels are indicated by * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

**Appendix-Table D.15. Research and Development Expenditures,
CEO Tenure, and Labor Intensity**

	(1)	(2)	(3)	(4)
CEO Tenure / 100	0.063 (0.040)	0.020 (0.018)	0.063 (0.040)	0.021 (0.018)
Industry Distress	0.006 (0.004)	0.008*** (0.003)	0.017 (0.017)	0.006 (0.010)
CEO Tenure × Industry Distress	−0.077** (0.035)	−0.067*** (0.025)	−0.079** (0.036)	−0.071*** (0.025)
Above Median Labor Intensity	−0.008 (0.006)		−0.008 (0.006)	
CEO Tenure × Above Median Labor Intensity	0.003 (0.051)	−0.007 (0.023)	0.003 (0.051)	−0.007 (0.023)
Industry Distress × Above Median Labor Intensity	−0.004 (0.005)	−0.007* (0.004)	−0.005 (0.005)	−0.007* (0.004)
CEO Tenure × Industry Distress × Above Median Labor Intensity	0.022 (0.049)	0.066** (0.031)	0.023 (0.049)	0.069** (0.031)
Controls	Yes	Yes	Yes	Yes
Control Interactions	No	No	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
State FE	Yes	No	Yes	No
Industry FE	Yes	No	Yes	No
Firm FE	No	Yes	No	Yes
Observations	14,355	14,305	14,355	14,305

Notes: This table reports coefficients from linear regressions where the dependent variable is research and development expenditures divided by assets at the beginning of the year. Control variables are beginning-of-year firm profitability, Ln assets, and Ln employees. Control interactions are interactions between each of the control variables and industry distress. Above median labor intensity is an indicator equal to one if the firm's industry's labor intensity is above the median for our sample. Industry distress is an indicator equal to one if the median firm of the industry had a stock price decline of at least 30% in the two years prior or the prior and current year. Ln assets is the natural logarithm of total assets. Ln employees is the natural logarithm of total employment. Firm profitability is operating income before depreciation divided by total assets. CEO tenure is the number of years the CEO has been in that position. We rescale tenure by dividing by 100. Firm characteristics are from the beginning of the year. More details about the data are in Section 2 and Appendix-Section B. Standard errors, clustered by firm, are shown in parentheses. Significance levels are indicated by * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Appendix-Table D.16. Layoff Propensity and CEO Tenure—Layoffs Around Holiday Season With Extended Layoff-Year Window

	Layoff-Years + Q1		Layoff-Years + Q1 + Q2	
	(1)	(2)	(3)	(4)
CEO Tenure / 100	−0.271*** (0.061)	−0.298*** (0.112)	−0.249*** (0.057)	−0.261** (0.106)
Controls	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
State FE	Yes	No	Yes	No
Industry FE	Yes	No	Yes	No
Firm FE	No	Yes	No	Yes
Observations	6,585	6,144	7,133	6,787

Notes: This table reports coefficients from linear regressions where the dependent variable is equal to one for years with at least one layoff in December and zero otherwise. Controls are industry distress, Ln assets, Ln employees and firm profitability. Layoff-Years + Q1 are CEO-firm-years with at least one layoff either in the CEO-firm-year or the three following months. Layoff-Years + Q1 + Q2 are CEO-firm-years with at least one layoff either in the CEO-firm-year or the six following months. Industry distress is an indicator equal to one if the median firm of the industry had a stock price decline of at least 30% in the two years prior or the prior and current year. Ln assets is the natural logarithm of total assets. Ln employees is the natural logarithm of total employment. Firm profitability is operating income before depreciation divided by total assets. CEO tenure is the number of years the CEO has been in that position. We rescale tenure by dividing by 100. Firm characteristics are from the beginning of the year. More details about the data are in Section 2 and Appendix-Section B. More details about the data are in Section 2 and Appendix-Section B. Standard errors, clustered by firm, are shown in parentheses. Significance levels are indicated by * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Appendix-Table D.17. Layoff Propensity and CEO Tenure—Holiday Layoffs, IV Approach

	Full Sample		Layoff-Years	
	(1)	(2)	(3)	(4)
CEO $\widehat{\text{Tenure}} / 100$	−0.177*** (0.030)	−0.174*** (0.029)	−0.925*** (0.179)	−1.116*** (0.201)
Controls	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
State FE	Yes	No	Yes	No
Industry FE	Yes	No	Yes	No
Firm FE	No	Yes	No	Yes
Observations	34,683	34,565	5,722	5,174

Notes: This table reports coefficients from the second stage of two-stage linear regression. The dependent variable in the first stage is CEO tenure. The dependent variable in the second stage is equal to one for years with at least one layoff in December and zero otherwise. Controls are industry distress, Ln assets, Ln employees and firm profitability. CEO tenure proportion is defined as $(\text{Tenure}_{i,j,t} - \overline{\text{Tenure}_{i,j}}) / \max_t \text{Tenure}_{i,j,t}$ for CEO i at firm j in year t . The instrumented values of CEO tenure are used in the second stage. CEO tenure is the number of years the CEO has been in that position. We rescale tenure by dividing by 100. Control variables are from the beginning of the year and defined in Appendix-Section B.2. More details about the data are in Section 2 and Appendix-Section B. Standard errors, clustered by firm, are shown in parentheses. Significance levels are indicated by * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Appendix-Table D.18. Layoff Propensity and CEO Tenure—White-Collar Layoffs, IV Approach

	Full Sample		Layoff-Years
	(1)	(2)	(3)
CEO $\widehat{\text{Tenure}} / 100$	−0.060* (0.034)	−0.060* (0.033)	−0.043 (0.195)
Controls	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
State FE	Yes	No	Yes
Industry FE	Yes	No	Yes
Firm FE	No	Yes	No
Observations	34,683	34,565	5,722

Notes: This table reports coefficients from the second stage of two-stage linear regression. The dependent variable in the first stage is CEO tenure. The dependent variable in the second stage is equal to one for years with at least one white-collar layoff and zero otherwise. Controls are industry distress, Ln assets, Ln employees and firm profitability. CEO tenure proportion is defined as $(\text{Tenure}_{i,j,t} - \overline{\text{Tenure}_{i,j}}) / \max_t \text{Tenure}_{i,j,t}$ for CEO i at firm j in year t . The instrumented values of CEO tenure are used in the second stage. CEO tenure is the number of years the CEO has been in that position. We rescale tenure by dividing by 100. Control variables are from the beginning of the year and defined in Appendix-Section B.2. More details about the data are in Section 2 and Appendix-Section B. Standard errors, clustered by firm, are shown in parentheses. Significance levels are indicated by * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Appendix-Table D.19. Layoff Propensity and CEO Tenure—Geography, IV Approach

	HQ-State Layoff		HQ-City Layoff	
	(1)	(2)	(3)	(4)
CEO $\widehat{\text{Tenure}} / 100$	−0.091** (0.042)	−0.066** (0.028)	−0.245 (0.164)	−0.347 (0.480)
Sample	Full	Full	Layoff-Years	HQ-State Layoff-Years
Controls	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes
Firm FE	No	No	No	No
Observations	34,683	34,683	5,722	1,406

Notes: This table reports coefficients from the second stage of two-stage linear regression. The dependent variable in the first stage is CEO tenure. The dependent variables in the second stage are indicator variables that equal one if a firm makes at least one layoff at the respective geographic level in a given year and zero otherwise. Layoff-years are CEO-firm-years with at least one layoff. HQ-state layoff-years are CEO-firm-years with any layoff in the state of the firm's headquarters. Controls are industry distress, Ln assets, Ln employees and firm profitability. CEO tenure proportion is defined as $(\text{Tenure}_{i,j,t} - \overline{\text{Tenure}_{i,j}}) / \max_t \text{Tenure}_{i,j,t}$ for CEO i at firm j in year t . The instrumented values of CEO tenure are used in the second stage. CEO tenure is the number of years the CEO has been in that position. We rescale tenure by dividing by 100. Control variables are from the beginning of the year and defined in Appendix-Section B.2. More details about the data are in Section 2 and Appendix-Section B. Standard errors, clustered by firm, are shown in parentheses. Significance levels are indicated by * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Appendix-Table D.20. Layoff Propensity and CEO Tenure—Holiday Layoffs, Heterogeneity in Prosociality

	Full Sample	CEO Empathy / OC Sample	
	(1)	(2)	(3)
CEO Tenure / 100	−0.407*** (0.129)	−0.365** (0.180)	
CEO Tenure × Empathy			−0.771*** (0.278)
CEO Tenure × Overconfidence			−0.194 (0.215)
CEO Overconfidence		−0.025 (0.023)	−0.050* (0.029)
<i>p</i> -value for equality of coefficients on CEO Tenure × Empathy and CEO Tenure × Overconfidence			0.0608
Controls	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes
Observations	5,174	3,732	3,732

Notes: This table reports coefficients from linear regressions where the dependent variable is equal to one for years with at least one layoff in December and zero otherwise. Controls are industry distress, Ln assets, Ln employees and firm profitability. CEO Overconfidence is an indicator variable defined based on the Longholder measure of managerial overconfidence ([Malmendier and Tate 2005](#), [Malmendier, Pezone and Zheng 2023](#)). CEO Empathy is an indicator variable equal to one if the CEO is not classified as overconfident, motivated by [Kaplan, Klebanov and Sørensen \(2012\)](#) and [Kaplan, Sørensen and Zakolyukina \(2022\)](#) that identify overconfidence and empathy are primary distinguishing characteristics of managers. Industry distress is an indicator equal to one if the median firm of the industry had a stock price decline of at least 30% in the two years prior or the prior and current year. Ln assets is the natural logarithm of total assets. Ln employees is the natural logarithm of total employment. Firm profitability is operating income before depreciation divided by total assets. CEO tenure is the number of years the CEO has been in that position. We rescale tenure by dividing by 100. Firm characteristics are from the beginning of the year. More details about the data are in Section 2 and Appendix-Section B. Standard errors, clustered by firm, are shown in parentheses. Significance levels are indicated by * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Appendix-Table D.21. Layoff Propensity and CEO Tenure—Business Education CEOs

	(1)	(2)	(3)
CEO Tenure / 100	−0.187*** (0.039)	−0.154*** (0.048)	−0.159*** (0.047)
CEO with Business Education			−0.010 (0.008)
Sample	Full Sample	Degree Info Subsample	Degree Info Subsample
Controls	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes
Observations	34,565	27,722	27,722

Notes: This table reports coefficients from linear regressions where the dependent variable is equal to one for significant layoff events, defined as CEO-firm-years with at least three layoff events or where 1% of the firm is laid off and 0 otherwise. Controls are industry distress, Ln assets, Ln employees and firm profitability. Column (1) reprints column (3) of Table 3. Column (2) re-estimates column (1) using the subsample of CEOs for whom information on educational background is available. Column (3) adds a control for whether the CEO has a business education. Industry distress is an indicator equal to one if the median firm of the industry had a stock price decline of at least 30% in the two years prior or the prior and current year. Ln assets is the natural logarithm of total assets. Ln employees is the natural logarithm of total employment. Firm profitability is operating income before depreciation divided by total assets. CEO tenure is the number of years the CEO has been in that position. We rescale tenure by dividing by 100. Firm characteristics are from the beginning of the year. More details about the data are in Section 2 and Appendix-Section B. Standard errors, clustered by firm, are shown in parentheses. Significance levels are indicated by * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.